

Capacitación de secuenciación genómica y análisis bioinformático

Control de Calidad y Ensamblaje

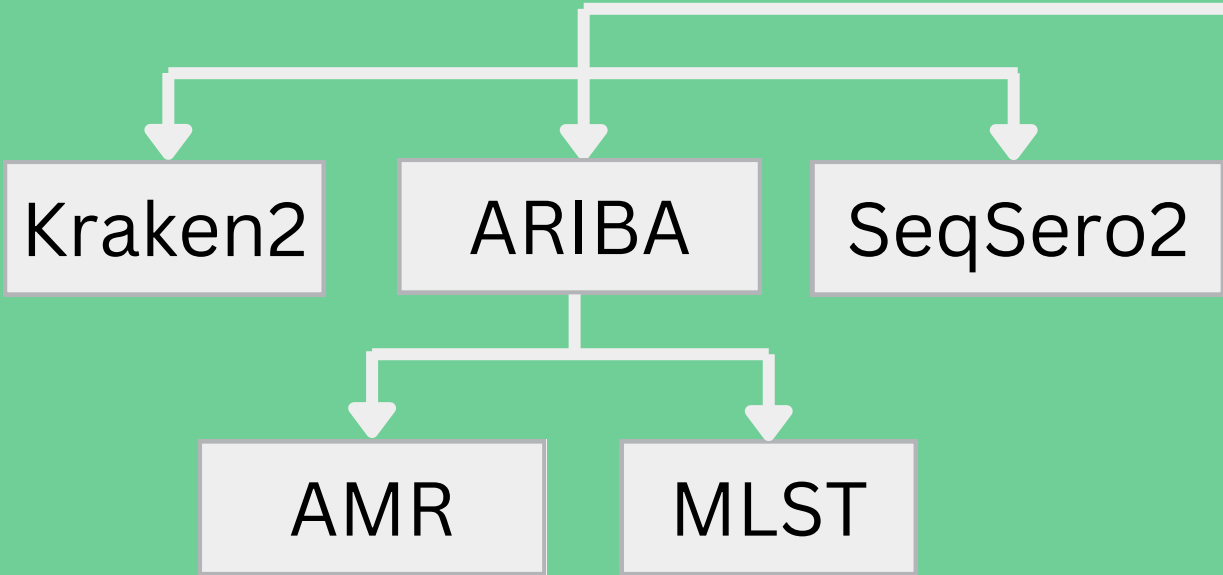
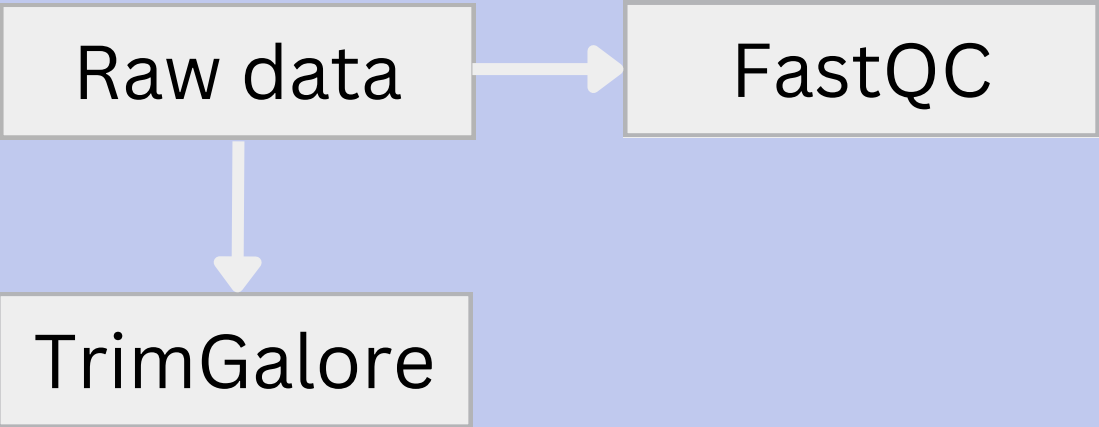
MSc. Melany Calderón Osorno

mcaldersono@inciensa.sa.cr

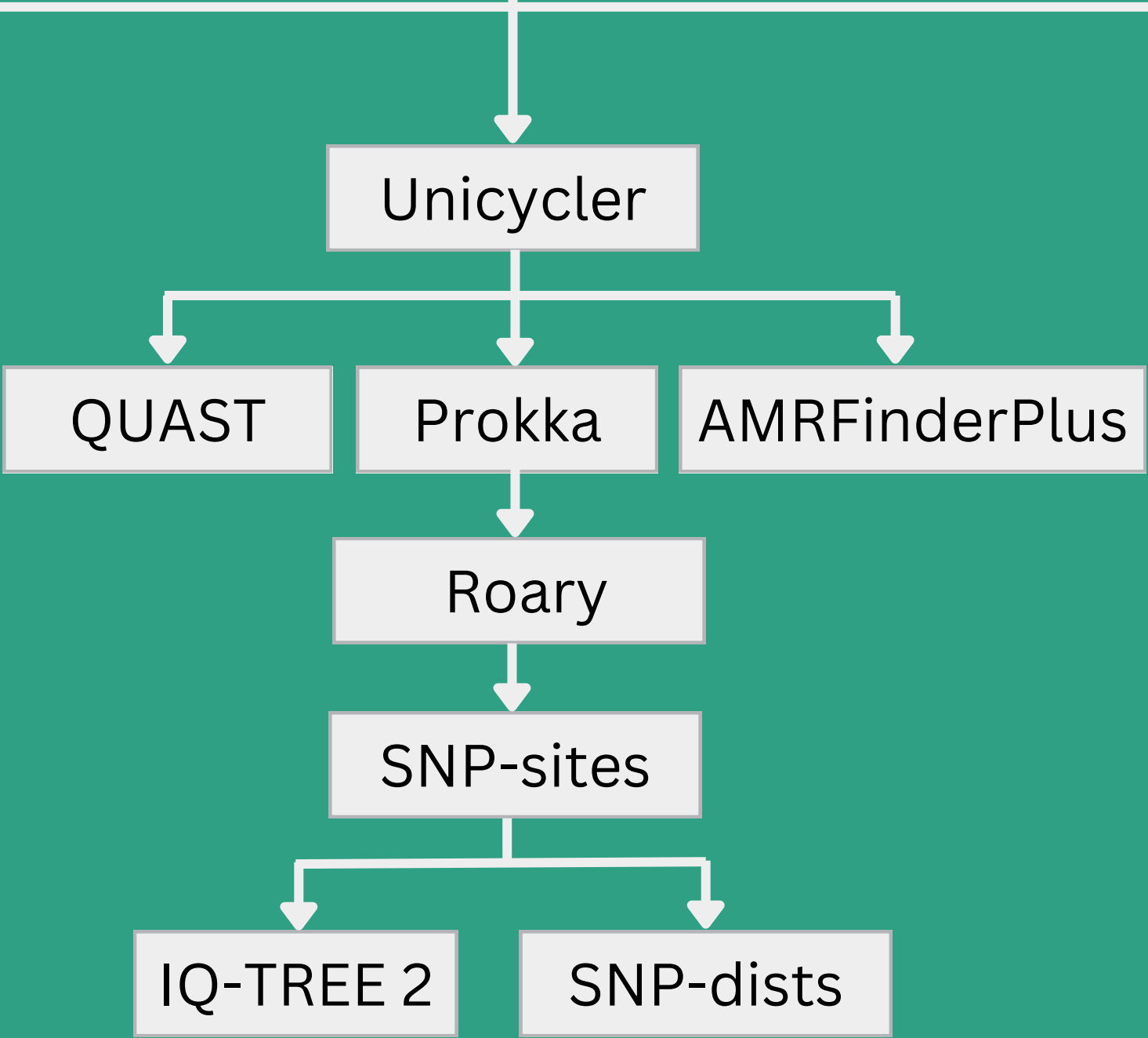
Laboratorio de genómica y biología molecular

Junio, 2026

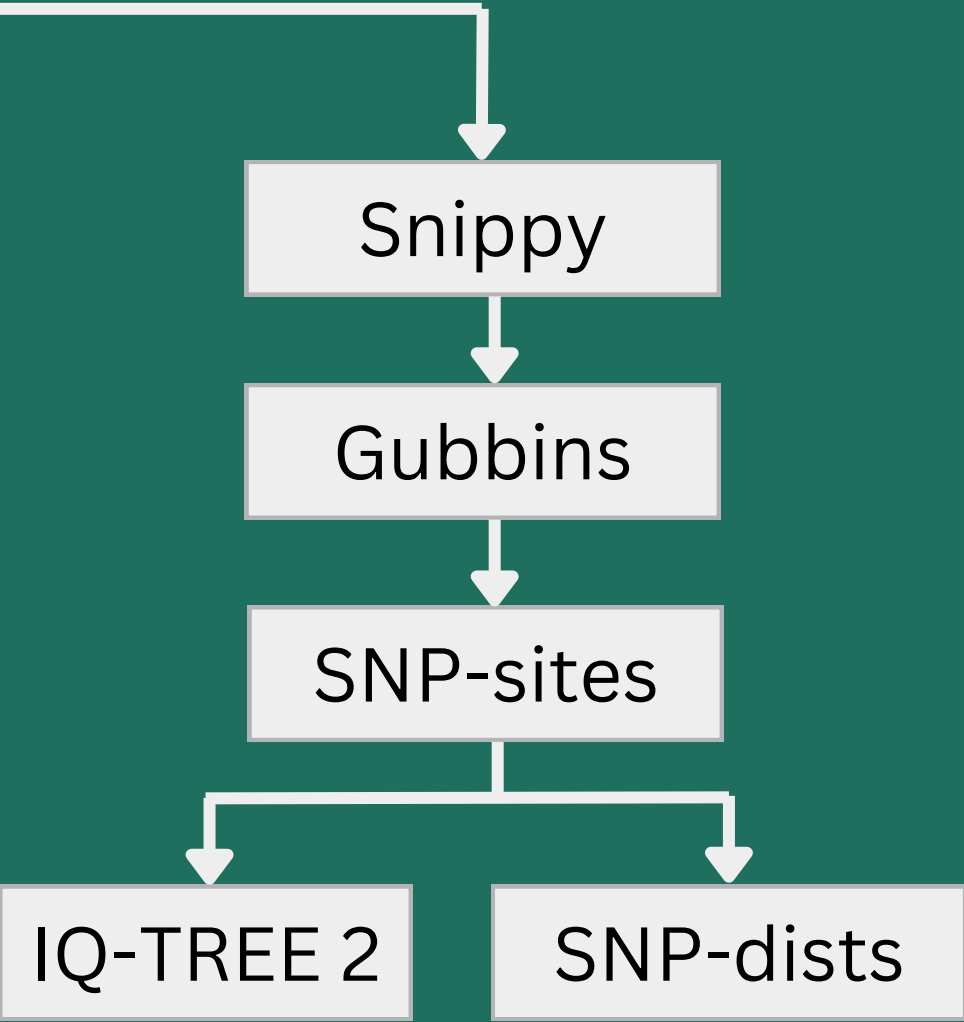
Preprocesamiento



Análisis a partir de lecturas

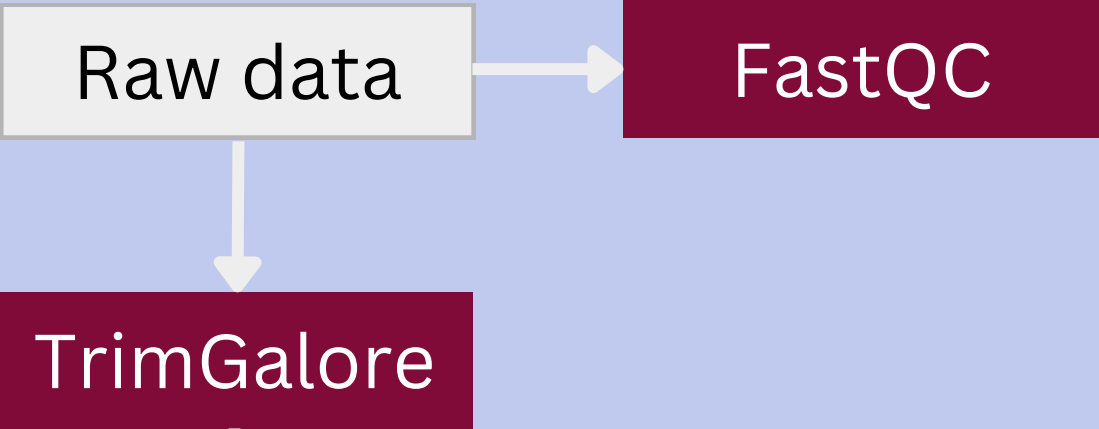


Análisis a partir de ensamblaje de novo

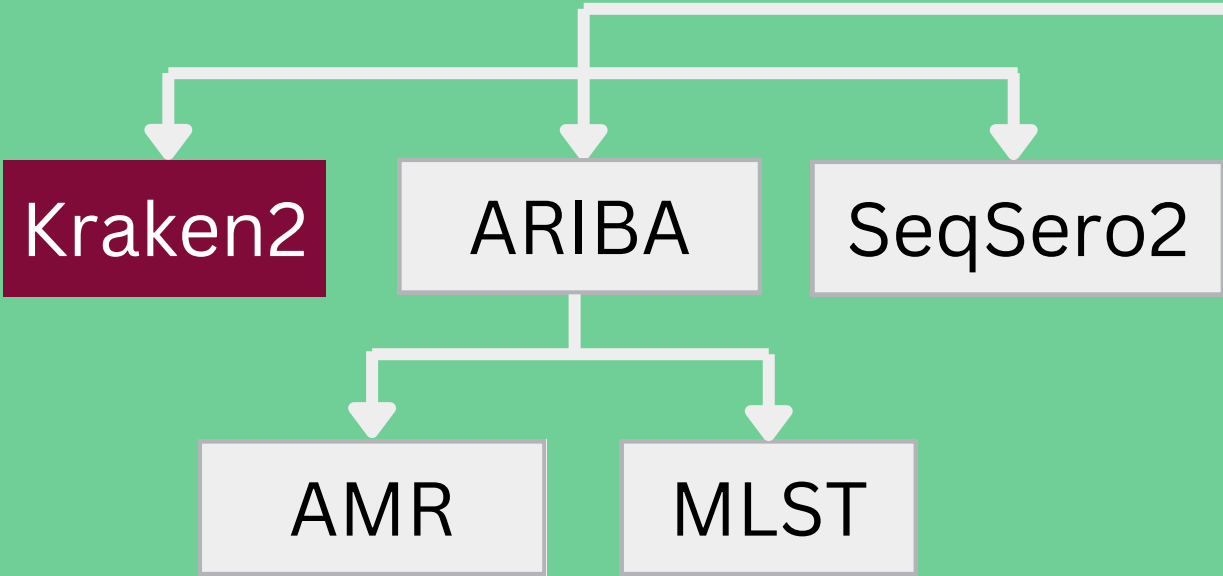


Análisis a partir de ensamblaje de referencia

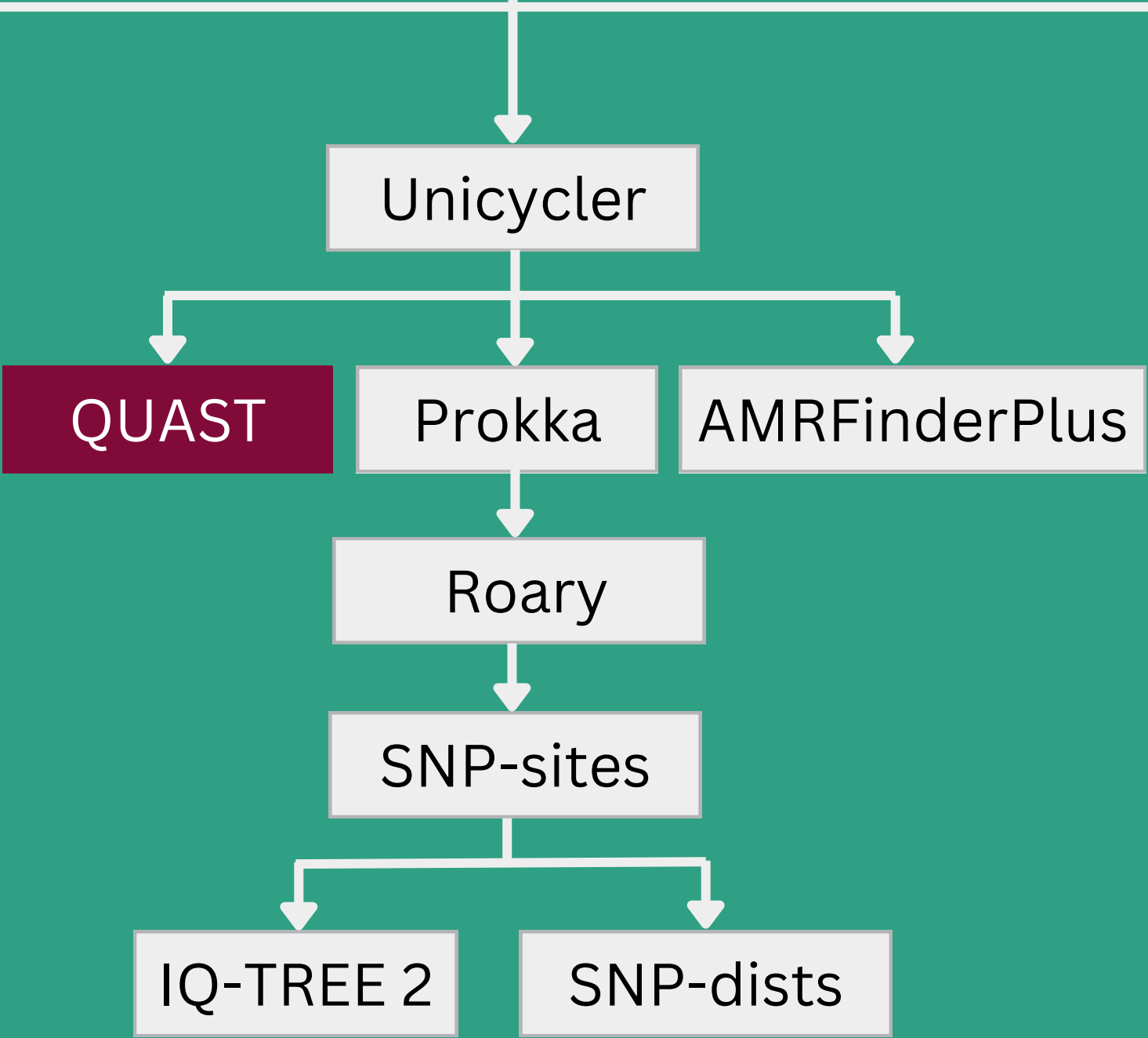
Preprocesamiento



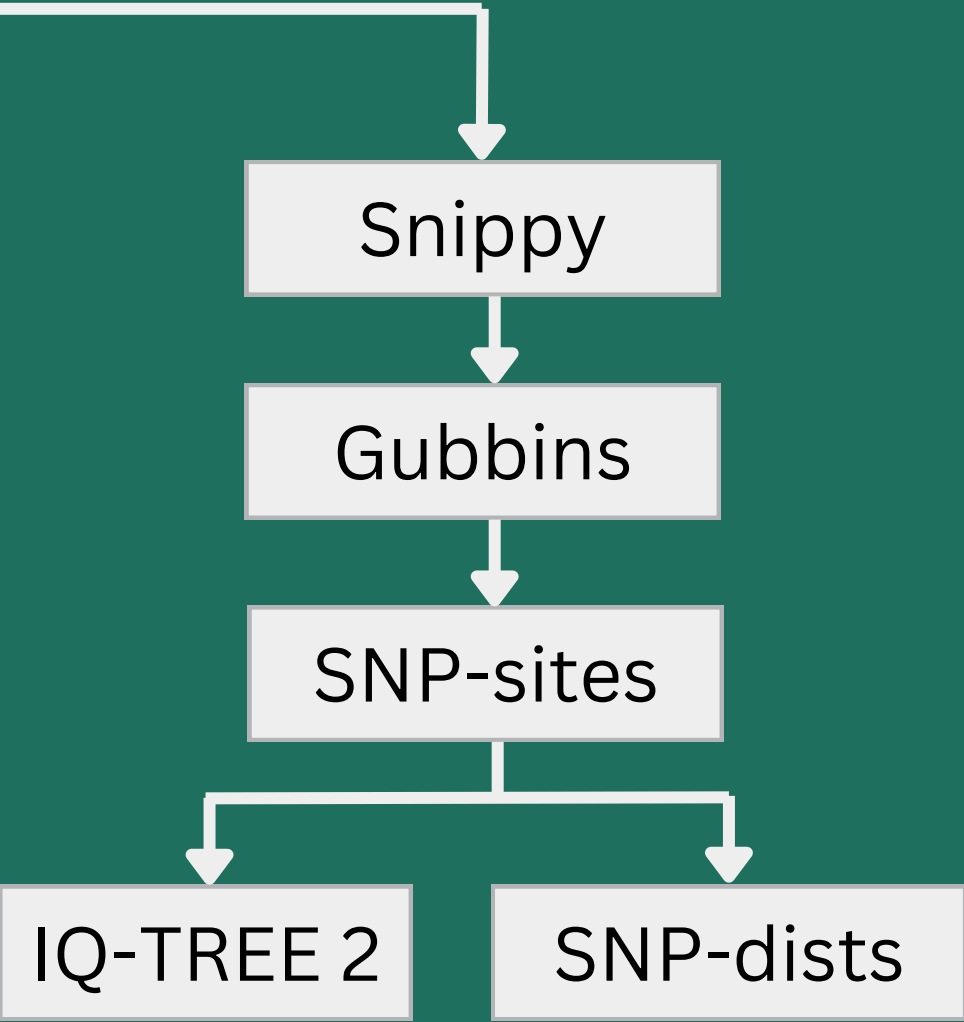
Control de calidad



Análisis a partir de lecturas



Análisis a partir de ensamblaje de novo

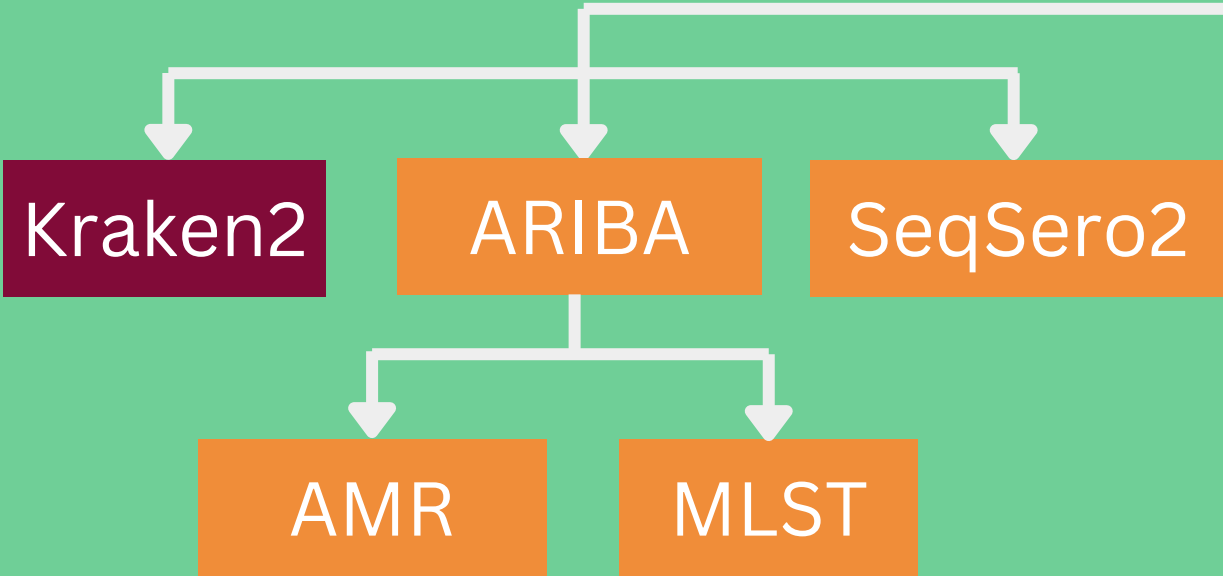
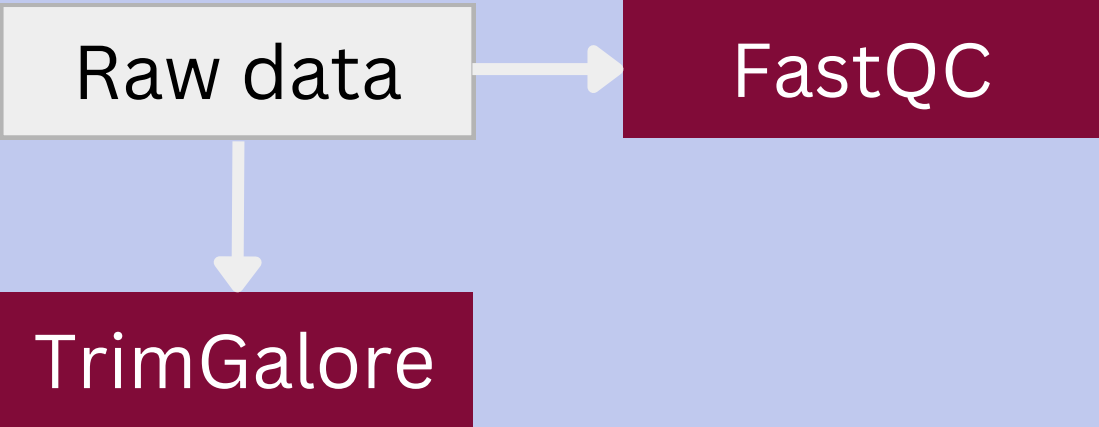


Análisis a partir de ensamblaje de referencia

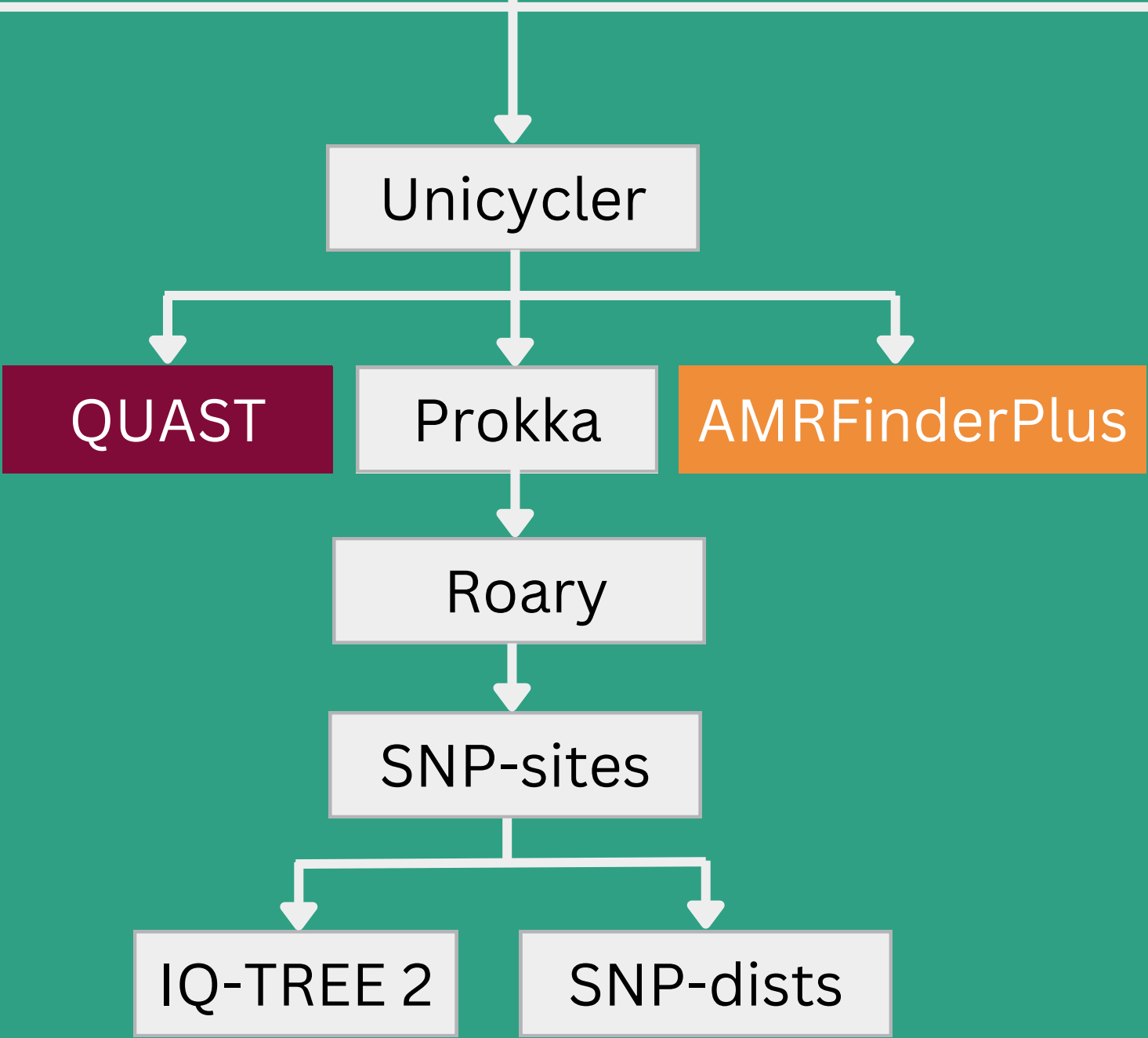
Preprocesamiento

Control de calidad

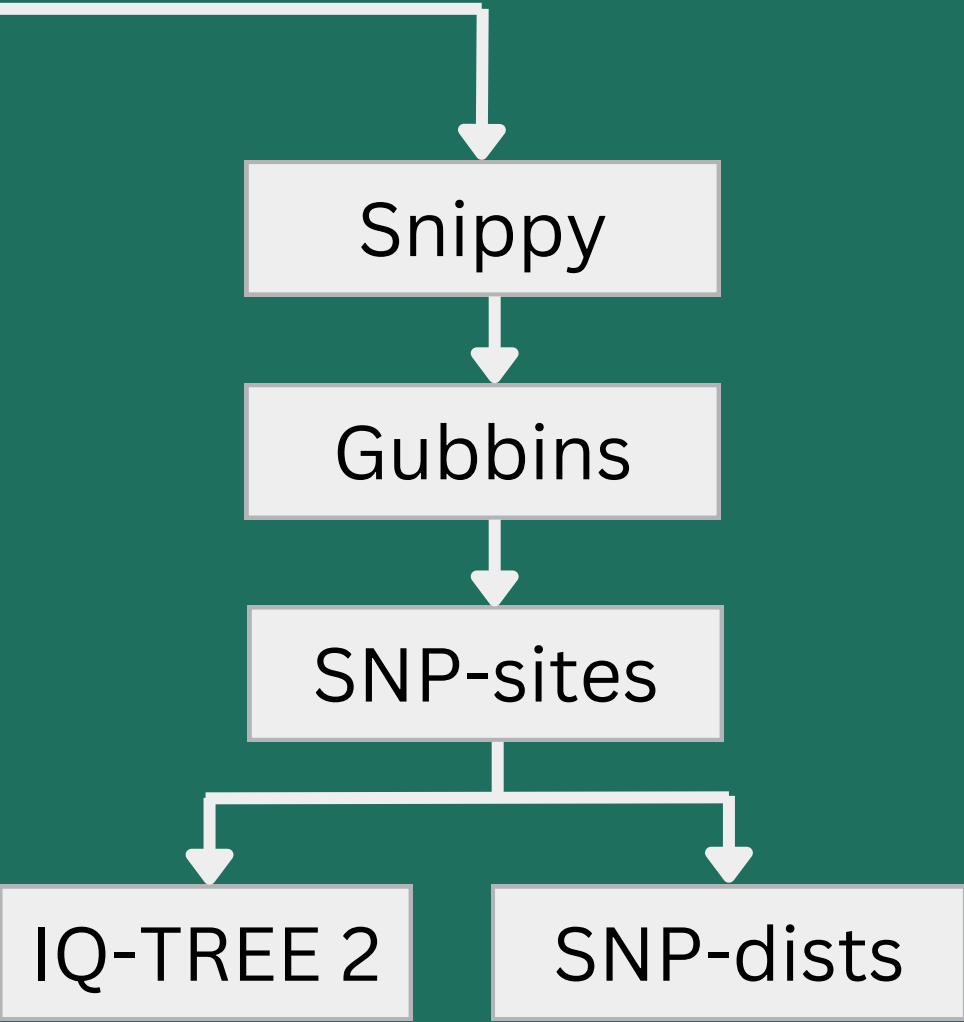
Caracterización



Análisis a partir de lecturas



Análisis a partir de ensamblaje de novo



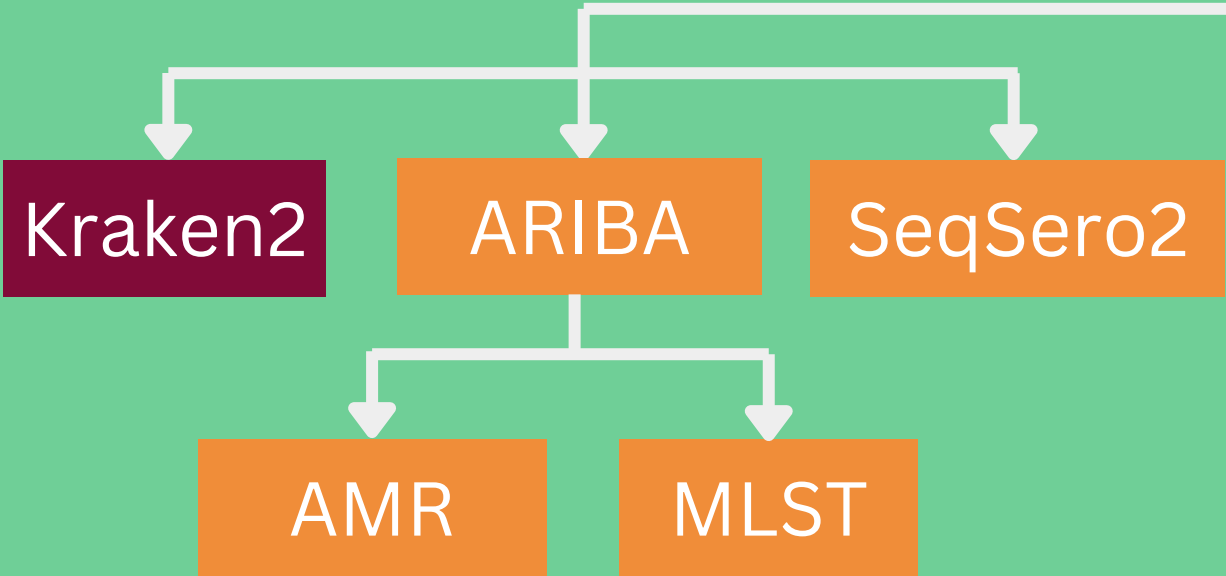
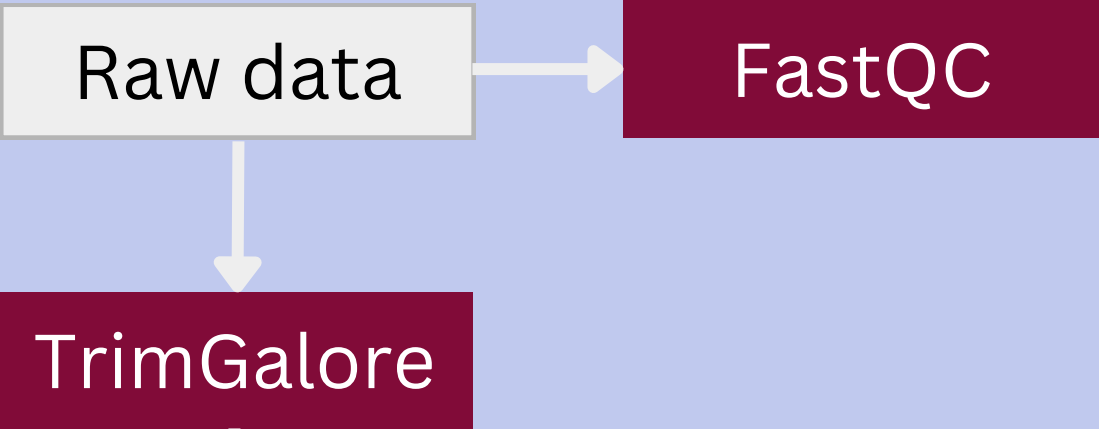
Análisis a partir de ensamblaje de referencia

Preprocesamiento

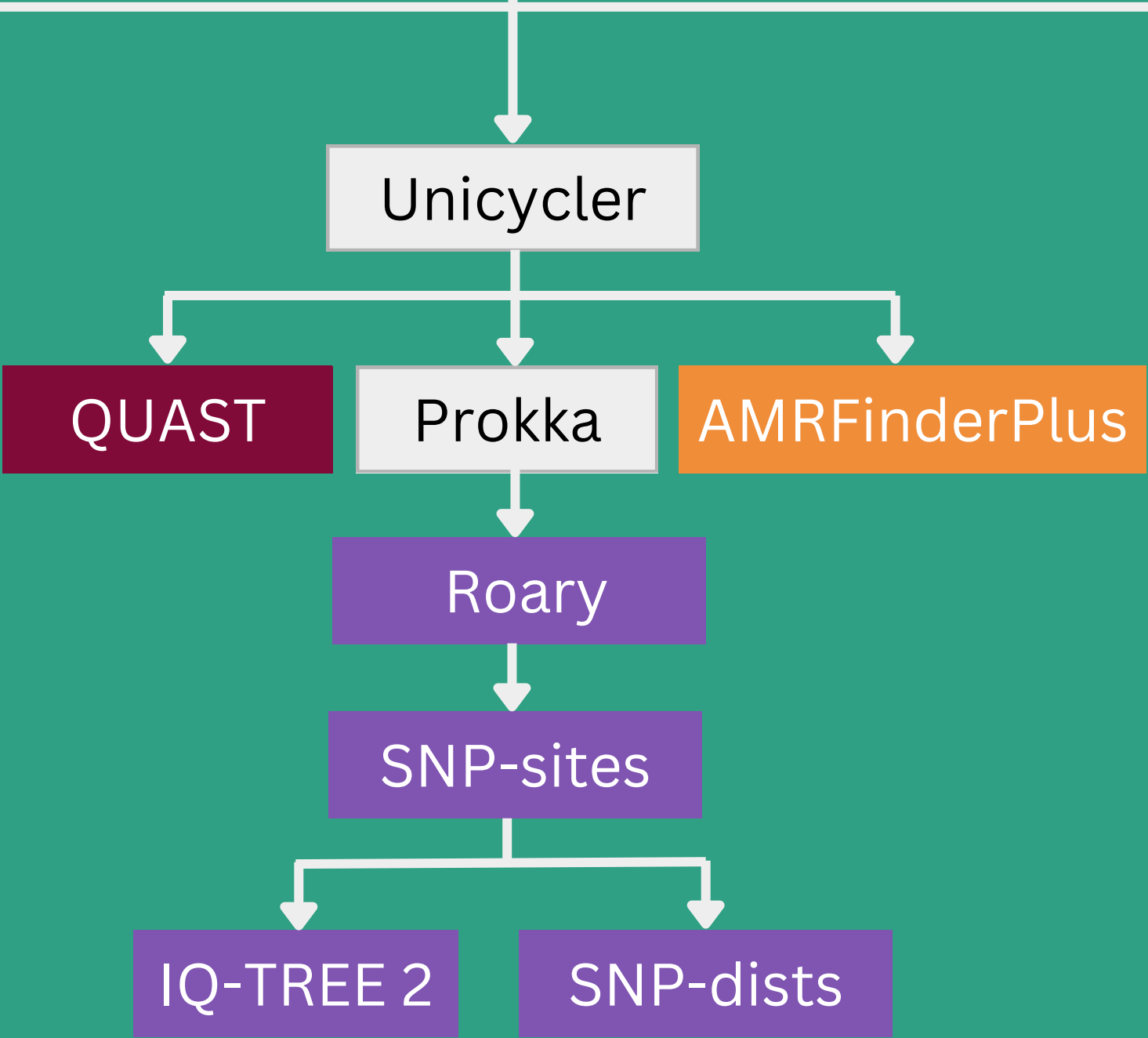
Control de calidad

Caracterización

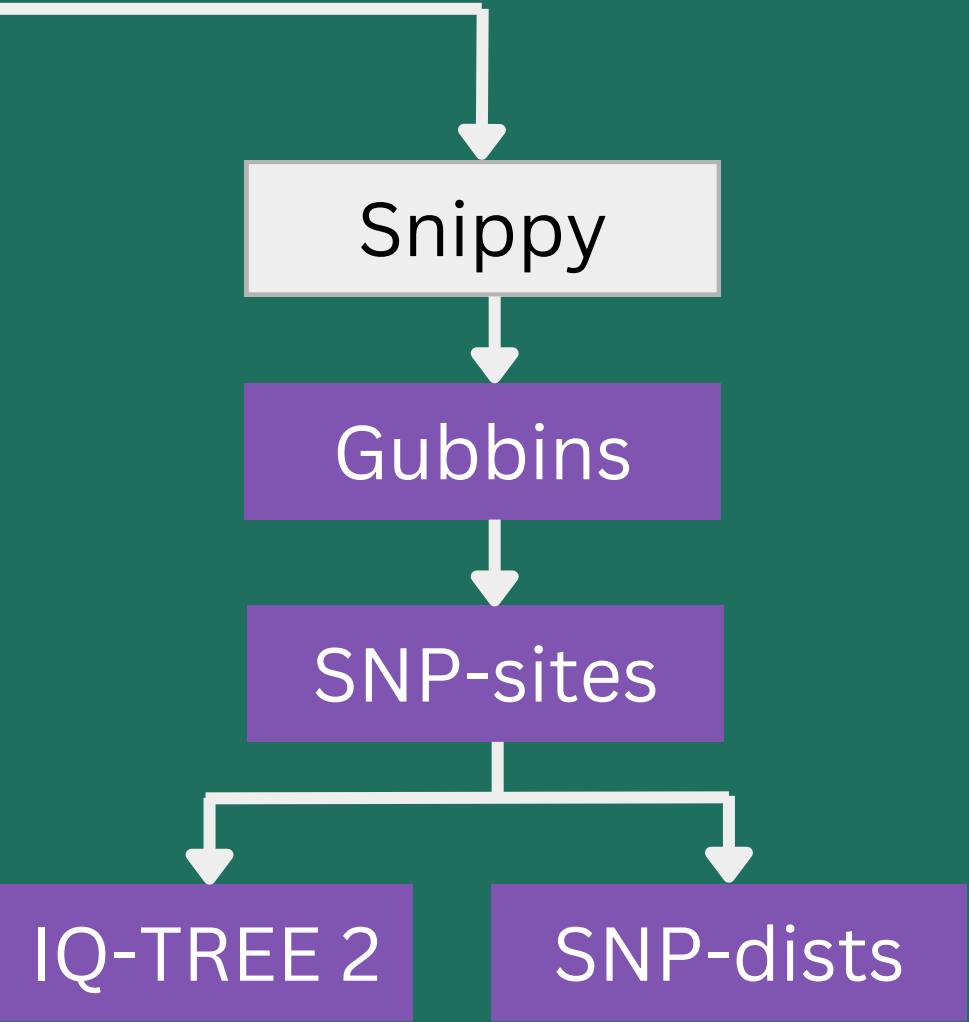
Comparación



Análisis a partir de lecturas



Análisis a partir de ensamblaje de novo



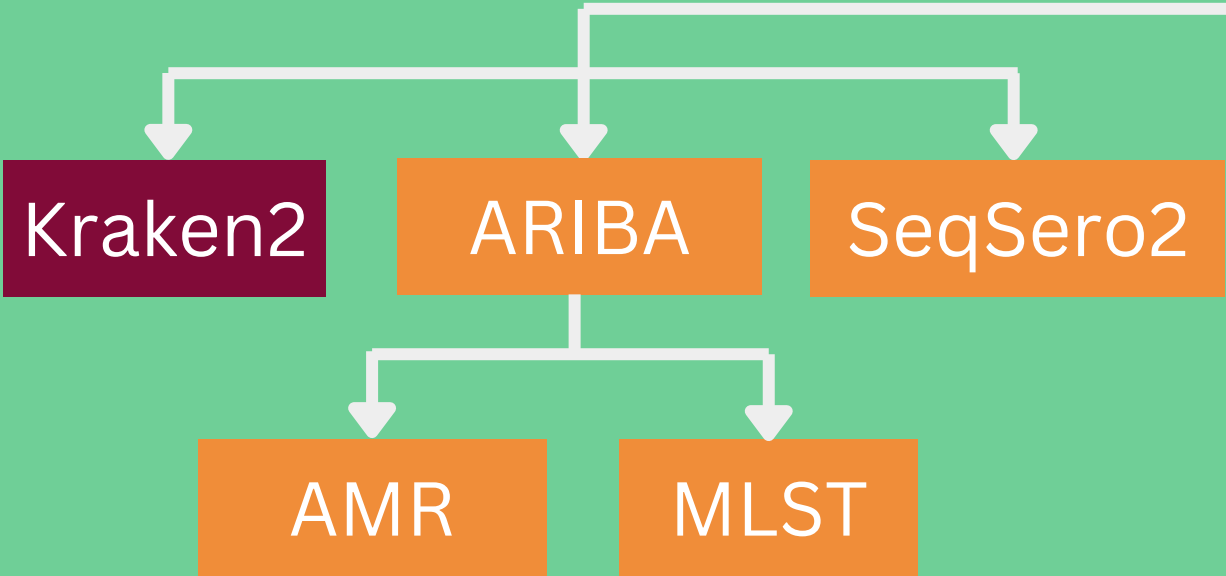
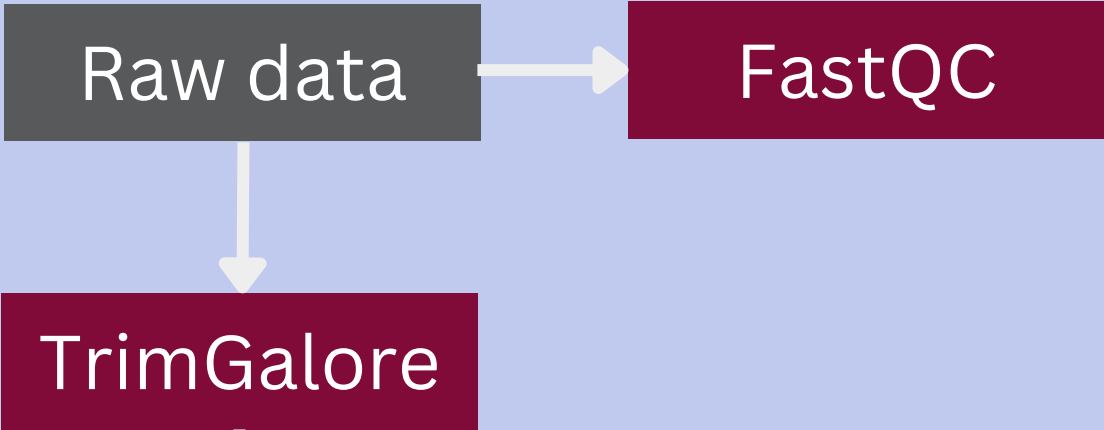
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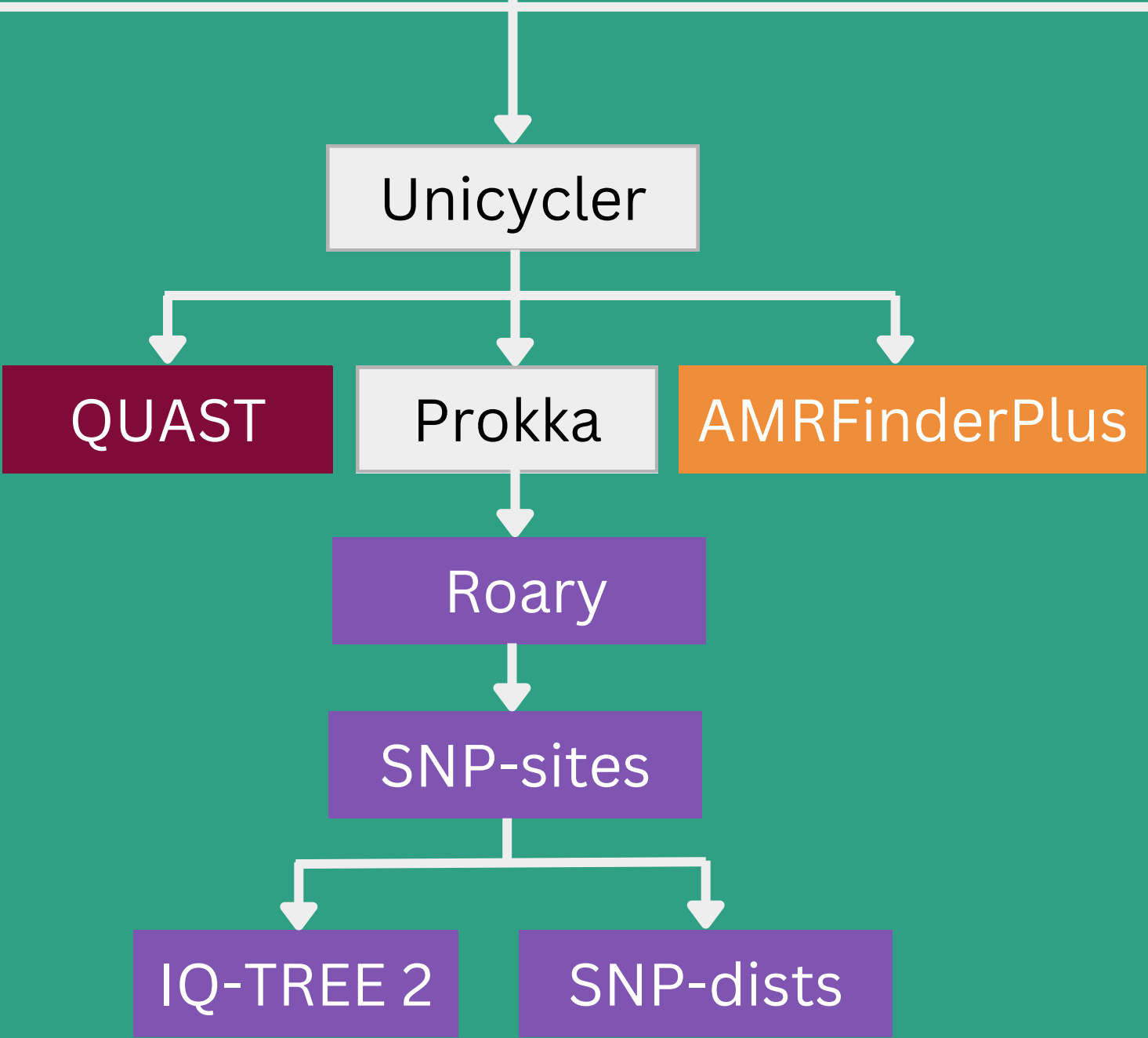
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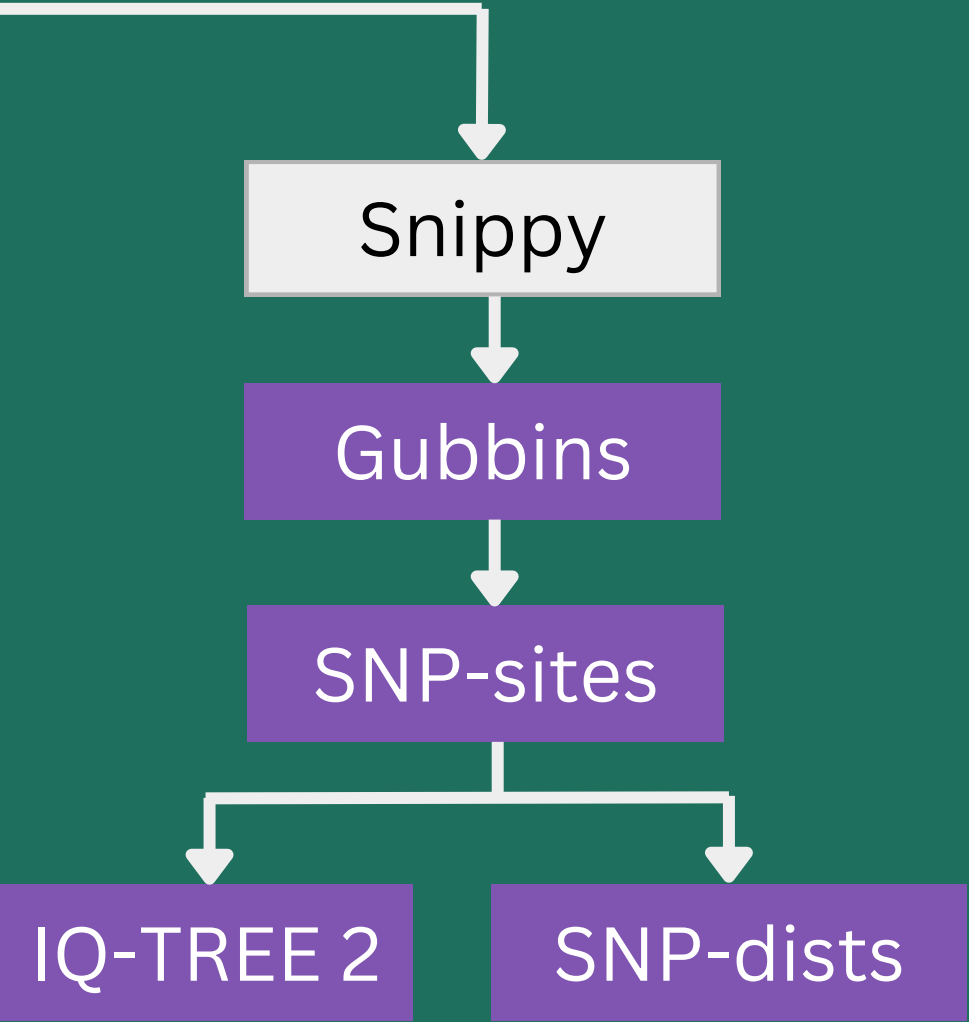
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Análisis a partir de lecturas



Análisis a partir de ensamblaje de novo



Análisis a partir de ensamblaje de referencia

Secuenciación: formato de read

[illegible]



Secuenciación: Calidades

Los nucleótidos y los valores de calidad corresponden uno a uno:

Read 1	Nucleótidos	AGCTCTGGTGACCCATGGGCAGCTGCTAGGGA
	Valores de Calidad	 HHHHHHHHHHHHHHHGGCGC5FEFFFGHHHHH

Un valor de calidad es una codificación ASCII de un número, Q

donde $Q = -10 \log_{10} p$

donde p es la estimación del secuenciador de la probabilidad de que el nucleótido en esa posición haya sido identificado **incorrectamente**.



Secuenciación: Calidades

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Puntaje calidad Phred

donde p es la estimación del secuenciador de la probabilidad de que el nucleótido en esa posición haya sido identificado **incorrectamente**.

Secuenciación: Calidades



ASCII_BASE=33 Illumina, Ion Torrent, PacBio and Sanger

Q	P_error	ASCII	Q	P_error	ASCII	Q	P_error	ASCII	Q	P_error	ASCII
0	1.00000	33 !	11	0.07943	44 ,	22	0.00631	55 7	33	0.00050	66 B
1	0.79433	34 "	12	0.06310	45 -	23	0.00501	56 8	34	0.00040	67 C
2	0.63096	35 #	13	0.05012	46 .	24	0.00398	57 9	35	0.00032	68 D
3	0.50119	36 \$	14	0.03981	47 /	25	0.00316	58 :	36	0.00025	69 E
4	0.39811	37 %	15	0.03162	48 0	26	0.00251	59 ;	37	0.00020	70 F
5	0.31623	38 &	16	0.02512	49 1	27	0.00200	60 <	38	0.00016	71 G
6	0.25119	39 '	17	0.01995	50 2	28	0.00158	61 =	39	0.00013	72 H
7	0.19953	40 (	18	0.01585	51 3	29	0.00126	62 >	40	0.00010	73 I
8	0.15849	41)	19	0.01259	52 4	30	0.00100	63 ?	41	0.00008	74 J
9	0.12589	42 *	20	0.01000	53 5	31	0.00079	64 @	42	0.00006	75 K
10	0.10000	43 +	21	0.00794	54 6	32	0.00063	65 A			

ASCII_BASE=64 Old Illumina

Q	P_error	ASCII	Q	P_error	ASCII	Q	P_error	ASCII	Q	P_error	ASCII
0	1.00000	64 @	11	0.07943	75 K	22	0.00631	86 V	33	0.00050	97 a
1	0.79433	65 A	12	0.06310	76 L	23	0.00501	87 W	34	0.00040	98 b
2	0.63096	66 B	13	0.05012	77 M	24	0.00398	88 X	35	0.00032	99 c
3	0.50119	67 C	14	0.03981	78 N	25	0.00316	89 Y	36	0.00025	100 d
4	0.39811	68 D	15	0.03162	79 O	26	0.00251	90 Z	37	0.00020	101 e
5	0.31623	69 E	16	0.02512	80 P	27	0.00200	91 [	38	0.00016	102 f
6	0.25119	70 F	17	0.01995	81 Q	28	0.00158	92 \	39	0.00013	103 g
7	0.19953	71 G	18	0.01585	82 R	29	0.00126	93]	40	0.00010	104 h
8	0.15849	72 H	19	0.01259	83 S	30	0.00100	94 ^	41	0.00008	105 i
9	0.12589	73 I	20	0.01000	84 T	31	0.00079	95 _	42	0.00006	106 j
10	0.10000	74 J	21	0.00794	85 U	32	0.00063	96 `			

Table 1: Quality Scores and Base Calling Accuracy

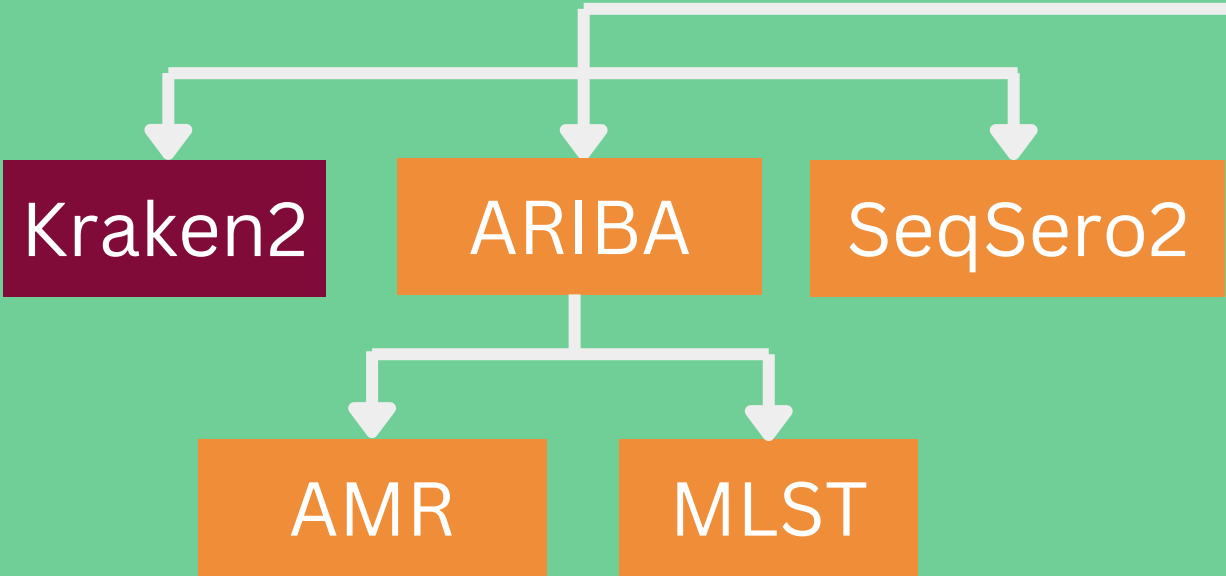
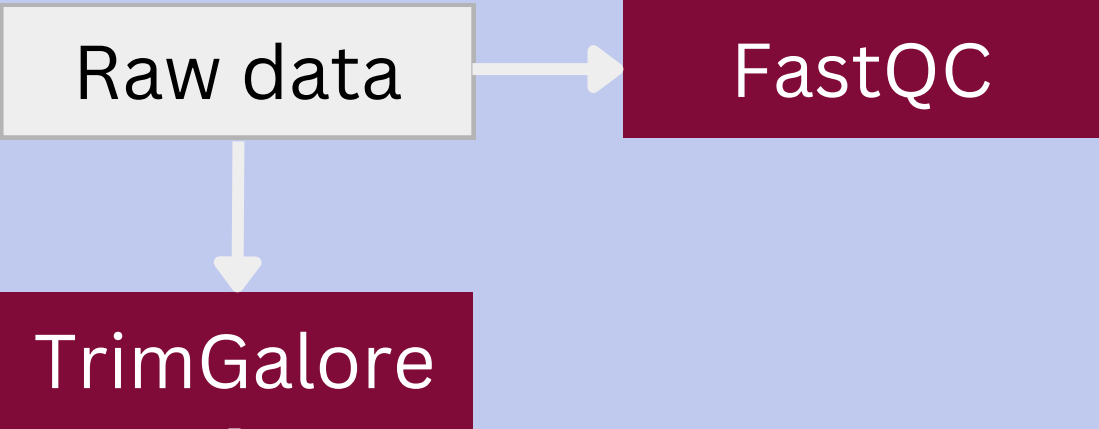
Phred Quality Score	Probability of Incorrect Base Call	Base Call Accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1,000	99.9%
40	1 in 10,000	99.99%
50	1 in 100,000	99.999%

Preprocesamiento

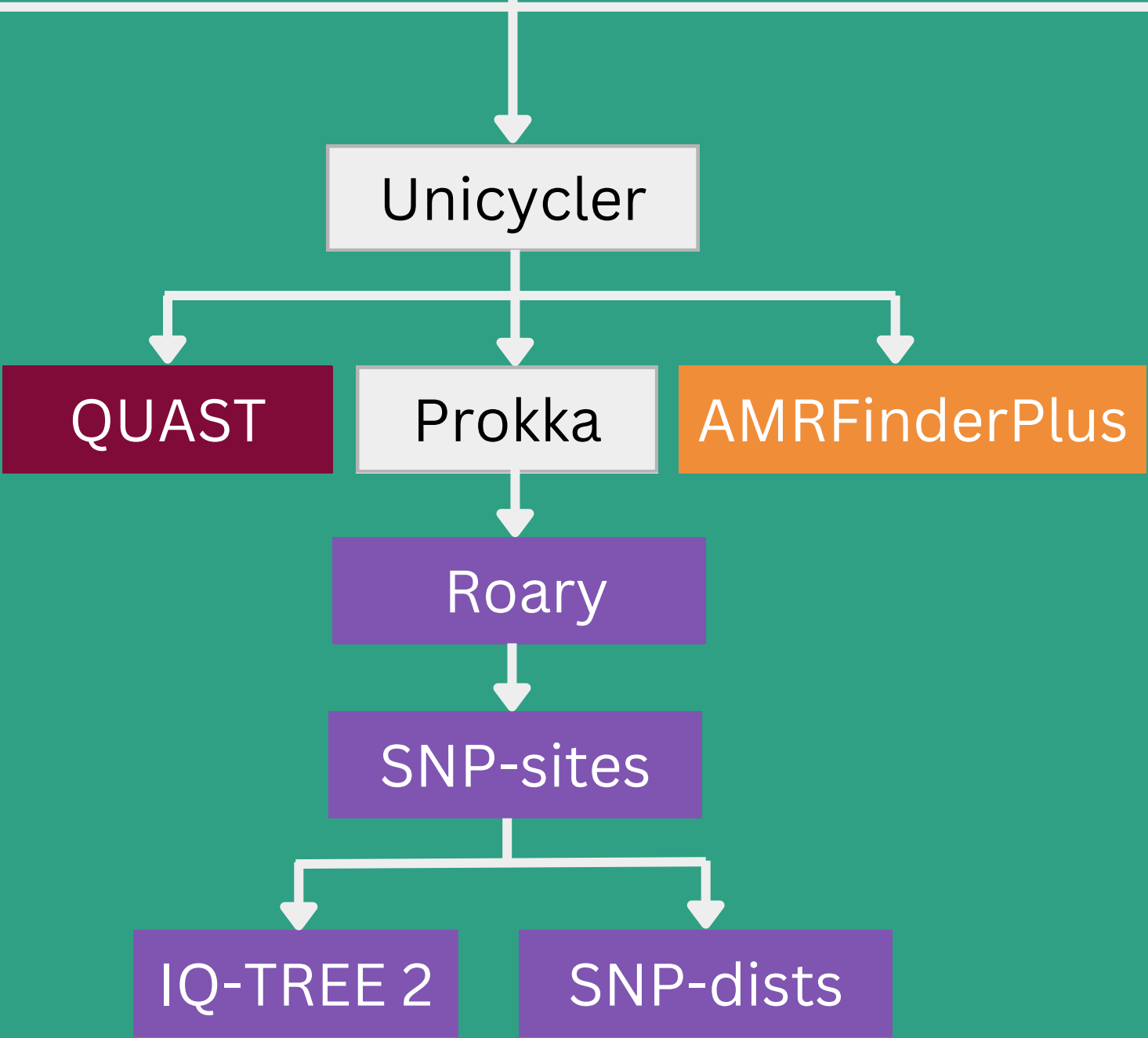
Control de calidad

Caracterización

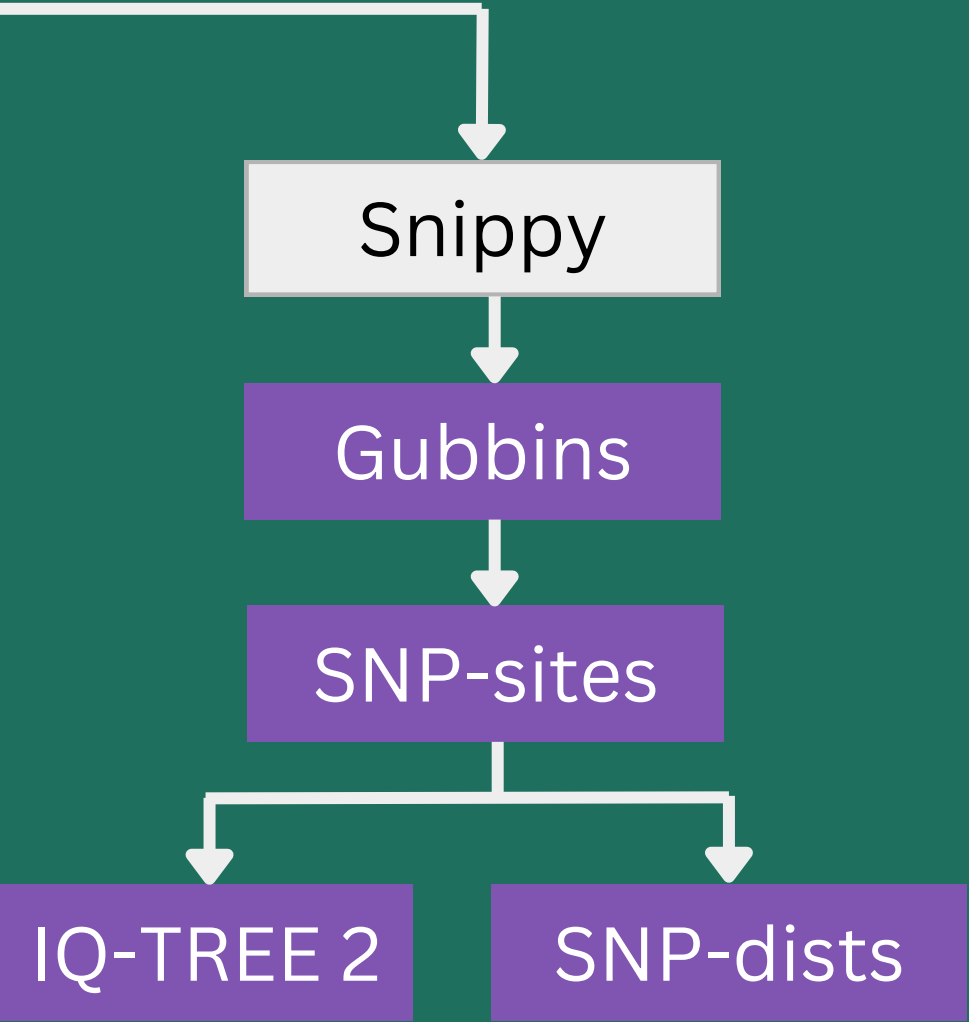
Comparación



Análisis a partir de lecturas

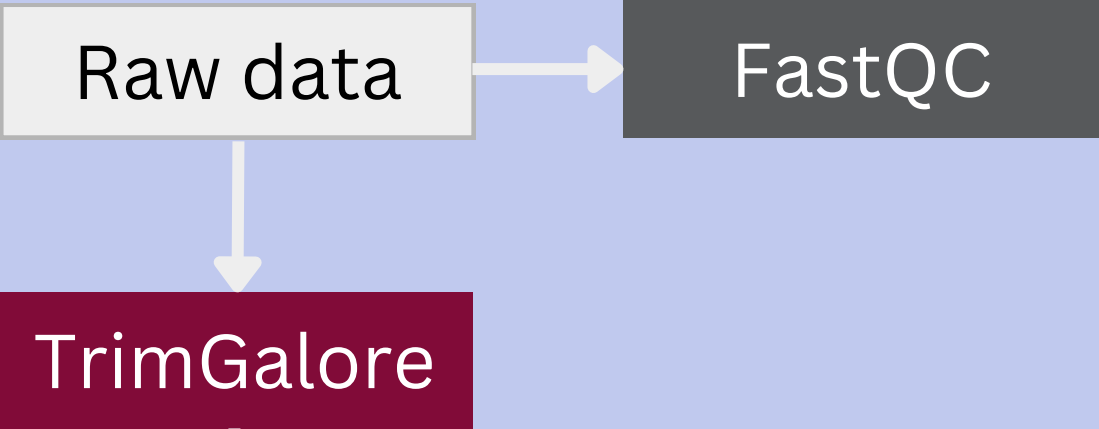


Análisis a partir de ensamblaje de novo



Análisis a partir de ensamblaje de referencia

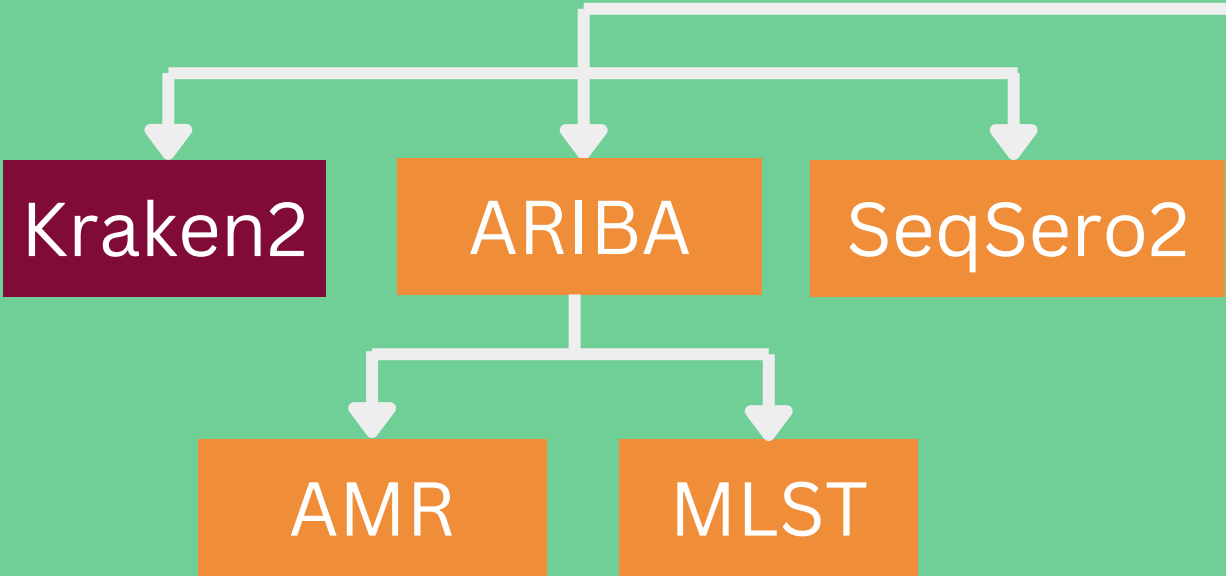
Preprocesamiento



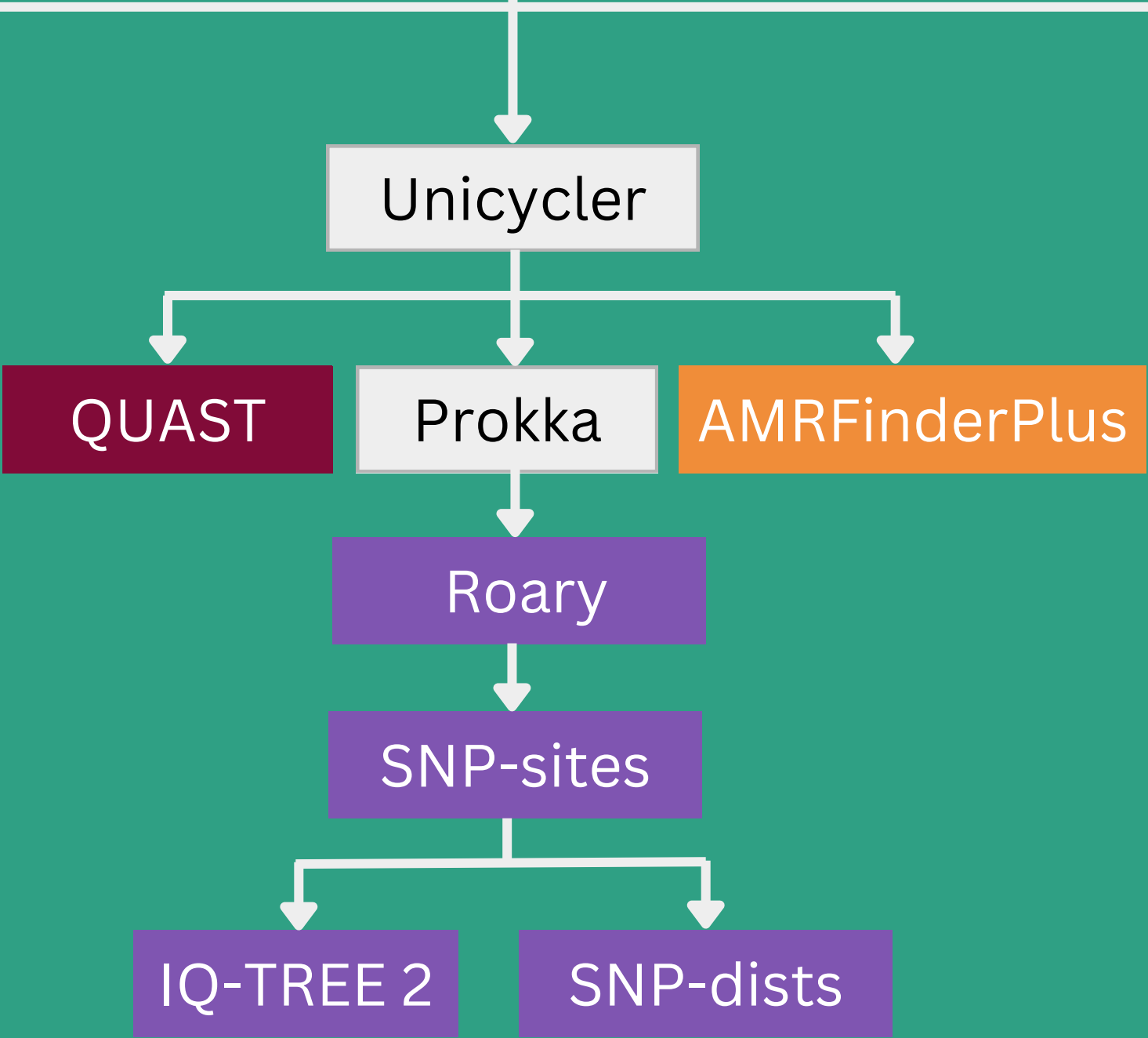
Control de calidad

Caracterización

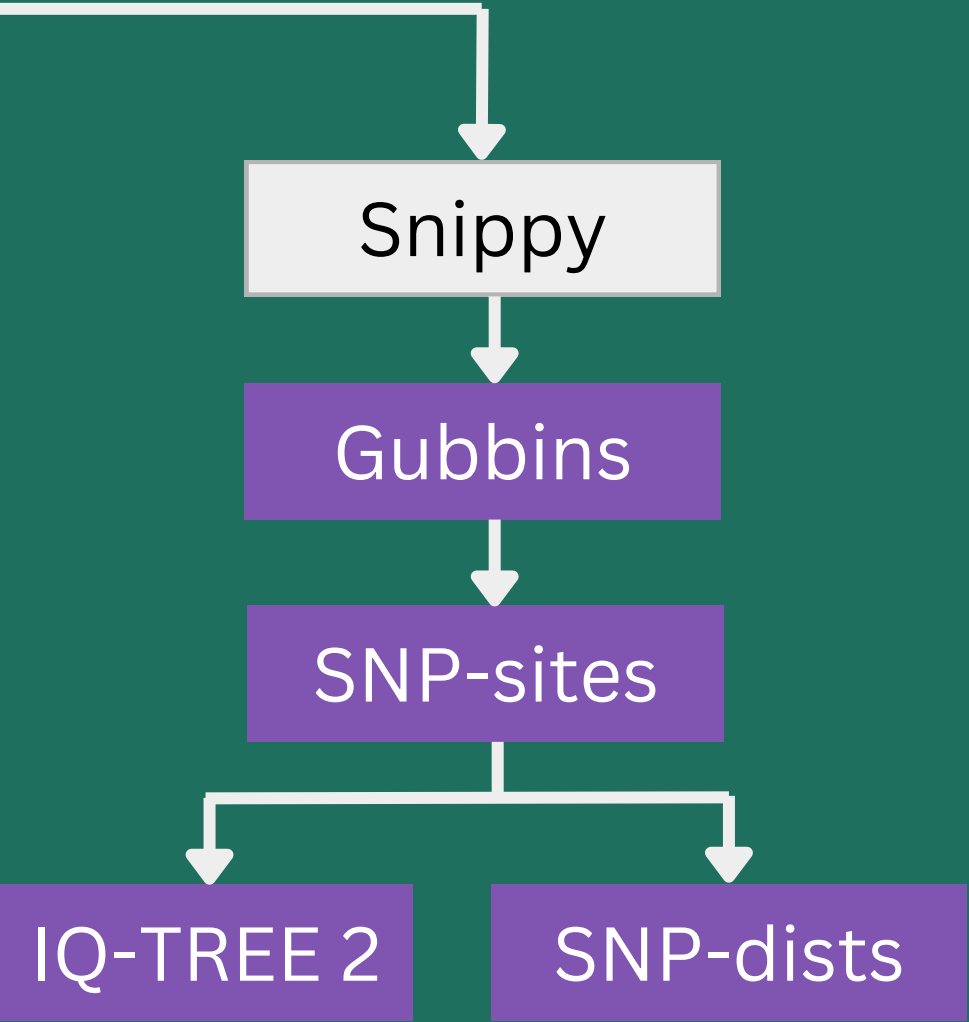
Comparación



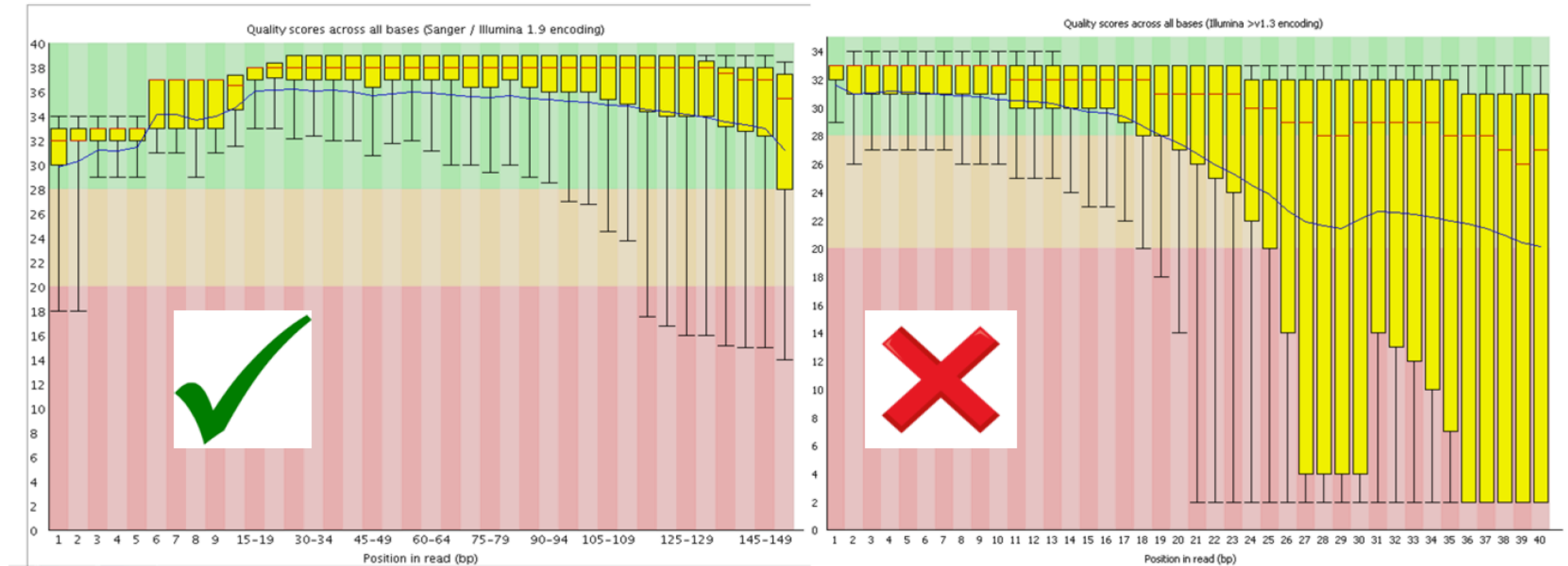
Análisis a partir de lecturas



Análisis a partir de ensamblaje de novo

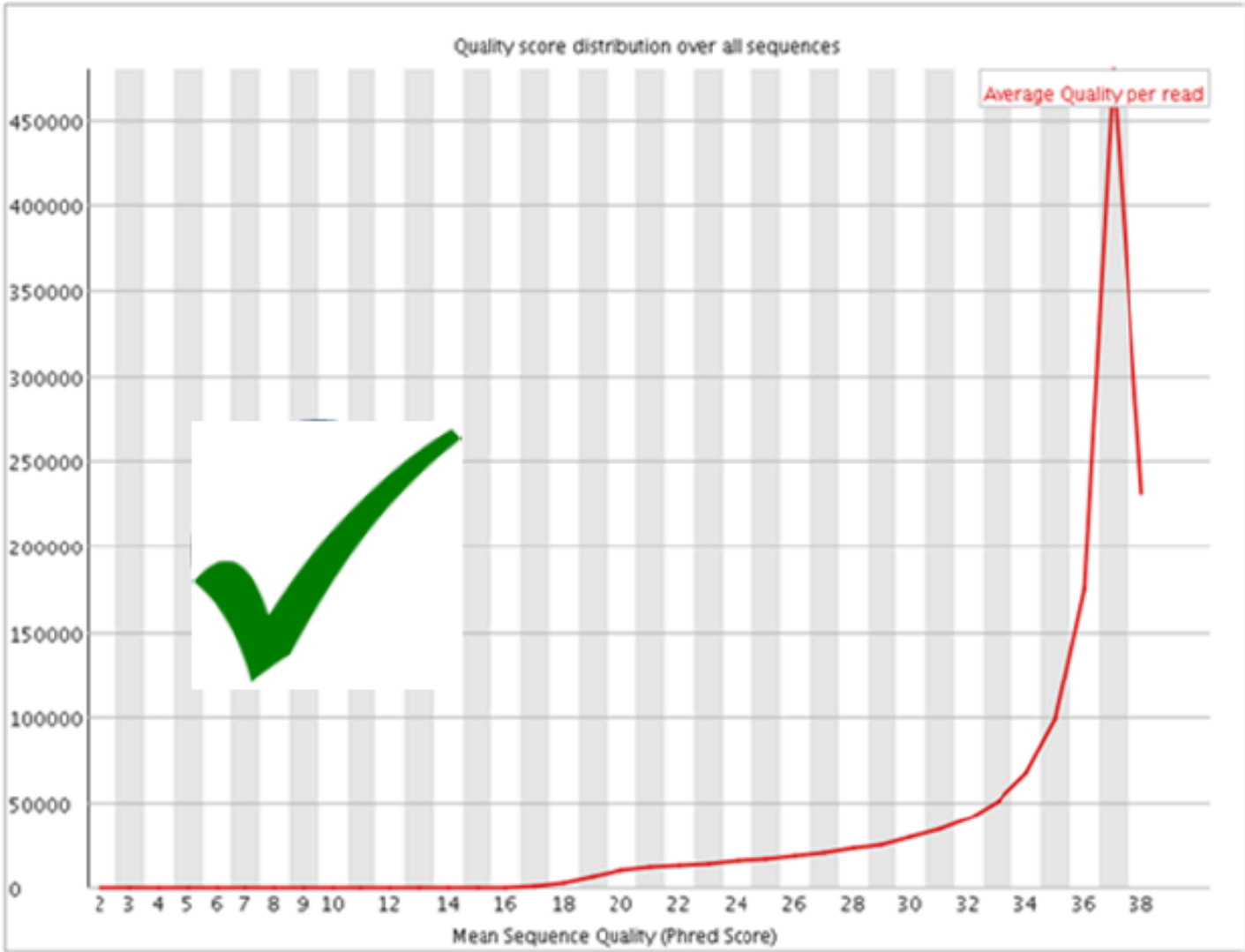


Análisis a partir de ensamblaje de referencia

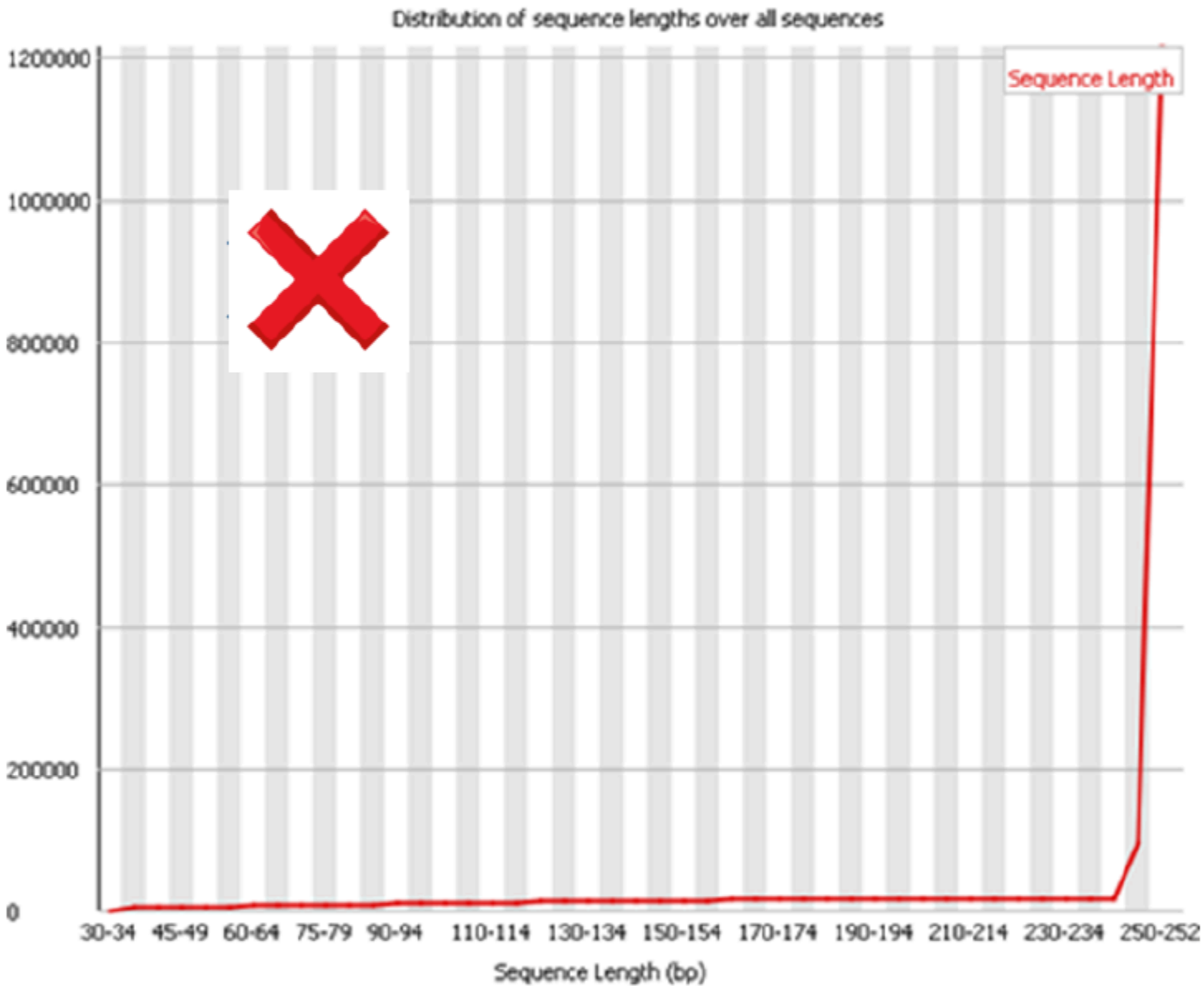
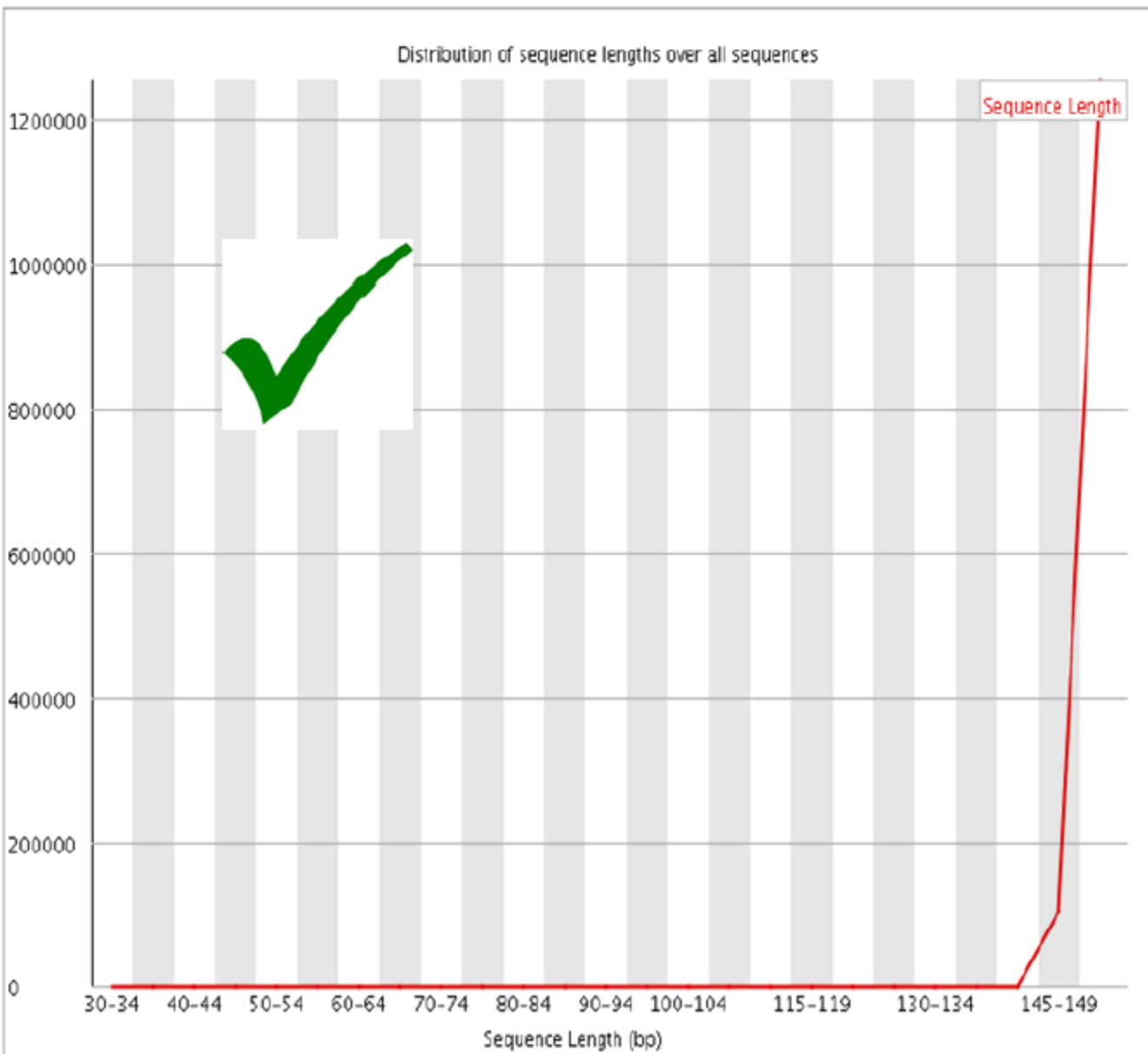


Figures 6 & 7. Example of sequences with passing per base sequence quality (left) and non-passing per base sequence quality (right).

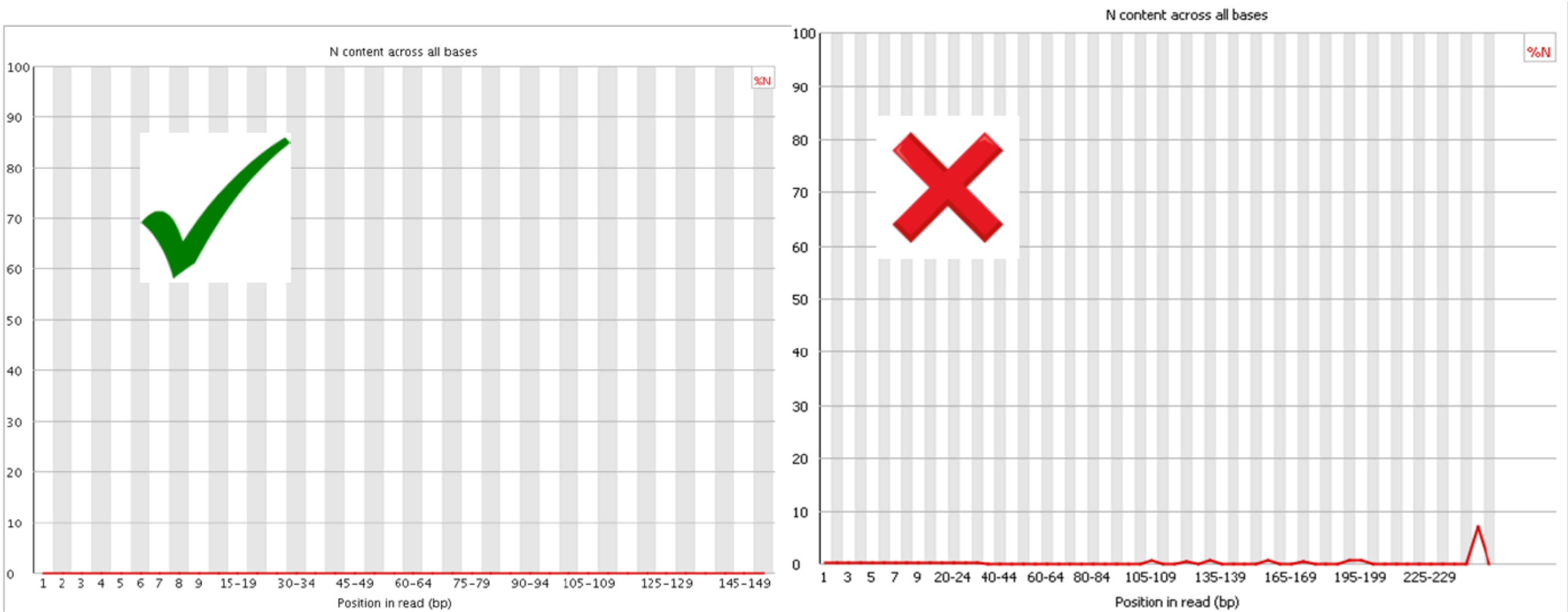
- ❌ El cuartil menor de cualquier base es menor de 5 o si la mediana de cualquier base es menor de 20



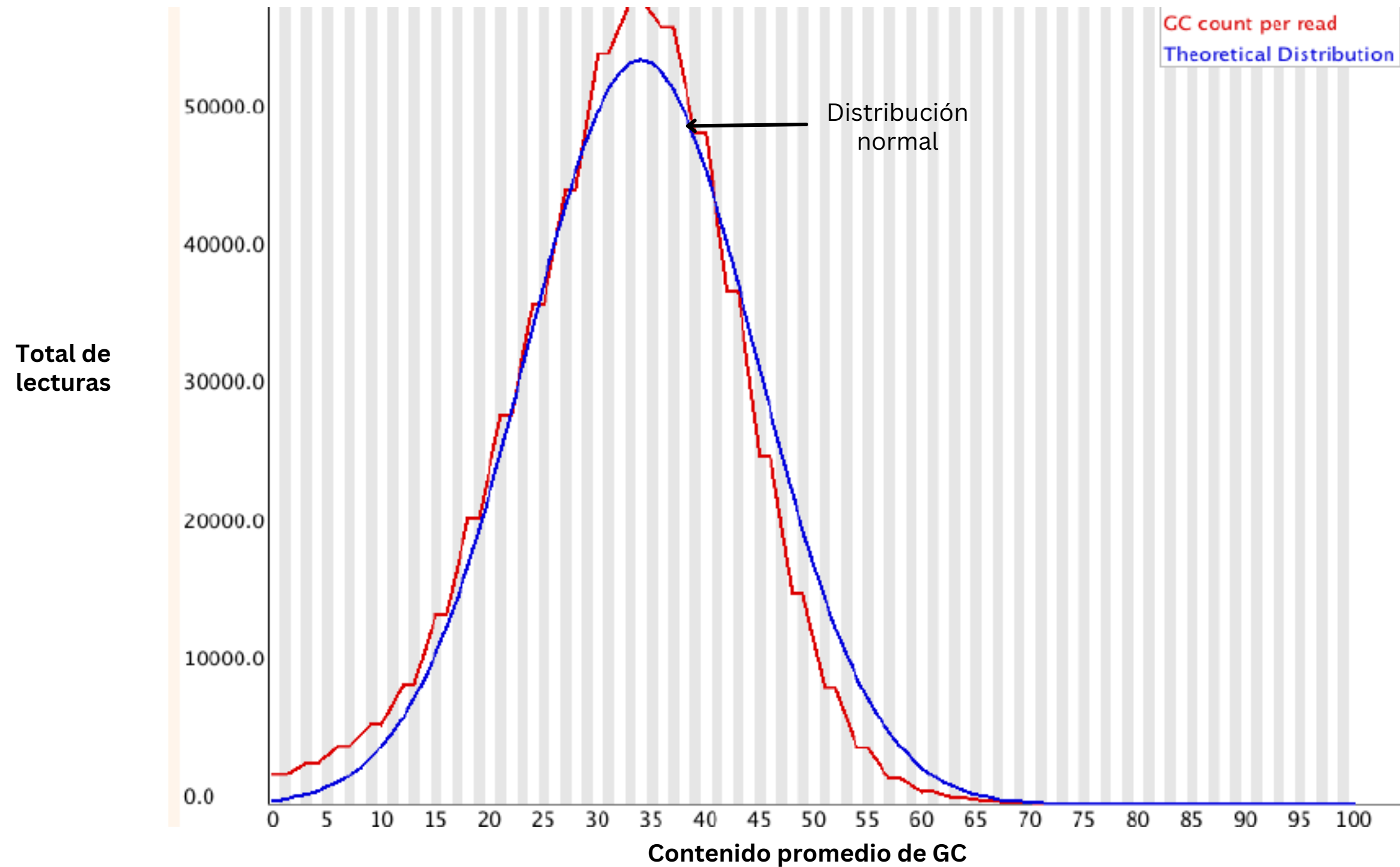
Figures 8 & 9. Examples of sequences with passing quality score distribution over all sequences (left,) and non-passing quality score distribution over all sequences (right).



Figures 12 & 13. Examples of the “Sequence Length Distribution” graphs: A sequence with ideal distribution of sequence length (left), and the graph for a sequence with less than ideal sequence length distribution (right).



Figures 10 & 11. Example of “Per base N content” graph for a sequence with ideal N content across all bases (left) and an example of a graph with a higher percentage of N content (right).



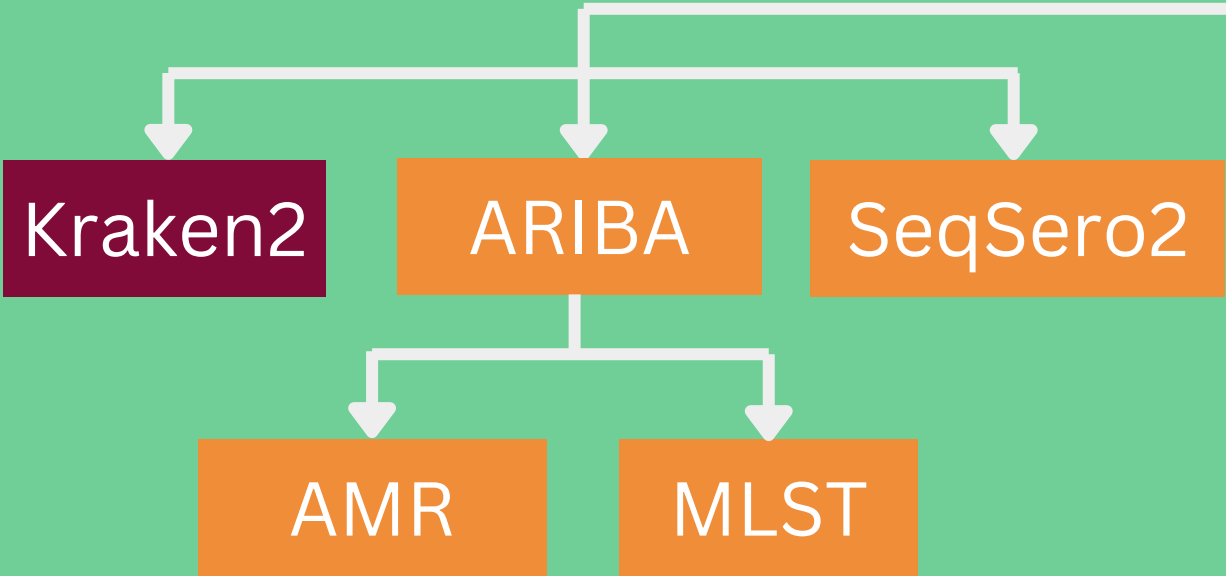
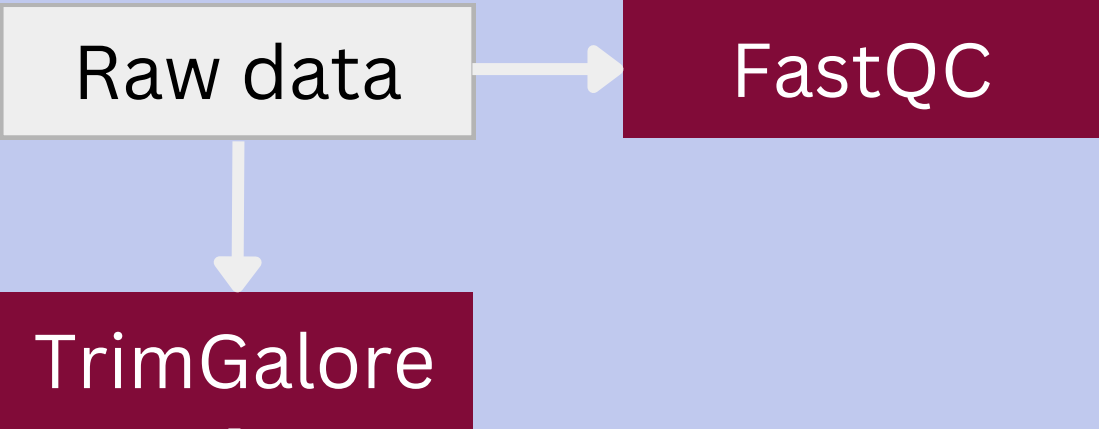
- ⚠ Si la suma de las desviaciones de la distribución normal representa más del 15 % de las lecturas.
- ✖ Si la suma de las desviaciones de la distribución normal representa más del 30 % de las lecturas.

Preprocesamiento

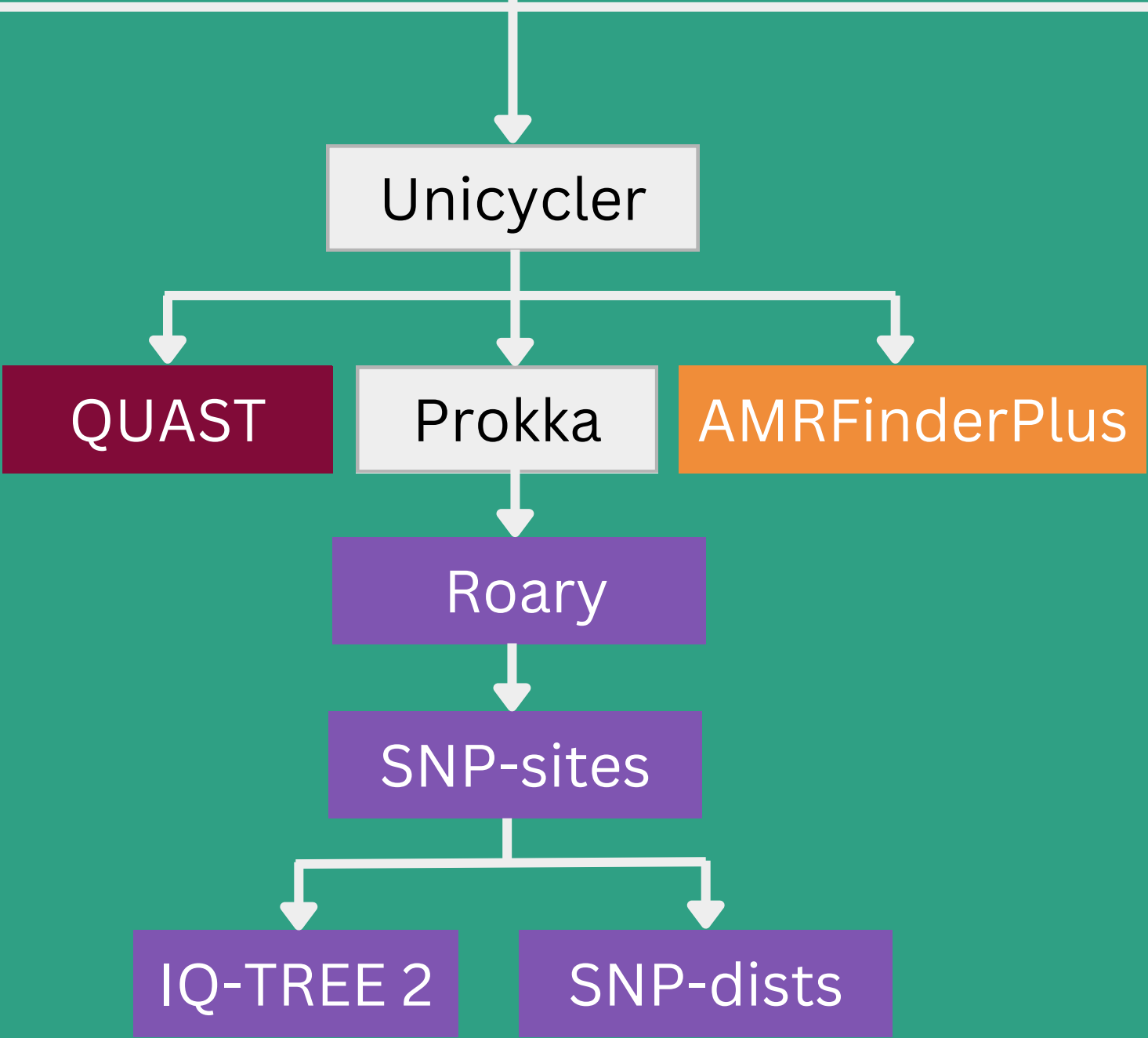
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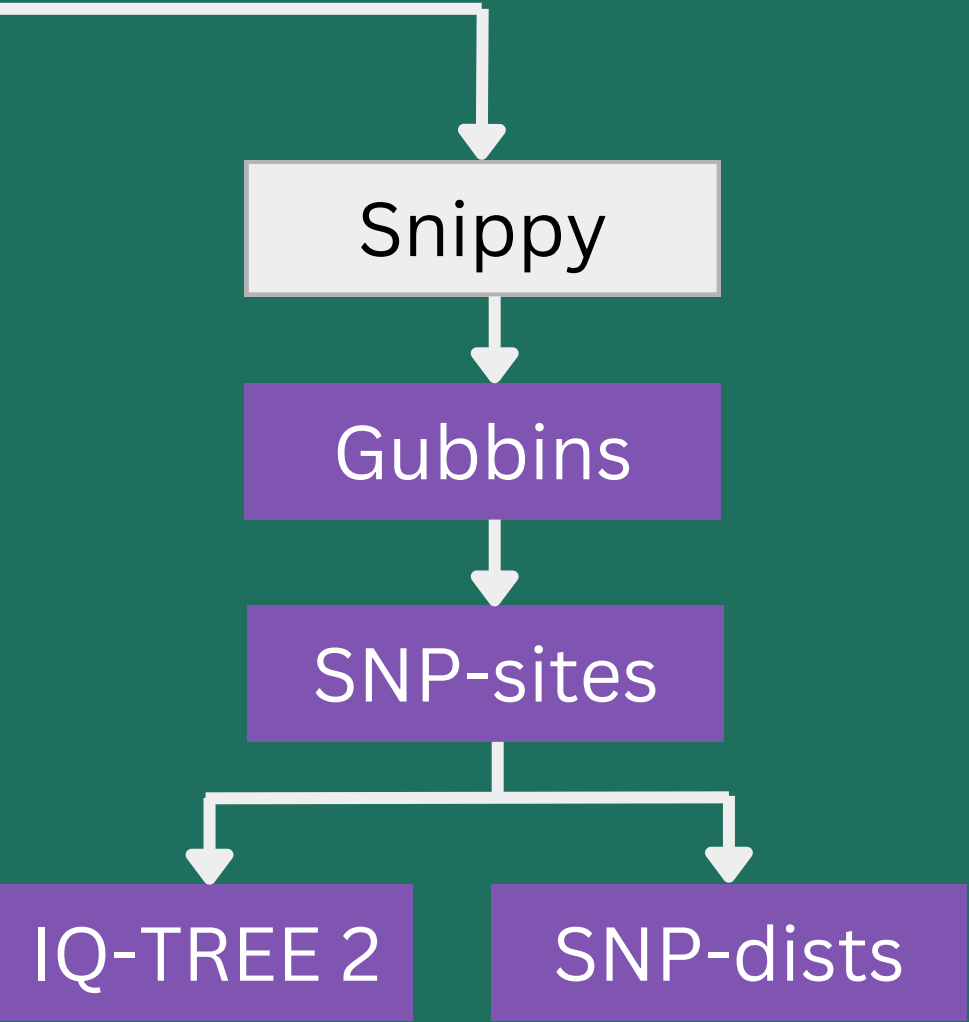
Comparación



Análisis a partir de lecturas



Análisis a partir de ensamblaje de novo



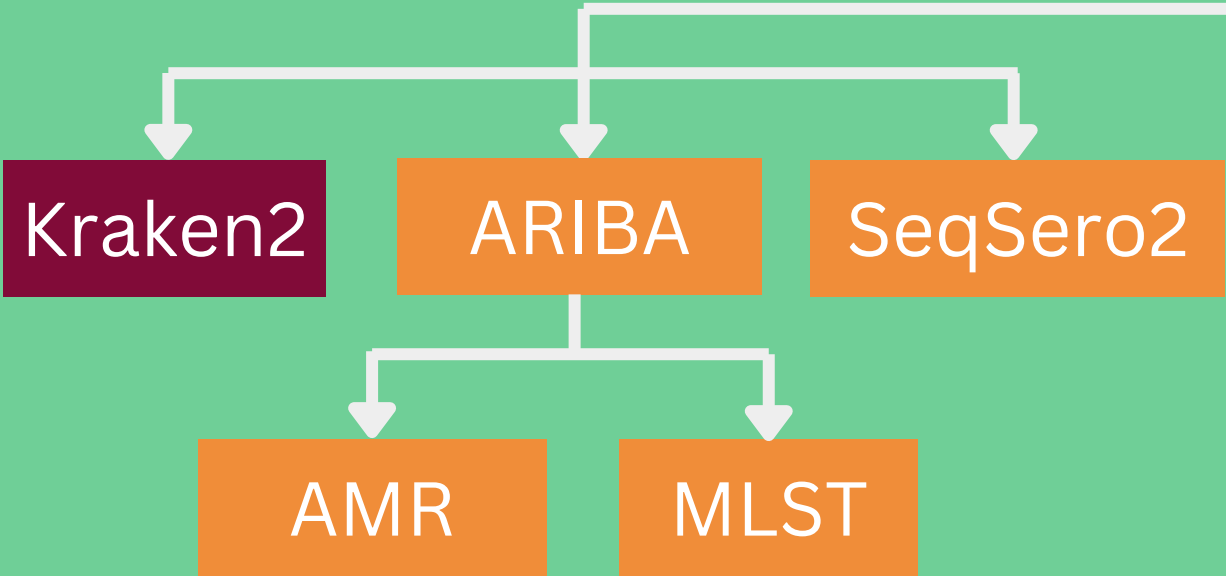
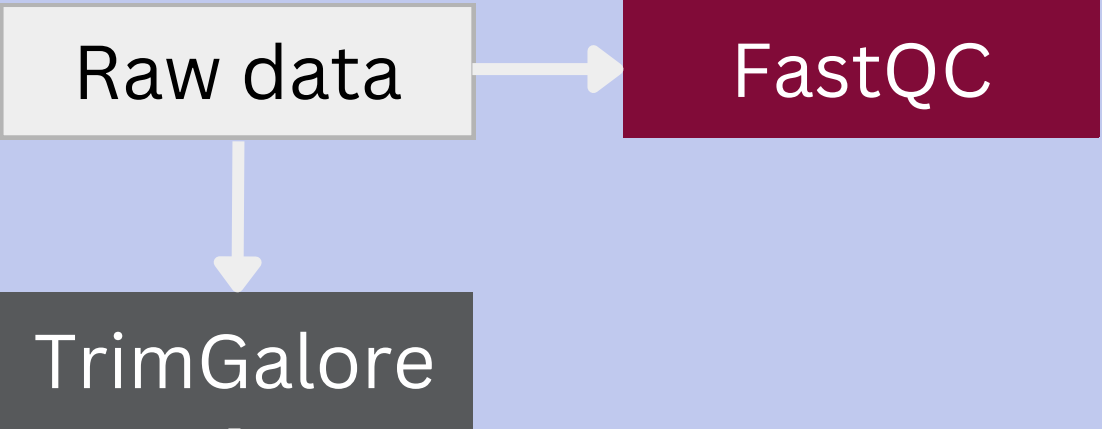
Análisis a partir de ensamblaje de referencia

Preprocesamiento

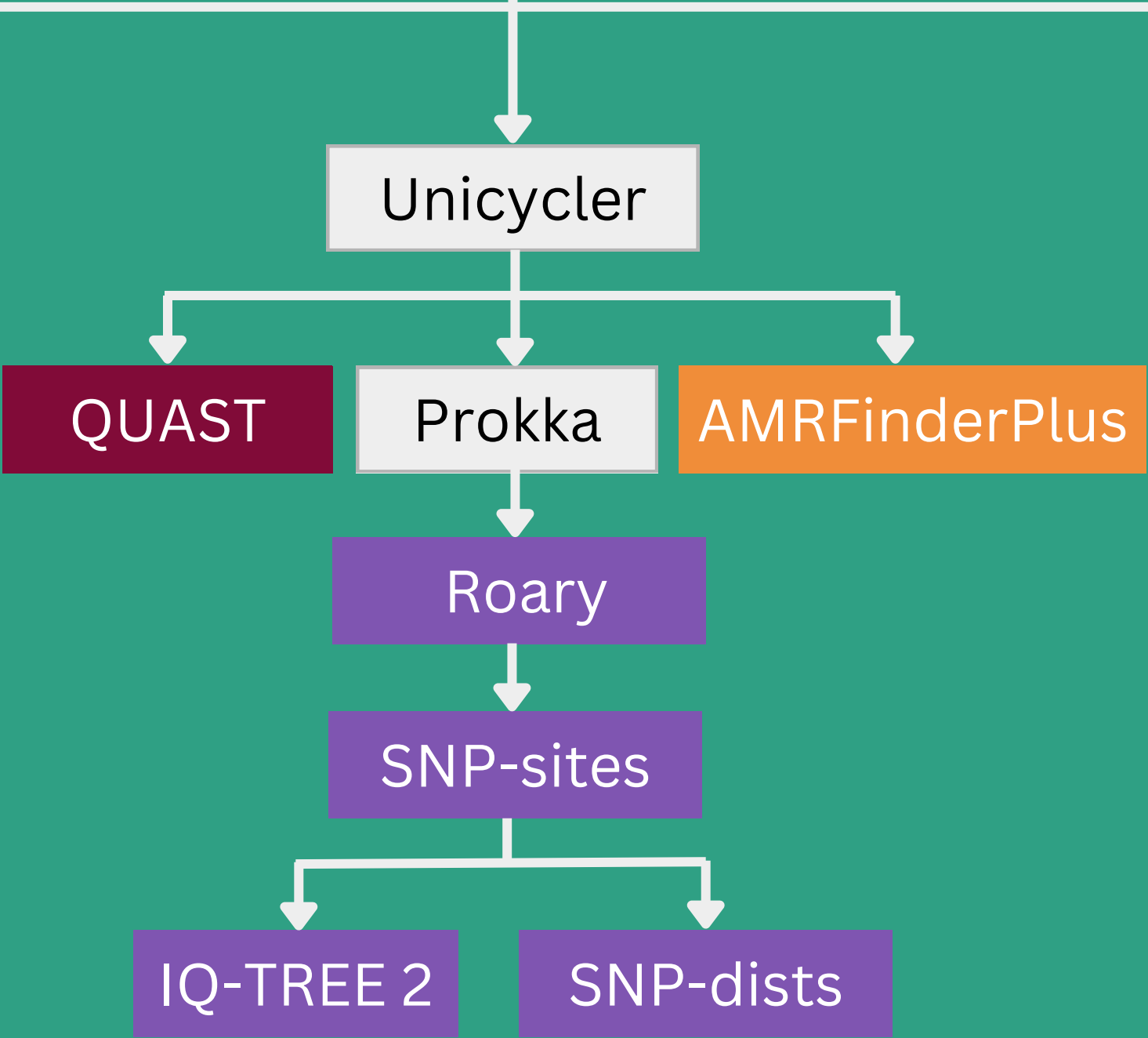
Control de calidad

Caracterización

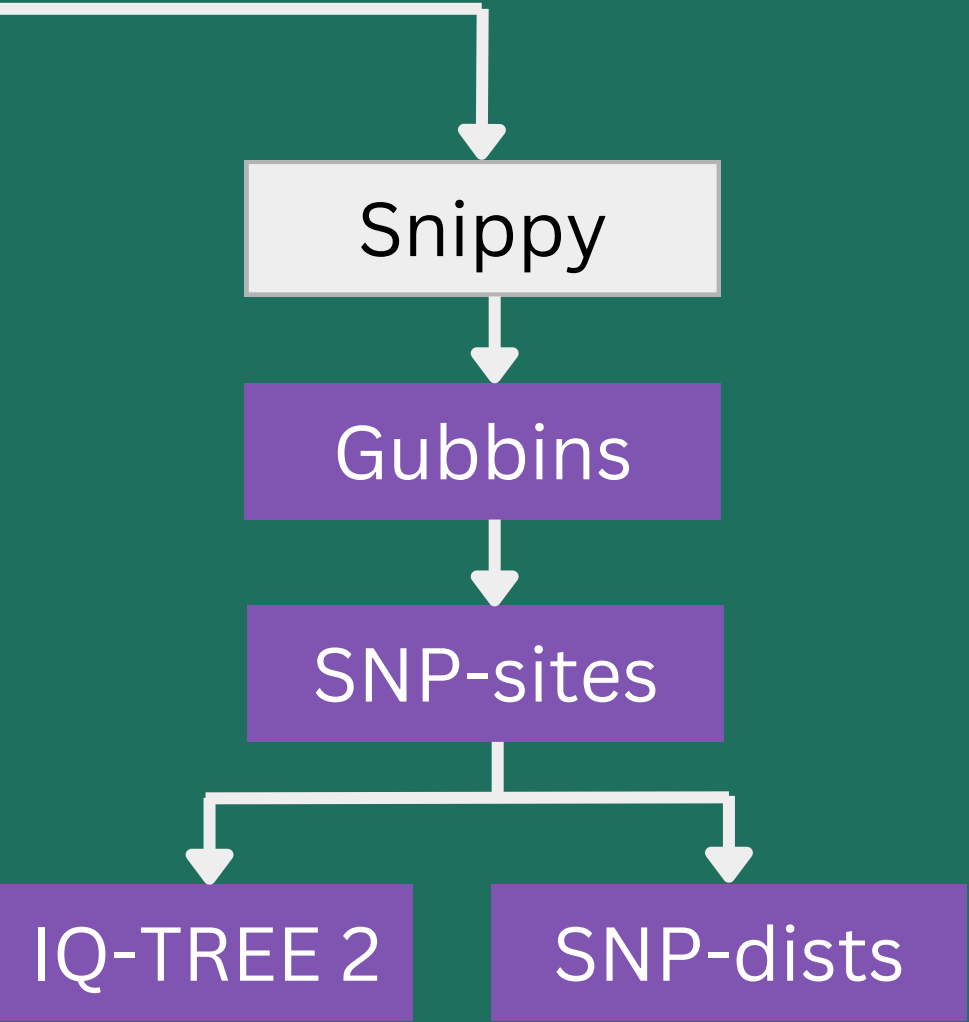
Comparación



Análisis a partir de lecturas



Análisis a partir de ensamblaje de novo



Análisis a partir de ensamblaje de referencia

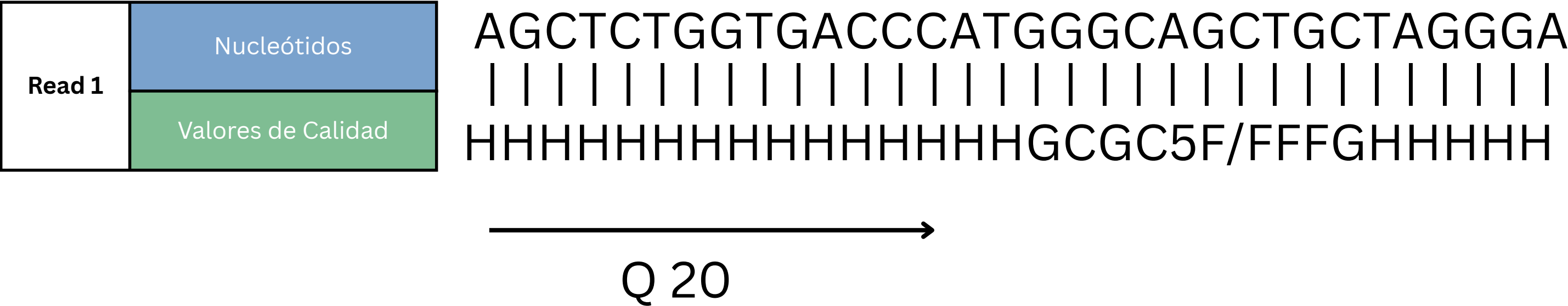


1. Quality Trimming

Read 1	Nucleótidos	AGCTCTGGTGACCCATGGGCGAGCTGCTAGGGA
	Valores de Calidad	 HHHHHHHHHHHHHHHGGCGC5F/FFFFGHHHHH

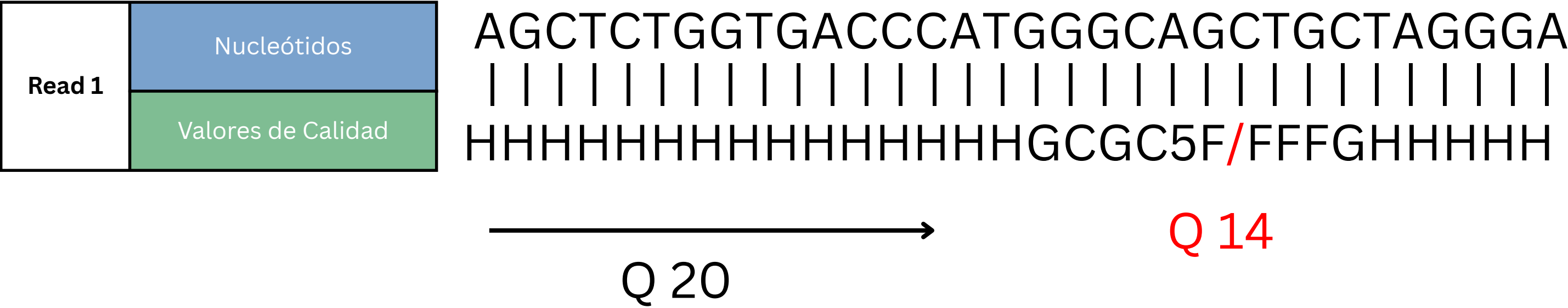


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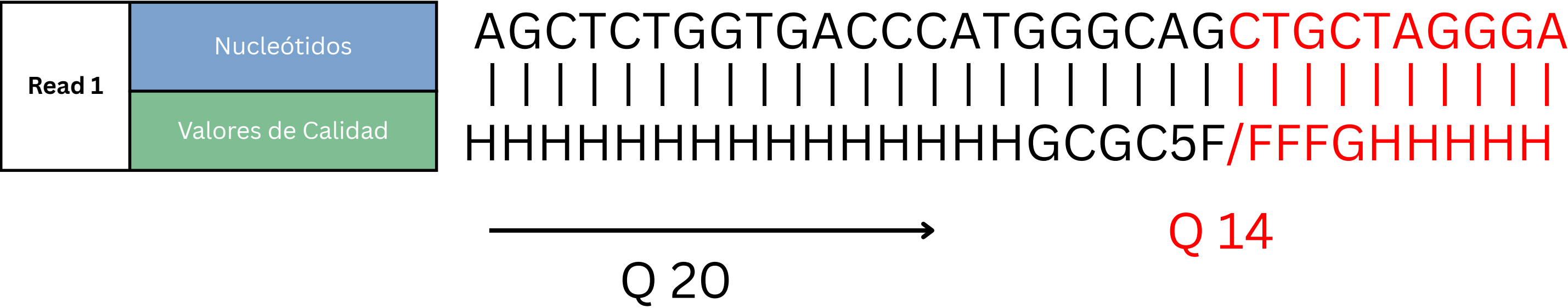


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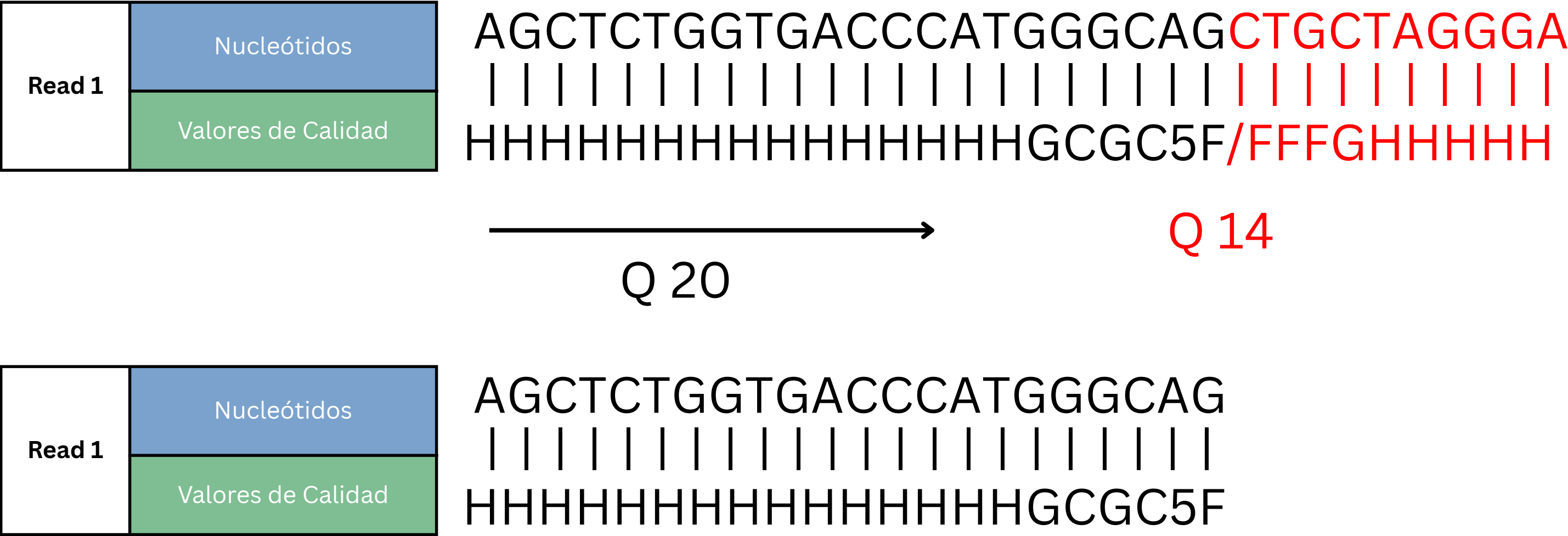


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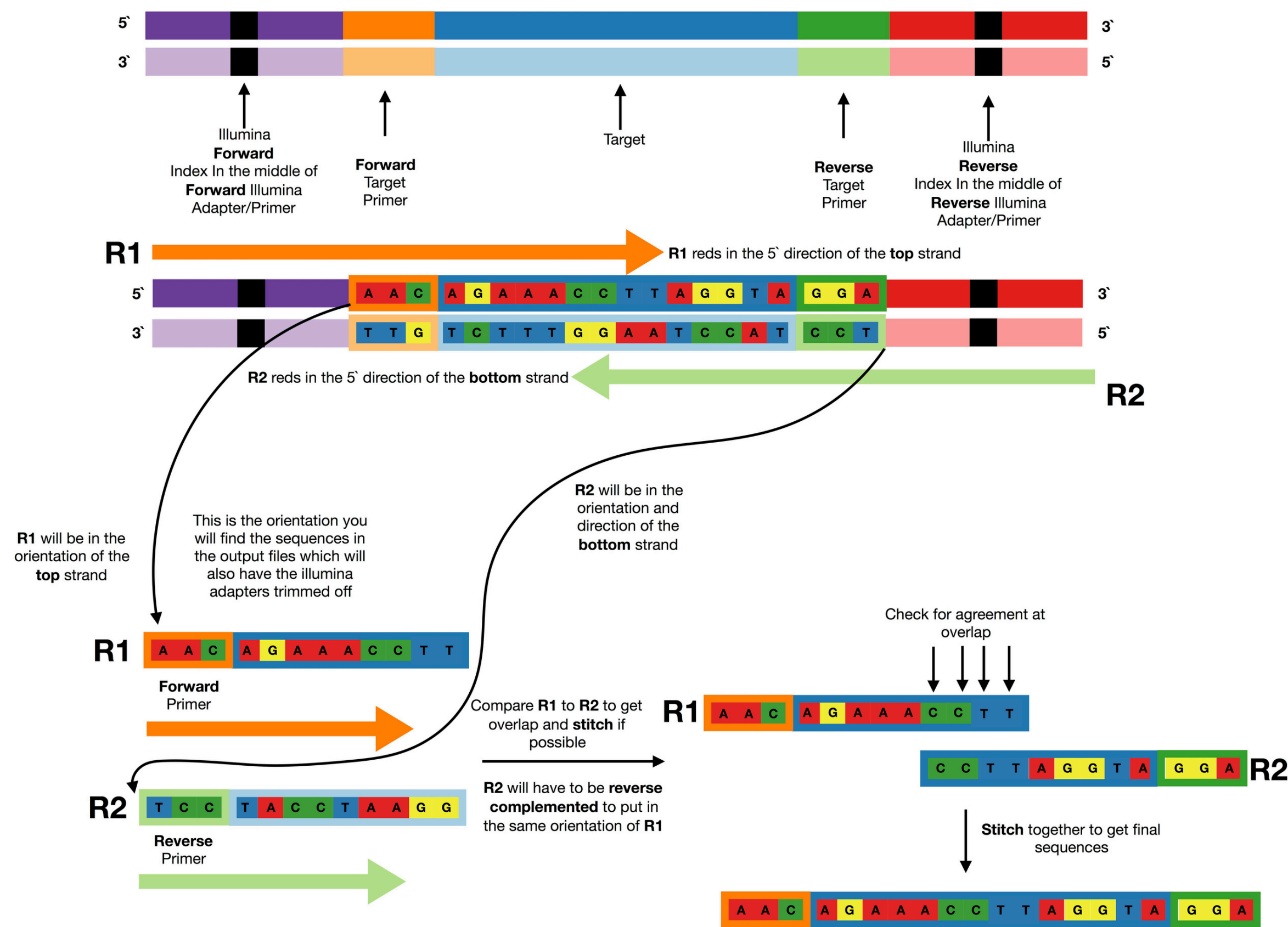


1. Quality Trimming





2. Adapter Trimming





2. Adapter Trimming

AGATCGGAAGAGC **Illumina**



2. Adapter Trimming

AGATCGGAAGAGC Illumina

AGATCG**GAAGAGC**

Adaptador

| | | | |

GAAGAGCAGCTGTCAT

Read



2. Adapter Trimming

AGATCGGAAGAGC Illumina

CCAGTCCT---CT	Adaptador
CCAGTCCTTTCCTGAGAGT	Read

Mayor cantidad de matches

2. Adapter Trimming

AGATCGGAAGAGC Illumina

CCAGTCCT---CT	Adaptador
CCAGTCCTTTCCT GAGAGT	Read

Mayor cantidad de matches

CCAGTCCTCT	Adaptador
X	
CCAGTCCTTTCCT GAGAGT	Read

Mínimo error



2. Adapter Trimming

AGATCGGAAGAGC Illumina

CCAGTCCTCT Adaptador

| | | | | | | | X |

CCAGTCCTTTCCTGAGAGT Read

Mayor cantidad de matches

Mínimo error



2. Adapter Trimming

AGATCGGAAGAGC Illumina

CCAGTCCTCT Adaptador

| | | | | | | | X |

CCAGTCCTTTCCTGAGAGT Read

Read trimeado

CCTGAGAGT

Mayor cantidad de matches

Mínimo error



3. Removing Short Sequences

AGATCGGAAGAGC **r1**

AGTCAATGA **r2**

CGATCGGAAGAG **r3**

Sample_1.fastq

AGATCGGAAGAGC **r1**

AGTCAATGA **r2**

CGATCGGAAGAG **r3**

Sample_2.fastq



3. Removing Short Sequences

AGATCGGAAGAGC **r1**

AGTCAATGA **r2**

CGATCGGAAGAG **r3**

Sample_1.fastq

len 10

AGATCGGAAGAGC **r1**

AGTCAATGA **r2**

CGATCGGAAGAG **r3**

Sample_2.fastq



3. Removing Short Sequences

AGATCGGAAGAGC **r1**

CGATCGGAAGAG **r3**

Sample_1.fastq

len 10

AGATCGGAAGAGC **r1**

CGATCGGAAGAG **r3**

Sample_2.fastq



3. Removing Short Sequences

AGATCGGAAGAGC **r1**

CGATCGGAAGAG **r3**

Sample_1.fastq

len 10

AGATCGGAAGAGC **r1**

CGATCGGA **r3**

Sample_2.fastq

3. Removing Short Sequences

AGATCGGAAGAGC **r1**

CGATCGGAAGAG **r3**

Sample_1.fastq

len 10

AGATCGGAAGAGC **r1**

CGATCGGA **r3**

Sample_2.fastq

3. Removing Short Sequences

AGATCGGAAGAGC **r1**

Sample_1.fastq

len 10

AGATCGGAAGAGC **r1**

Sample_2.fastq

3. Removing Short Sequences

AGATCGGAAGAGC **r1**

CGATCGGAAGAG **r3**

Sample_1.fastq

AGATCGGAAGAGC **r1**

CGATCGGA **r3**

Sample_2.fastq

len 10

--retain_unpaired

3. Removing Short Sequences

AGATCGGAAGAGC **r1**

Sample_1.fastq

len 10

--retain_unpaired

AGATCGGAAGAGC **r1**

Sample_2.fastq

CGATCGGAAGAG **r3**

Sample_unpaired.fastq



4. Specialised Trimming

AGCTCTGGTGACCCATGGGCAGCTGCTAGGGA

--hardtrim5 INT



4. Specialised Trimming

AGCTCTGGTGACCCATGGGCGAGCTGCTAGGGA

--hardtrim5 20

4. Specialised Trimming

AGCTCTGGTGACCCATGGGCAGCTGCTAGGGA

--hardtrim5 20

4. Specialised Trimming

AGCTCTGGTGACCCATGGGCAGCTGCTAGGGA

--hardtrim5 20

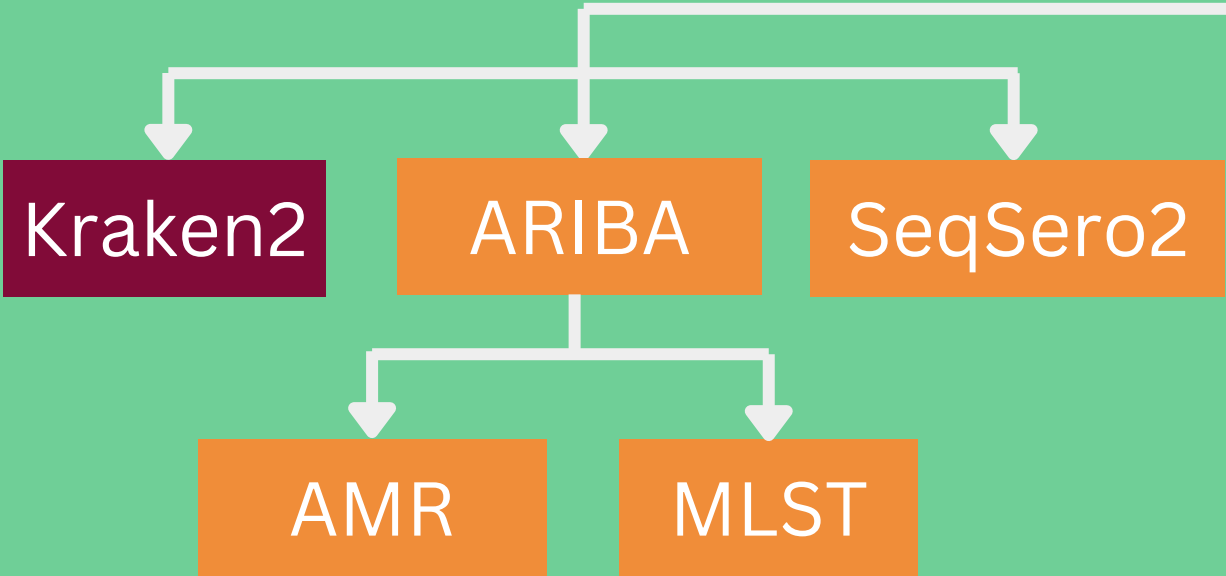
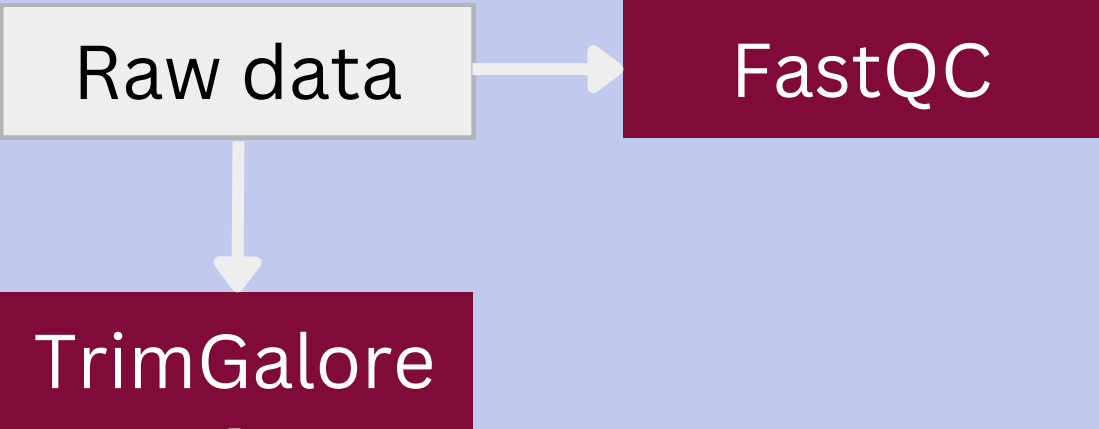
AGCTCTGGTGACCCATGGGC

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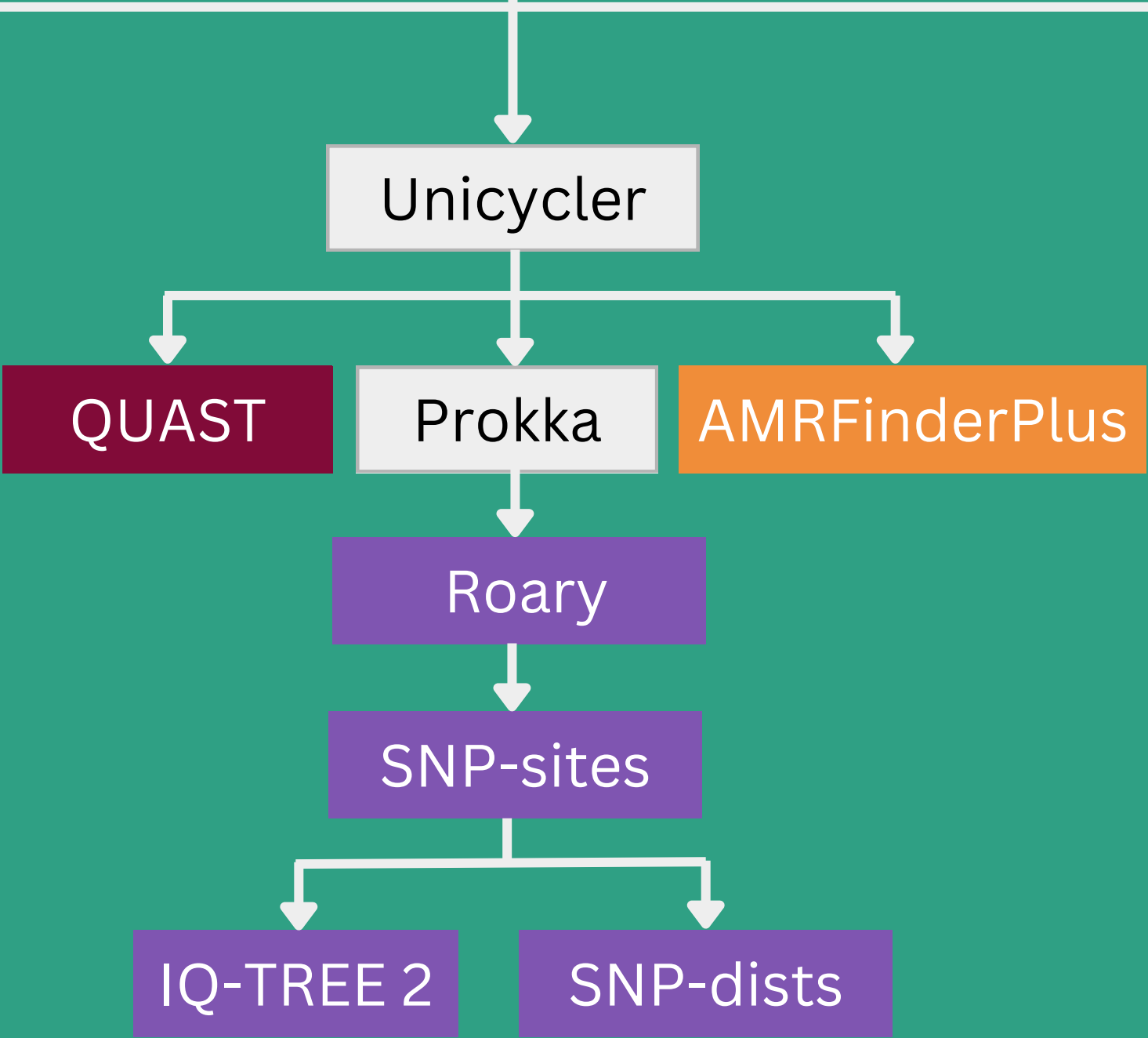
Control de calidad

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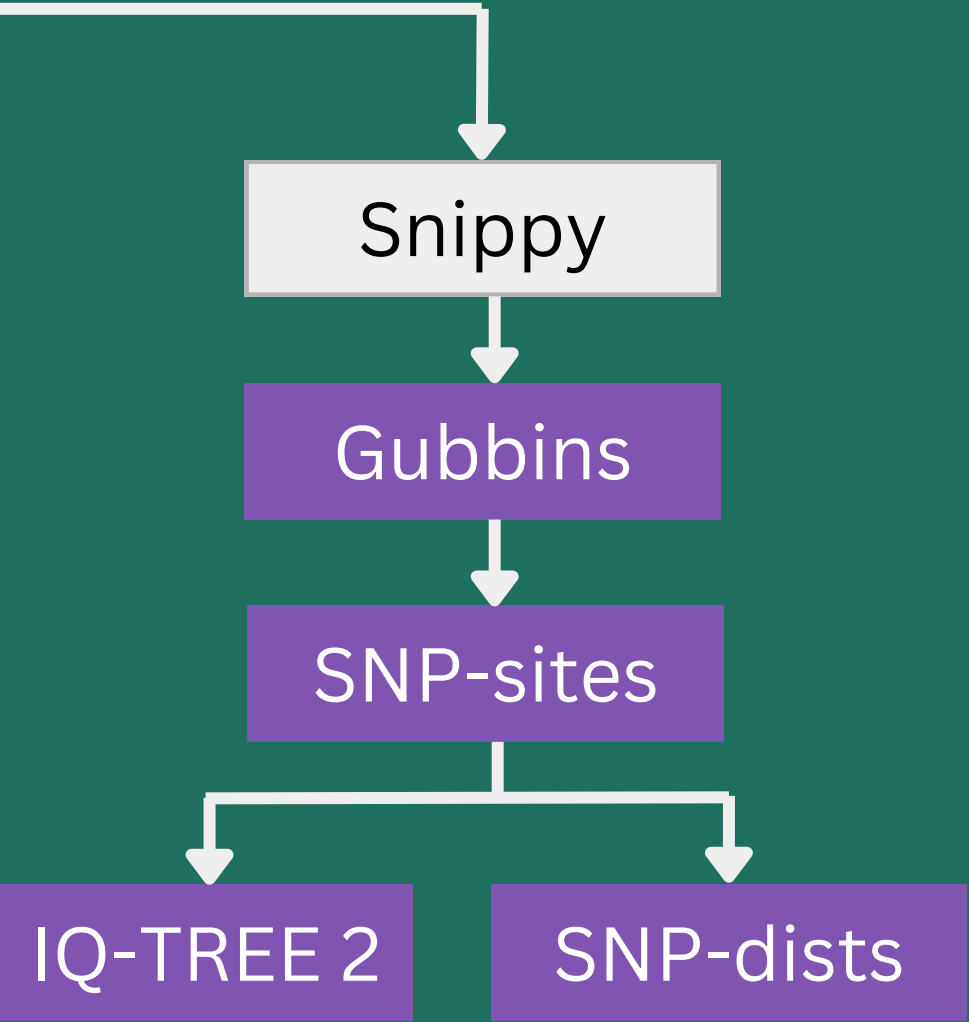
Comparación



Análisis a partir de lecturas



Análisis a partir de ensamblaje de novo



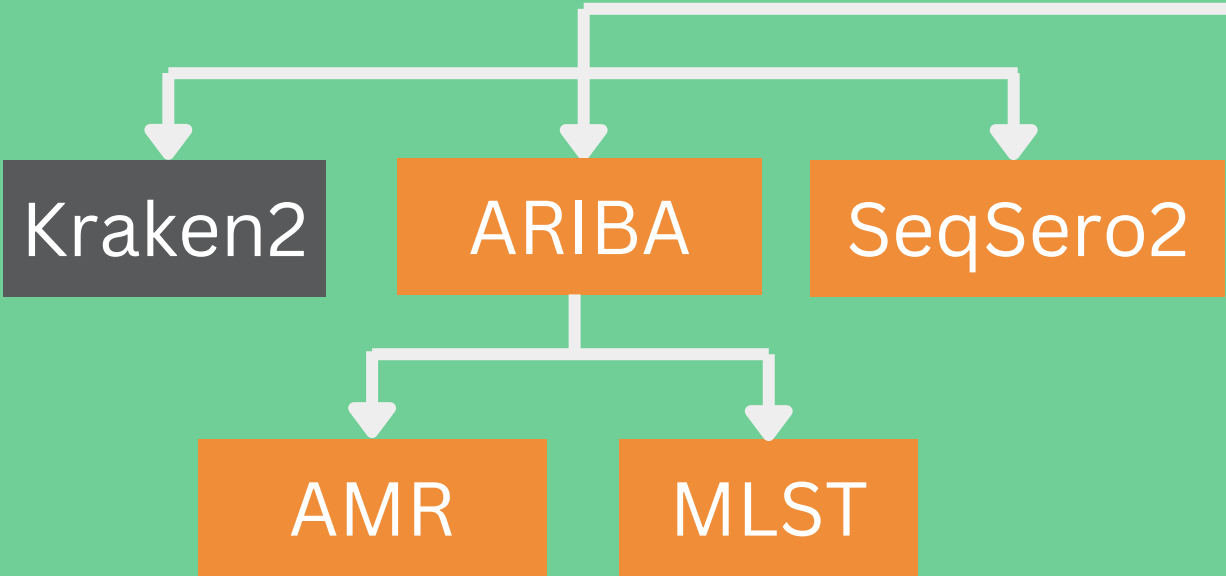
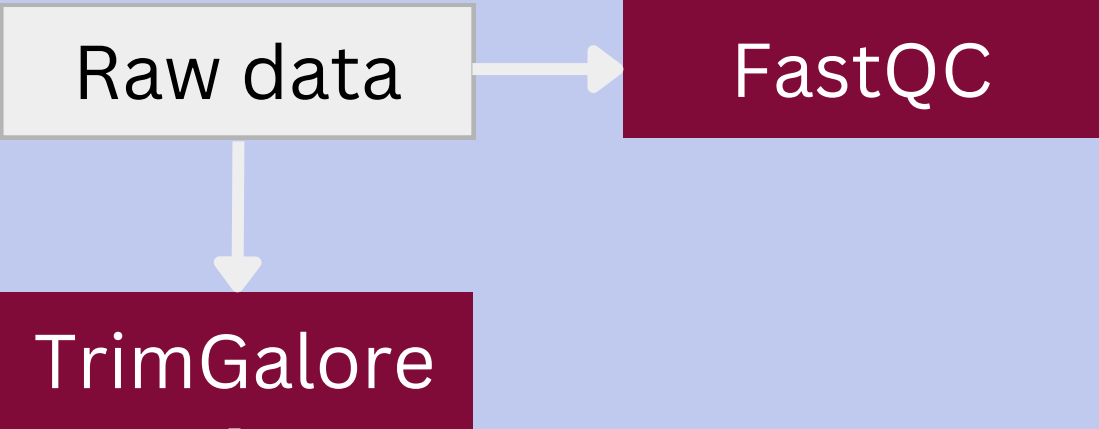
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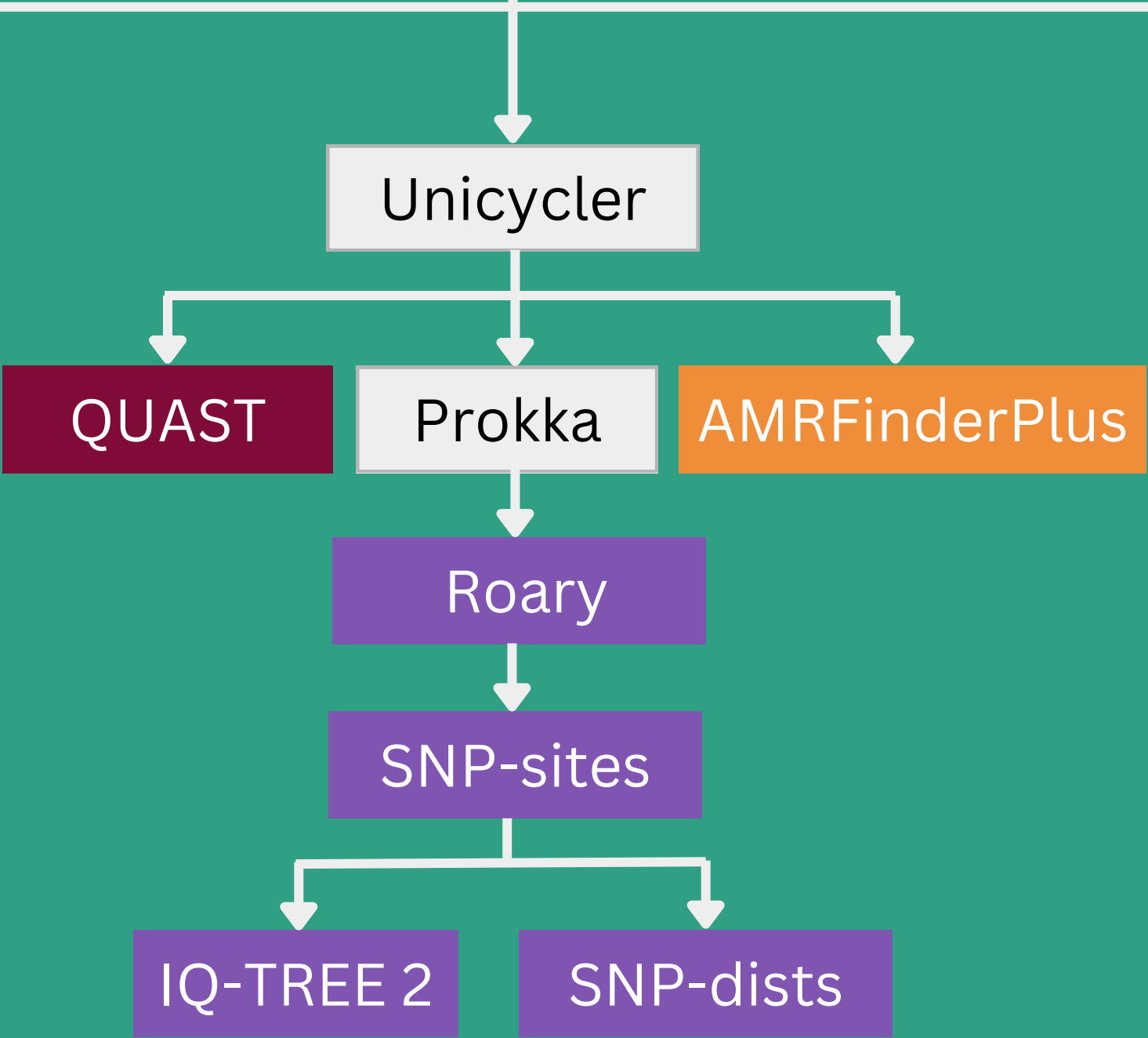
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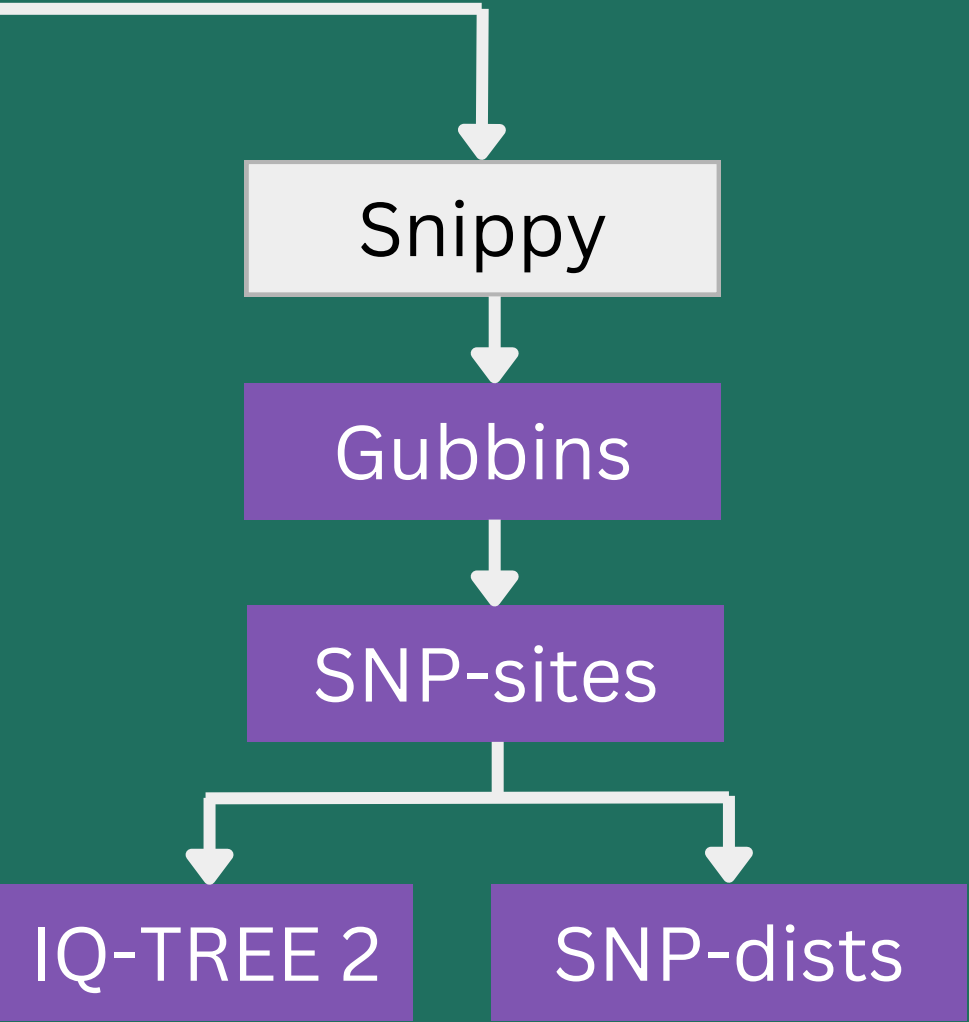
Comparación



Análisis a partir de lecturas



Análisis a partir de ensamblaje de novo



Análisis a partir de ensamblaje de referencia



AGCTCTGGTGACCCATGGGCAGCTGCTAGGGA

AGCTCT
GCTCTG



k-mers	Tax ID
AGCTCT	9606



Kraken2

Kraken2-Minimizer-based subsampling

ref ATGCTAGCTAGCT



Kraken2

Kraken2-Minimizer-based subsampling

ref **ATGCTAG**CTAGCT

k-7



Kraken2

Kraken2-Minimizer-based subsampling

ref **ATGCTAG**CTAGCT

k-7

k-mers

Subcadenas de longitud fija **k**
extraídas de una secuencia

Se recorren todas las posiciones
posibles y se toman fragmentos
consecutivos de longitud **k**



Kraken2

Kraken2-Minimizer-based subsampling

ref **ATGCTAG**CTAGCT
k-7
ATGCTAG



Kraken2

Kraken2-Minimizer-based subsampling

ref **ATGCTAGC**TAGCT

k-7

ATGCTAG

TGCTAGC



Kraken2

Kraken2-Minimizer-based subsampling

	ref
	ATGCTAGCTAGCT
k-7	
	ATGCTAG
	TGCTAGC
	GCTAGCT



Kraken2

Kraken2-Minimizer-based subsampling

	ref	ATGCTAGCTAGCT
k-7		
	ATGCTAG	
	TGCTAGC	
	GCTAGCT	
	CTAGCTA	
	TAGCTAG	
	AGCTAGC	
	GCTAGCT	



Kraken2

Kraken2-Minimizer-based subsampling

ref ATGCTAGCTAGCT

k-7

ATGCTAG

TGCTAGC

GCTAGCT

CTAGCTA

TAGCTAG

AGCTAGC

GCTAGCT

Kraken1 DB



Kraken2

Kraken2-Minimizer-based subsampling

ref ATGCTAGCTAGCT

k-7

ATGCTAG

TGCTAGC

GCTAGCT

CTAGCTA

TAGCTAG

AGCTAGC

GCTAGCT

Kraken1 DB

Minimizers ($\ell=4$)

Es el elemento mínimo dentro de un conjunto de **ℓ -mers** generado a partir de una subsecuencia.



Kraken2

Kraken2-Minimizer-based subsampling

	ref	ATGCTAGCTAGCT	Minimizers ($\ell=4$)
k-7			
	ATGCTAG		Kraken2
	TGCTAGC		
	GCTAGCT	ATGCTAG	
	CTAGCTA		
	TAGCTAG		
	AGCTAGC		
	GCTAGCT		
	Kraken1 DB		



Kraken2

Kraken2-Minimizer-based subsampling

	ref	ATGCTAGCTAGCT	Minimizers ($\ell=4$)
k-7	ATGCTAG		AGCA
	TGCTAGC		AGCA
	GCTAGCT		AGCT
	CTAGCTA		AGCT
	TAGCTAG		AGCT
	AGCTAGC		AGCT
	GCTAGCT		AGCT

Kraken1 DB



Kraken2

Kraken2-Minimizer-based subsampling

	ref	ATGCTAGCTAGCT	Minimizers ($\ell=4$)
k-7			
	ATGCTAG		AGCA
	TGCTAGC		AGCA
	GCTAGCT		AGCT
	CTAGCTA		AGCT
	TAGCTAG		AGCT
	AGCTAGC		AGCT
	GCTAGCT		AGCT

Kraken1 DB



Kraken2

Kraken2-Minimizer-based subsampling

	ref	ATGCTAGCTAGCT	Minimizers ($\ell=4$)
k-7			
	ATGCTAG		
	TGCTAGC		AGCA
	GCTAGCT		AGCT
	CTAGCTA		
	TAGCTAG		
	AGCTAGC		
	GCTAGCT		
	Kraken1 DB		Kraken2 DB



Kraken2

Kraken2-Spaced seed usage

ref ATGCTAGCTAGCT

Minimizers ($\ell=4$)

AGCA

AGCT



Kraken2

Kraken2-Spaced seed usage

ref ATGCTAGCTAGCT

Minimizers ($\ell=4$)

AGCA

AGCT

Spaced seed



Kraken2

Kraken2-Spaced seed usage

Spaced seed

1110101



Kraken2

Kraken2-Spaced seed usage

Spaced seed

1110101

1: posiciones que se comparan

0: posiciones que se enmascaran



Kraken2

Kraken2-Spaced seed usage

Spaced seed

1110101

1: posiciones que se comparan

0: posiciones que se enmascaran

Esto permite que dos secuencias coincidan aunque tengan mutaciones en las posiciones ignoradas.



Kraken2

Kraken2-Spaced seed usage

ref ATGCTAGCTAGCTACGTACGTACGTACGTACGTACGTACGT

Minimizers ($\ell=12$)

ACGTACGTACGT



Kraken2

Kraken2-Spaced seed usage

ref ATGCTAGCTAGCTACGTACGTACGTACGTACGTACGTACGT

Minimizers ($\ell=12$)

s = 3

ACGTACGTACGT



Kraken2

Kraken2-Spaced seed usage

ref ATGCTAGCTAGCTACGTACGTACGTACGTACGTACGTACGT

Minimizers ($\ell=12$)

s = 3

ACGTACGTACGT

11111010101



Kraken2

Kraken2-Spaced seed usage

ref ATGCTAGCTAGCTACGTACGTACGTACGTACGTACGTACGT

Minimizers ($\ell=12$)

$s = 3$

ACGTACGTACGT

11111**010101**



Kraken2

Kraken2-Spaced seed usage

ref ATGCTAGCTAGCTACGTACGTACGTACGTACGTACGTACGT

Minimizers ($\ell=12$)

s = 3

ACGTACGTACGT

111111**010101**

ACGTAC-T-C-T



Kraken2

Kraken2-Spaced seed usage

ref ATGCTAGCTAGCTACGTACGTACGTACGTACGTACGTACGT

Minimizers ($\ell=12$)

s = 3

ACGTACGTACGT

111111**010101**

ACGTAC-T-C-T

Aumento en s

Aumento en sensibilidad

Tolera más mutaciones



Kraken2

Kraken2-Spaced seed usage

ref ATGCTAGCTAGCTACGTACGTACGTACGTACGTACGTACGT

Minimizers ($\ell=12$)

s = 3

ACGTACGTACGT

111111**010101**

ACGTAC-T-C-T

Aumento en s

Disminuye precisión

Aumenta riesgo de
coincidencias falsas



Kraken2

Kraken2-Spaced seed usage

ref ATGCTAGCTAGCTACGTACGTACGTACGTACGTACGTACGT

Minimizers ($\ell=12$)

$s = 3$

ACGTACGTACGT

111111**010101**

ACGTAC-T-C-T

Valores por default

$k = 35$

$\ell = 31$

$s = 7$



Kraken2

Kraken2-Compact hash table

ref ATGCTAGCTAGCTACGTACGTACGTACGTACGTACGTACGT

Minimizers ($\ell=12$)

$s = 3$

ACGTAC-T-C-T



Kraken2

Kraken2-Compact hash table

ref ATGCTAGCTAGCTACGTACGTACGTACGTACGTACGTACGT

Minimizers ($\ell=12$)

s = 3

hash

ACGTAC-T-C-T



Kraken2

Kraken2-Compact hash table

ref ATGCTAGCTAGCTACGTACGTACGTACGTACGTACGTACGT

Minimizers ($\ell=12$)

s = 3

ACGTAC-T-C-T

hash

Transforma una
secuencia en un número



Kraken2

Kraken2-Compact hash table

ref ATGCTAGCTAGCTACGTACGTACGTACGTACGTACGTACGT

Minimizers ($\ell=12$)

s = 3

hash

ACGTAC-T-C-T

hash(ACGTAC-T-C-T)



Kraken2

Kraken2-Compact hash table

ref ATGCTAGCTAGCTACGTACGTACGTACGTACGTACGTACGT

Minimizers ($\ell=12$)

s = 3

hash

ACGTAC-T-C-T

hash(ACGTAC-T-C-T)

1001101 taxon 562 (E. coli)



Kraken2

Kraken2-Compact hash table

ref ATGCTAGCTAGCTACGTACGTACGTACGTACGTACGTACGT

Minimizers ($\ell=12$)

s = 3

hash

ACGTAC-T-C-T

hash(ACGTAC-T-C-T)

1001101 taxon 562 (E. coli)

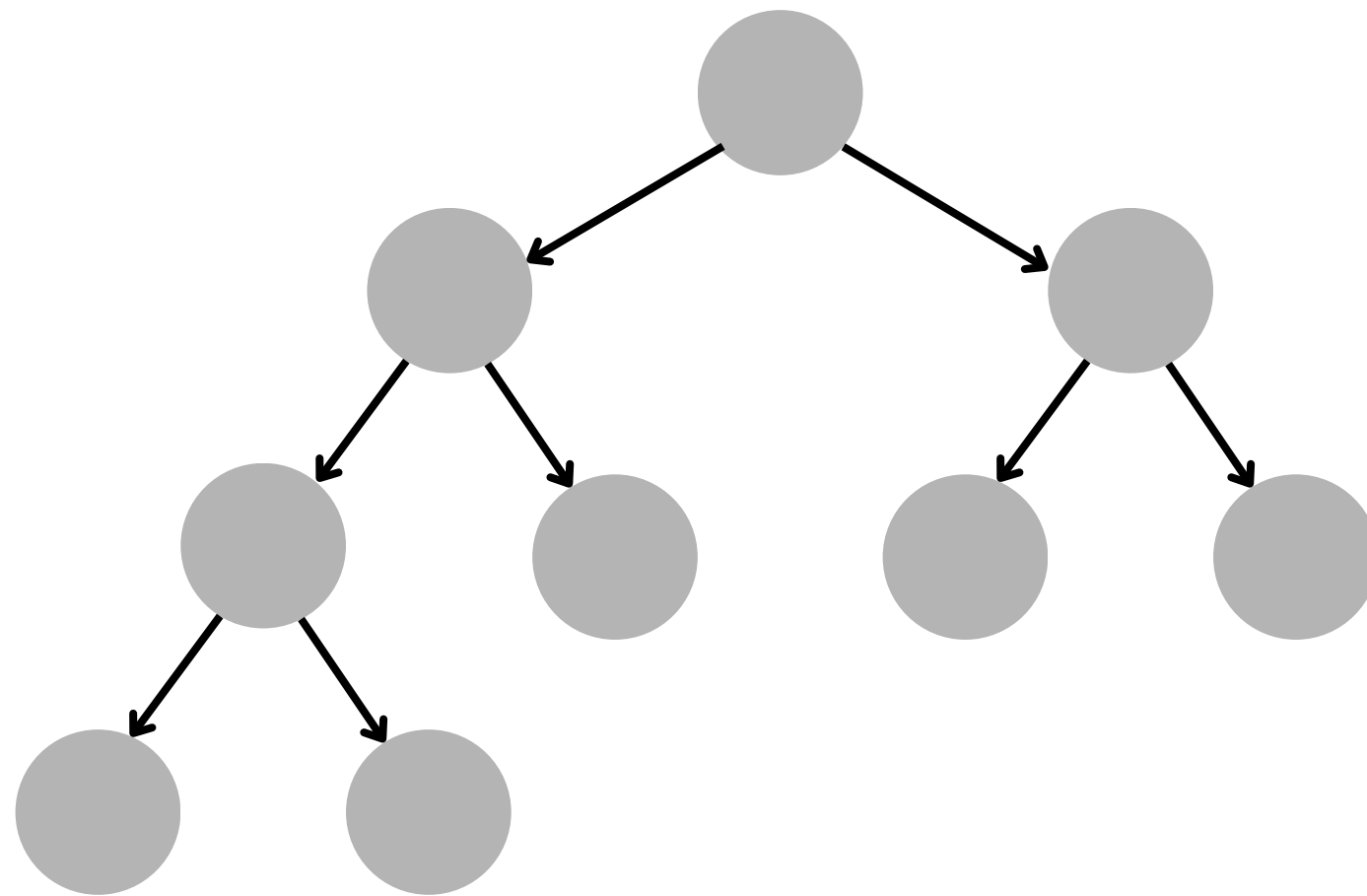
Taxonomic ID

LCA values



Kraken2

Kraken2-Compact hash table

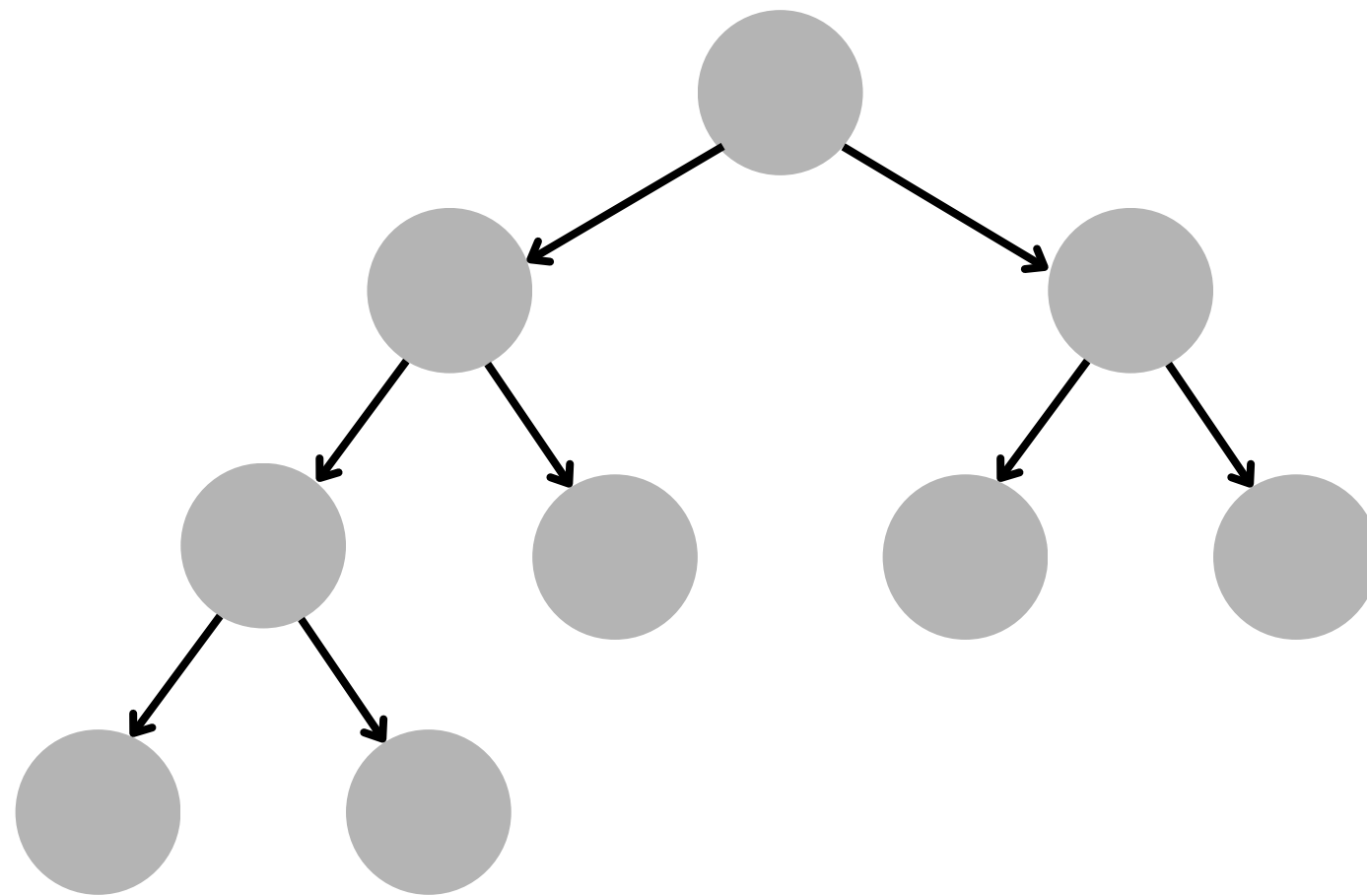


**LCA (Lowest
Common Ancestor)**



Kraken2

Kraken2-Compact hash table



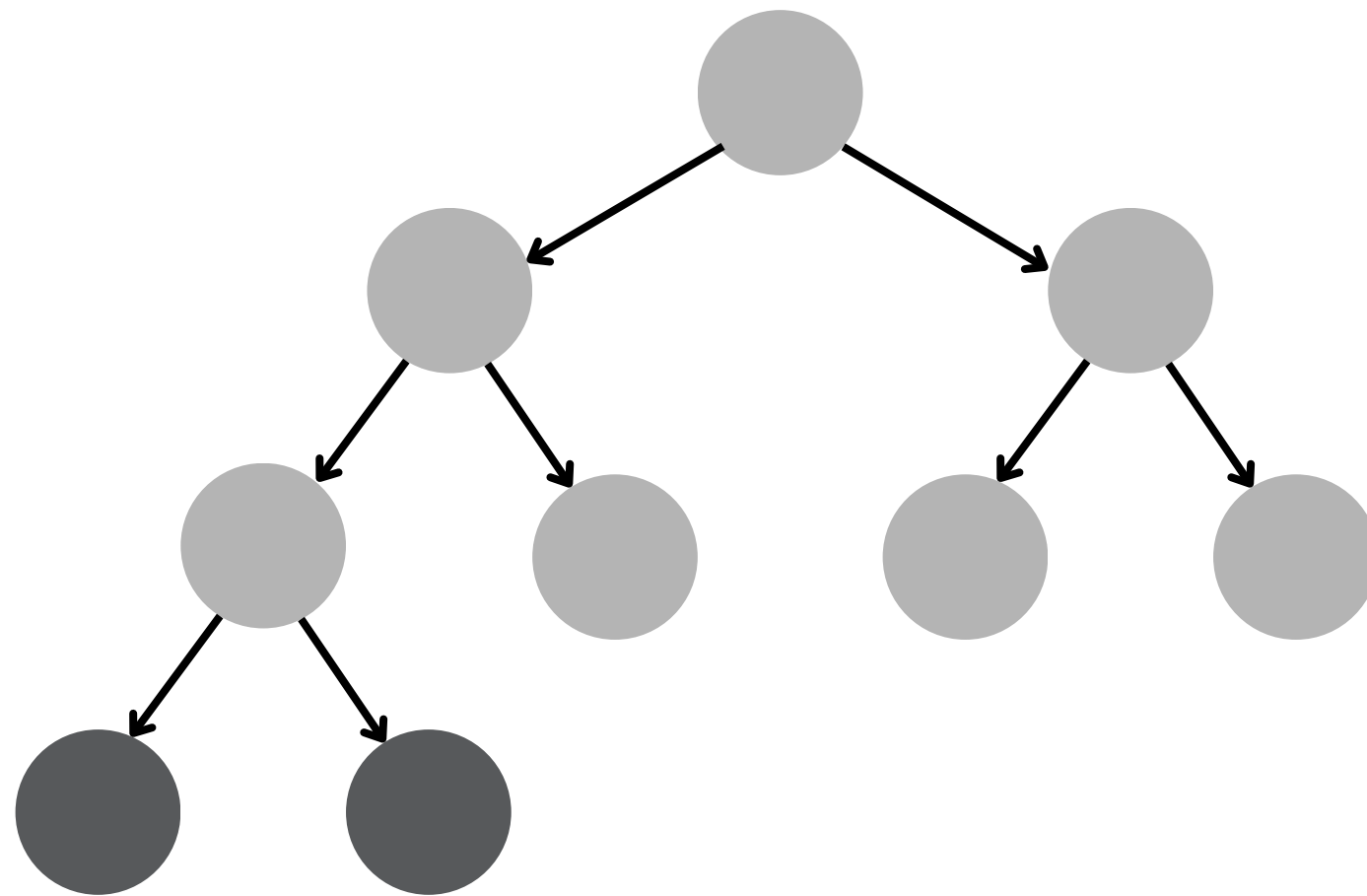
**LCA (Lowest
Common Ancestor)**

El ancestro común
compartido más
específico entre dos o
más nodos.



Kraken2

Kraken2-Compact hash table

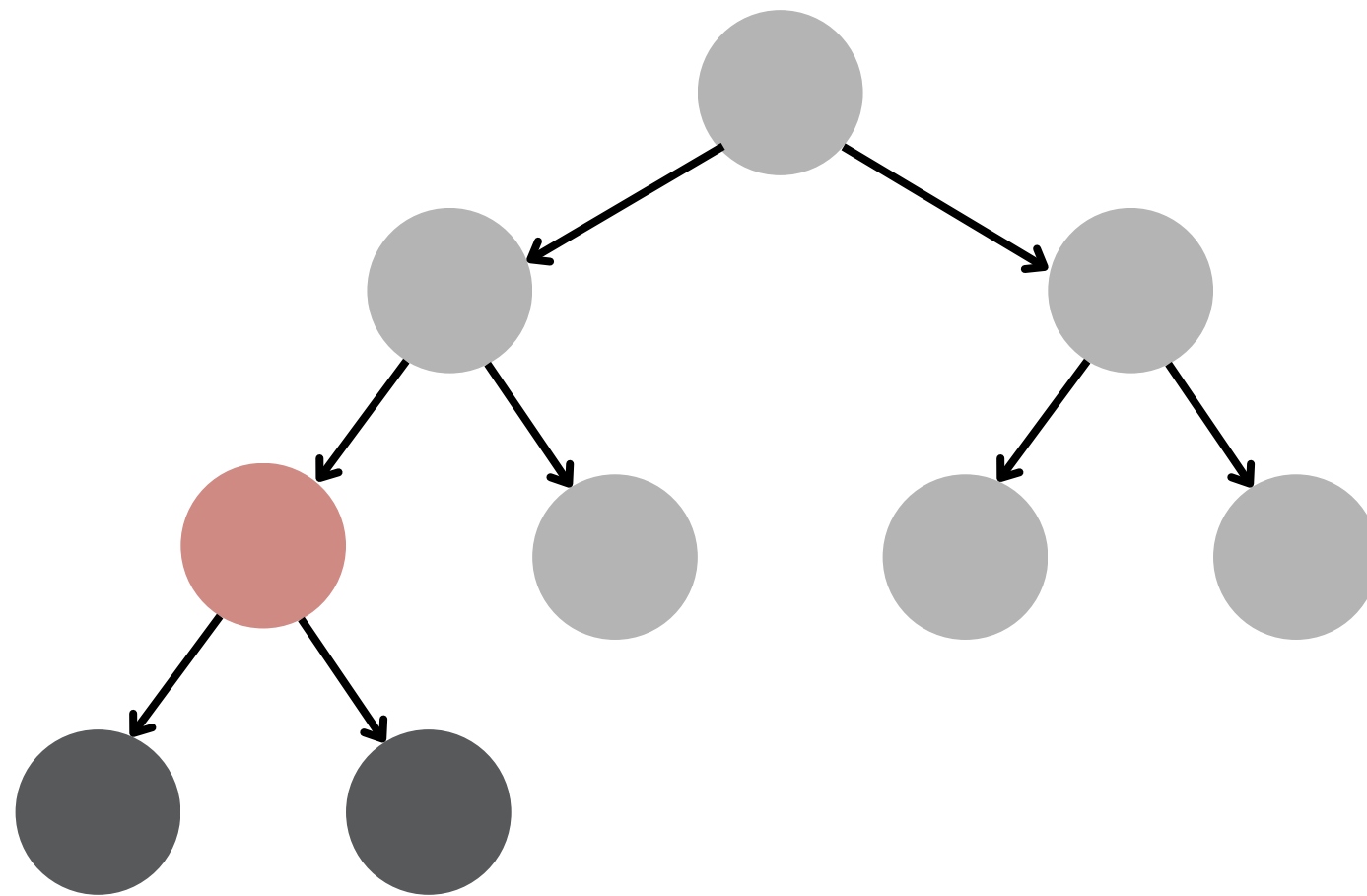


**LCA (Lowest
Common Ancestor)**



Kraken2

Kraken2-Compact hash table

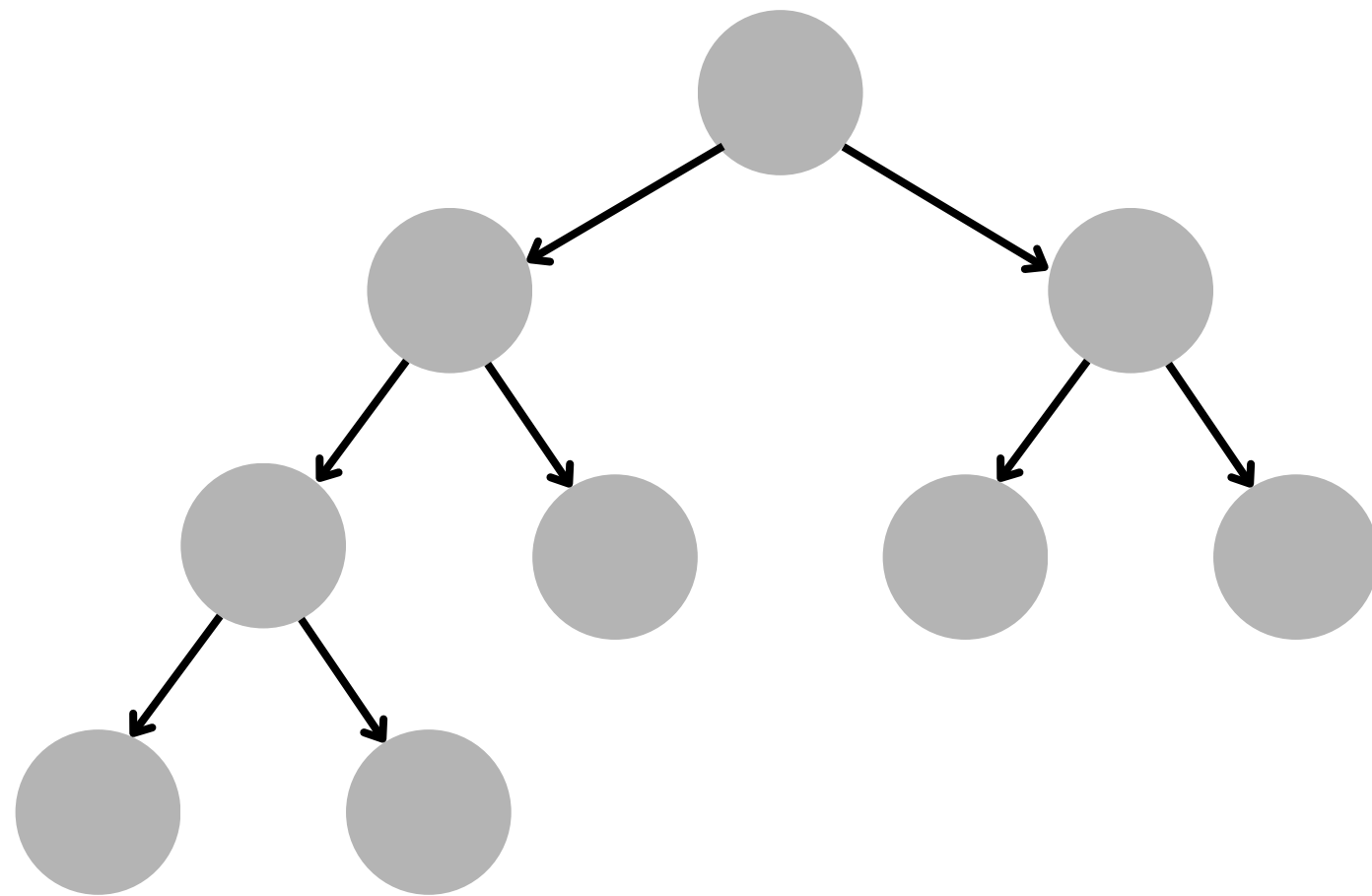


LCA (Lowest
Common Ancestor)



Kraken2

Kraken2-Compact hash table



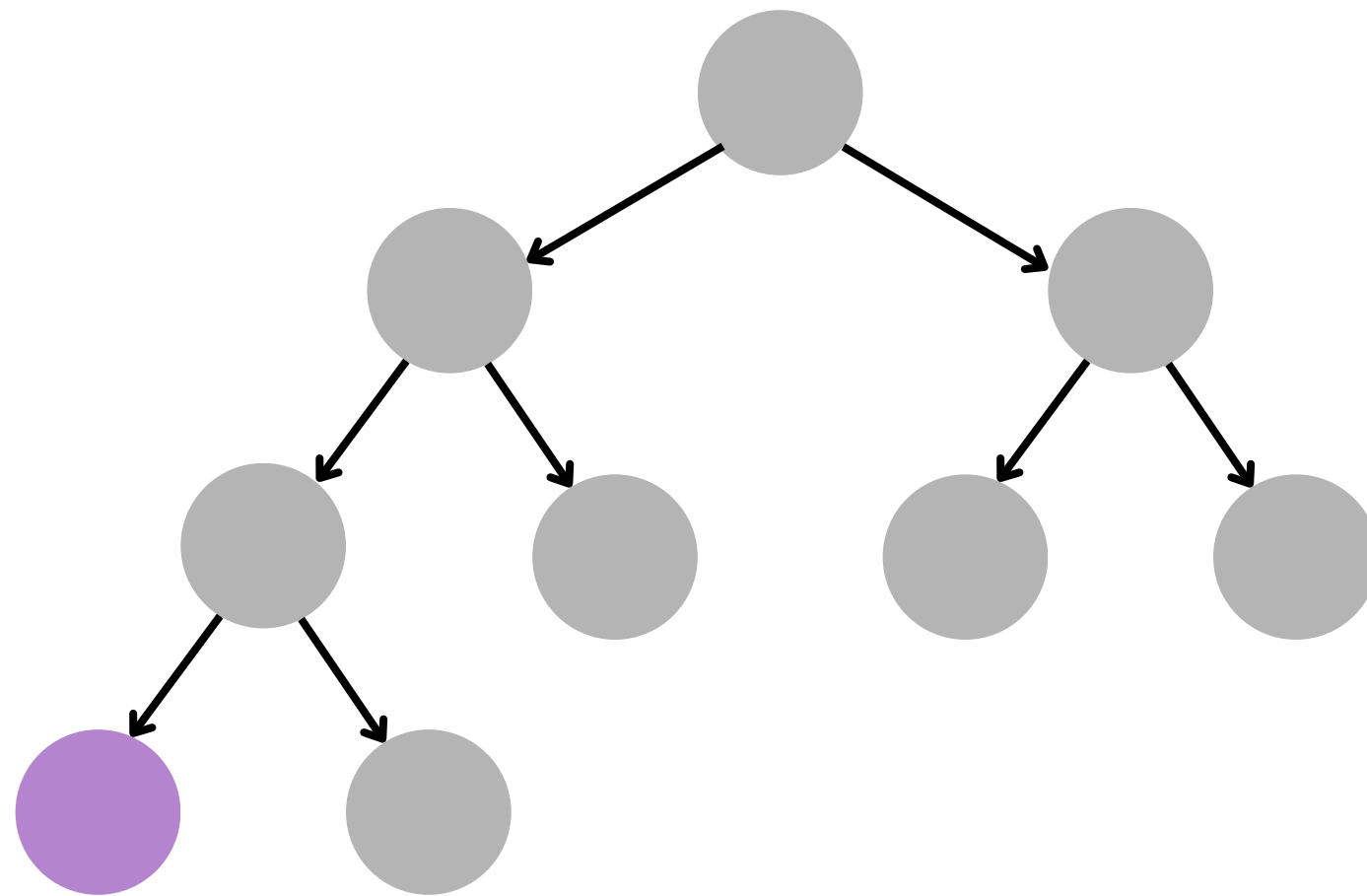
**LCA (Lowest
Common Ancestor)**





Kraken2

Kraken2-Compact hash table



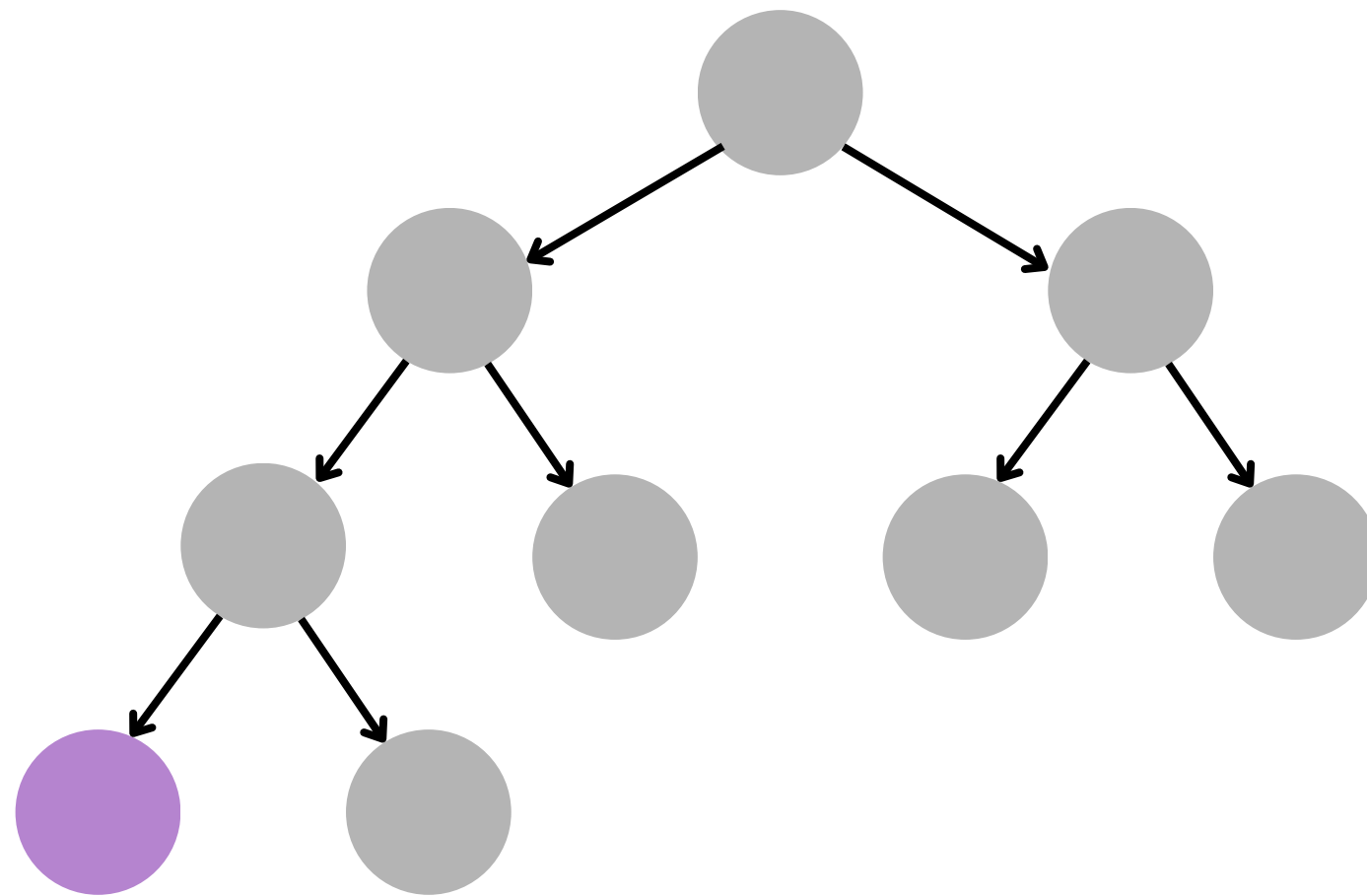
**LCA (Lowest
Common Ancestor)**





Kraken2

Kraken2-Compact hash table



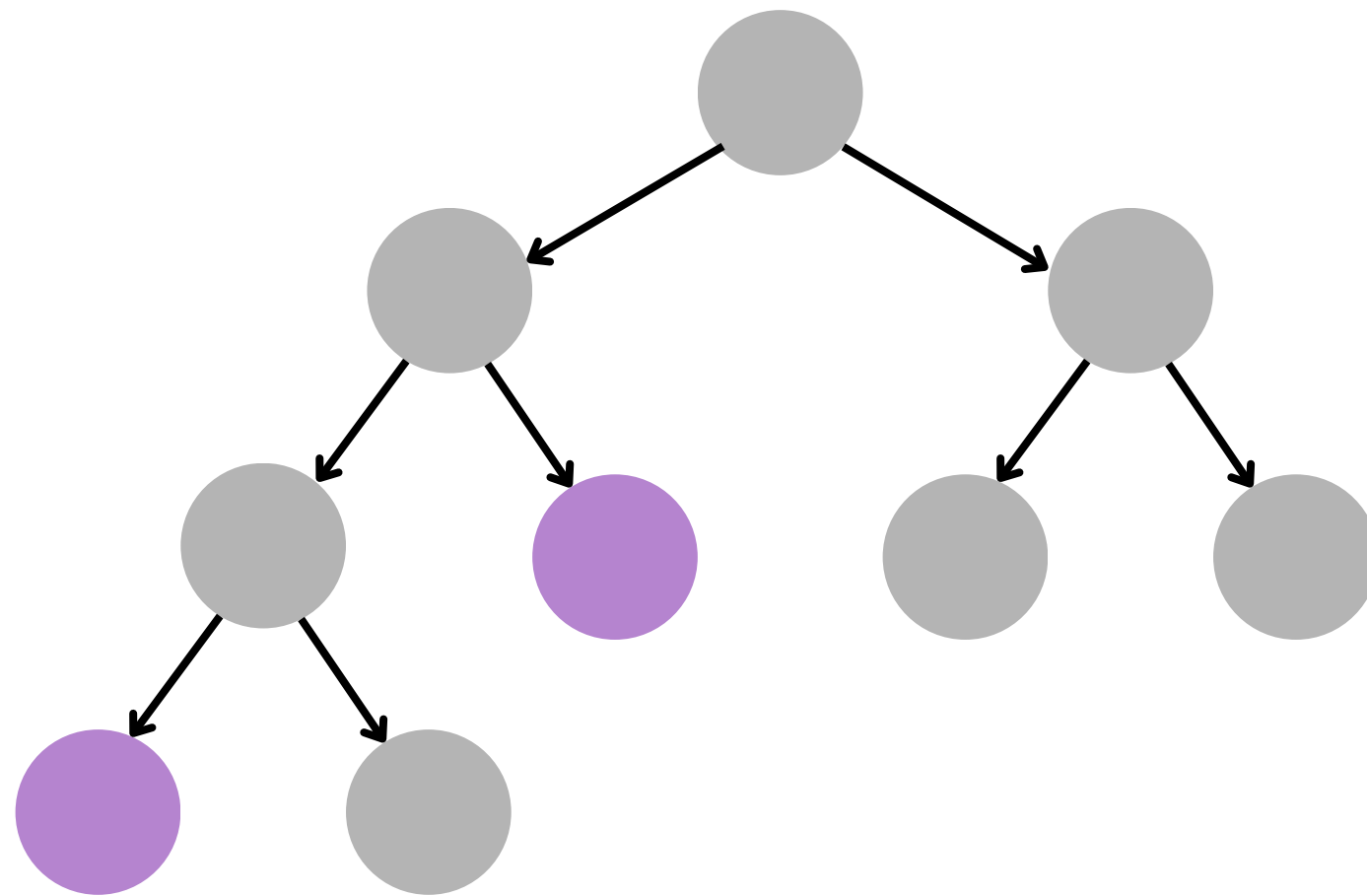
**LCA (Lowest
Common Ancestor)**





Kraken2

Kraken2-Compact hash table



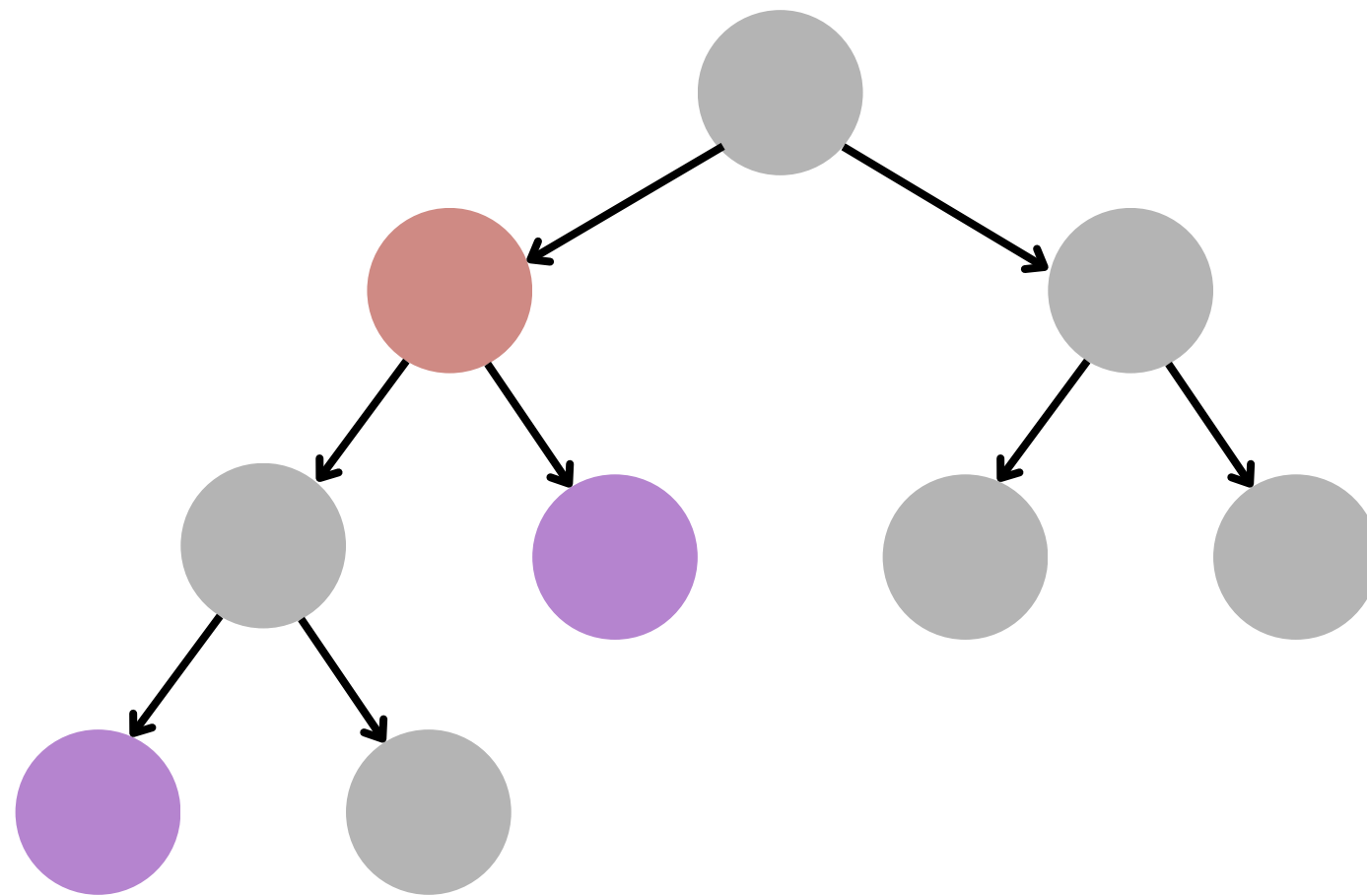
**LCA (Lowest
Common Ancestor)**





Kraken2

Kraken2-Compact hash table



**LCA (Lowest
Common Ancestor)**





Kraken2

Kraken2-Compact hash table

Hash code	Taxón
0010101	E. coli
1001101	Escherichia
1110001	Enterobacteriaceae
1010101	Bacteria



Kraken2

Kraken2-Compact hash table

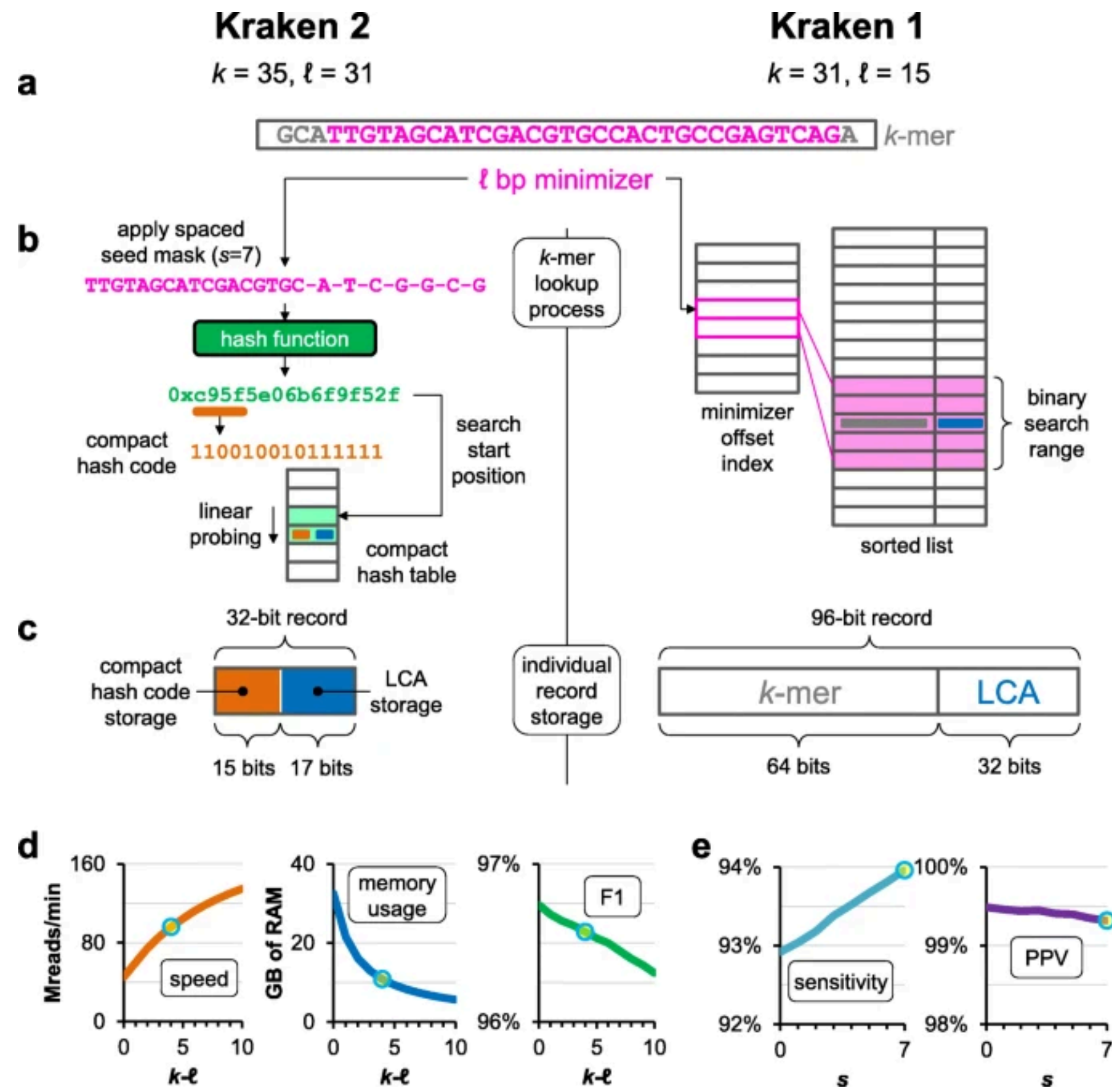
Base de datos de Kraken2

Hash code

LCA



Kraken2





Kraken2

Kraken2-Classification of a sequence fragment

1. Generación de k-mers

S ATGCTAGCTAGCTAGC

k-7 **ATGCTAG**
 TGCTAGC
 GCTAGCT
 CTAGCTA
 TAGCTAG
 AGCTAGC



Kraken2

Kraken2-Classification of a sequence fragment

2. Obtención del minimizer de cada k-mer

S ATGCTAGCTAGCTAGC

$\ell = 5$ **ATGCT**

$s = 2$ **CTAGC**

CTAGC

AGCTA

AGCTA

AGCTA



Kraken2

Kraken2-Classification of a sequence fragment

3. Obtención del hash

S ATGCTAGCTAGCTAGC

$\ell = 5$ **ATGCT** **1001101**

$s = 2$ **CTAGC**

CTAGC

AGCTA

AGCTA

AGCTA



Kraken2-Classification of a sequence fragment

4. Buscar minimizer en la CHT

S	ATGCTAGCTAGCTAGC	Hash	TaxID
		0010101	1239
	1001101	1001101	3585
		1110001	4617
		1010101	1279
		1110011	91061
		0101110	1280



Kraken2

Kraken2-Classification of a sequence fragment

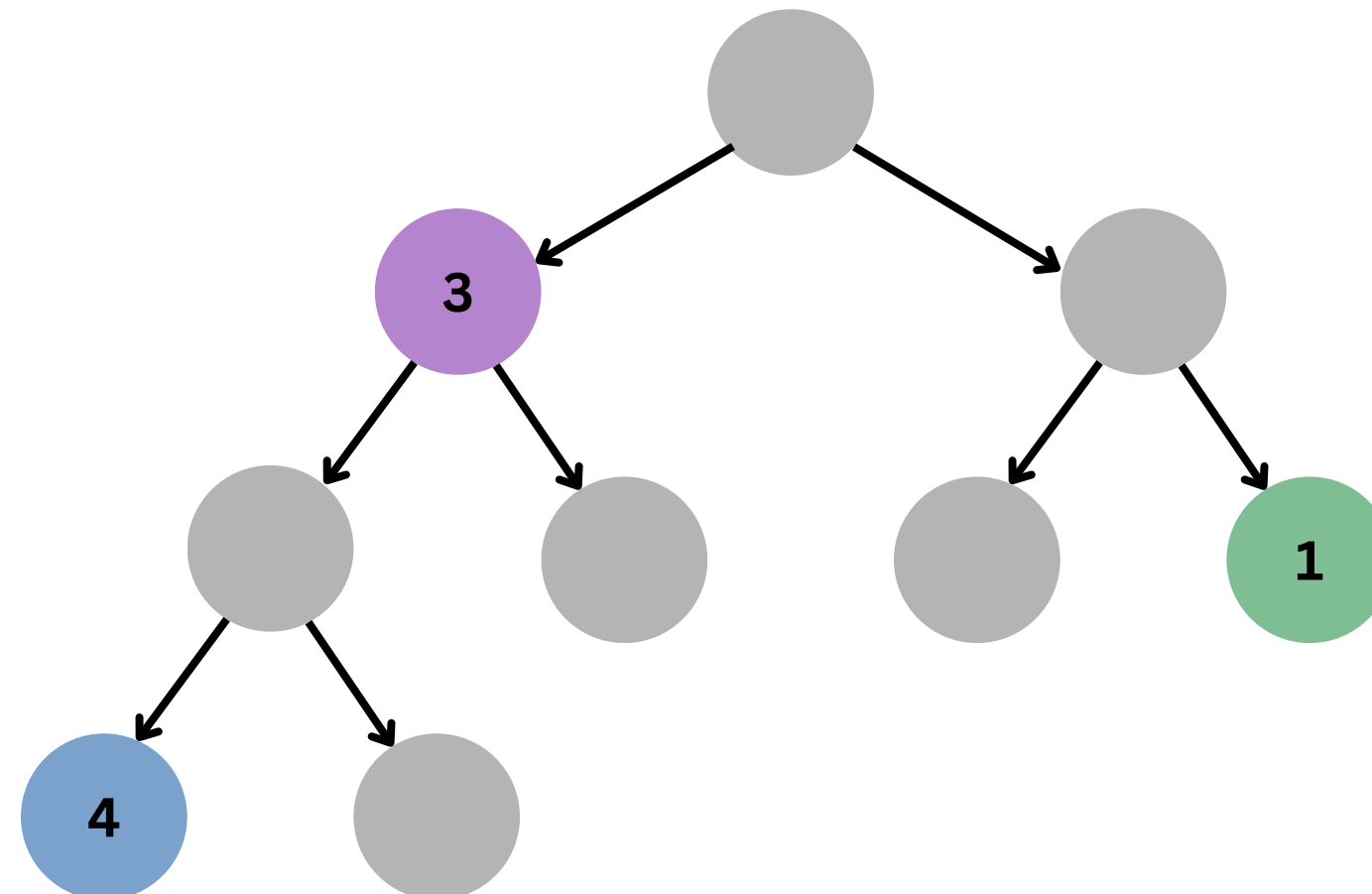
5. Asignación taxonómica mediante árbol de clasificación

S ATGCTAGCTAGCTAGC

0010101 0010101 1001101

1001101 0010101 1001101

1110001 0010101





Kraken2

Kraken2-Classification of a sequence fragment

5. Asignación taxonómica mediante árbol de clasificación

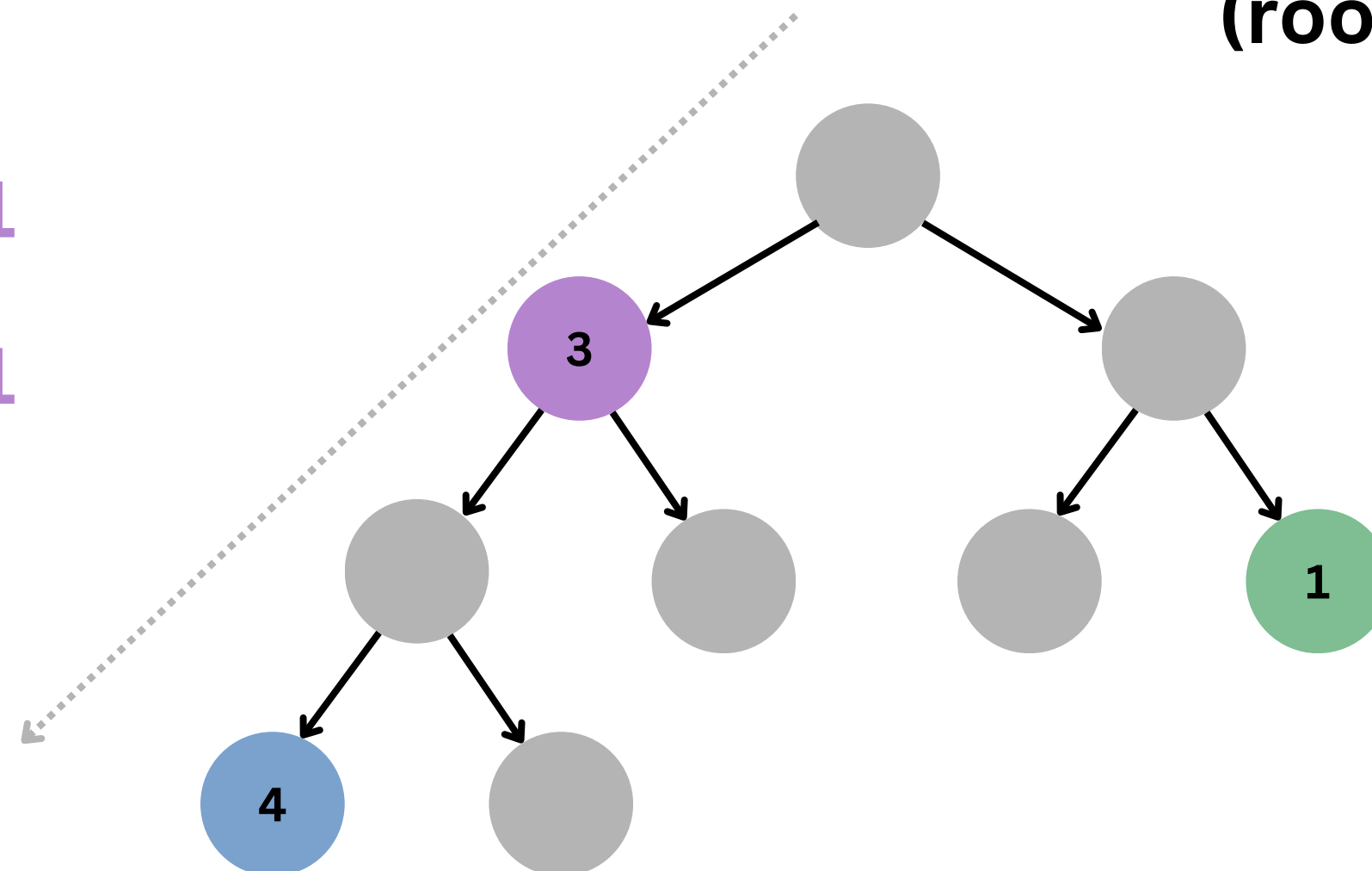
S ATGCTAGCTAGCTAGC

0010101 0010101 1001101

1001101 0010101 1001101

1110001 0010101

RTL
(root-to-leaf)





Kraken2

Kraken2-Classification of a sequence fragment

5. Asignación taxonómica mediante árbol de clasificación

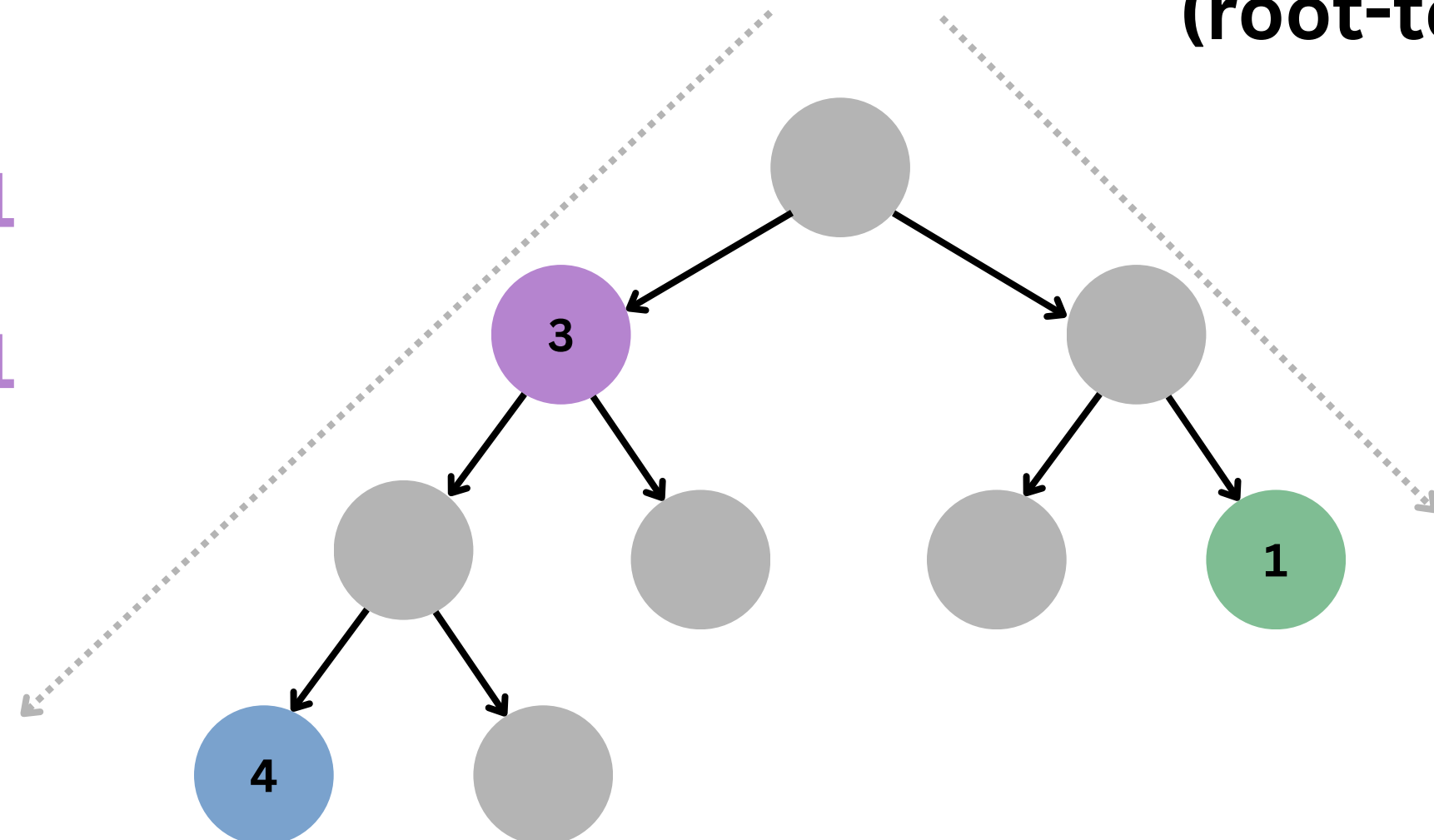
S ATGCTAGCTAGCTAGC

0010101 0010101 1001101

1001101 0010101 1001101

1110001 0010101

RTL
(root-to-leaf)





Kraken2

Kraken2-Classification of a sequence fragment

5. Asignación taxonómica mediante árbol de clasificación

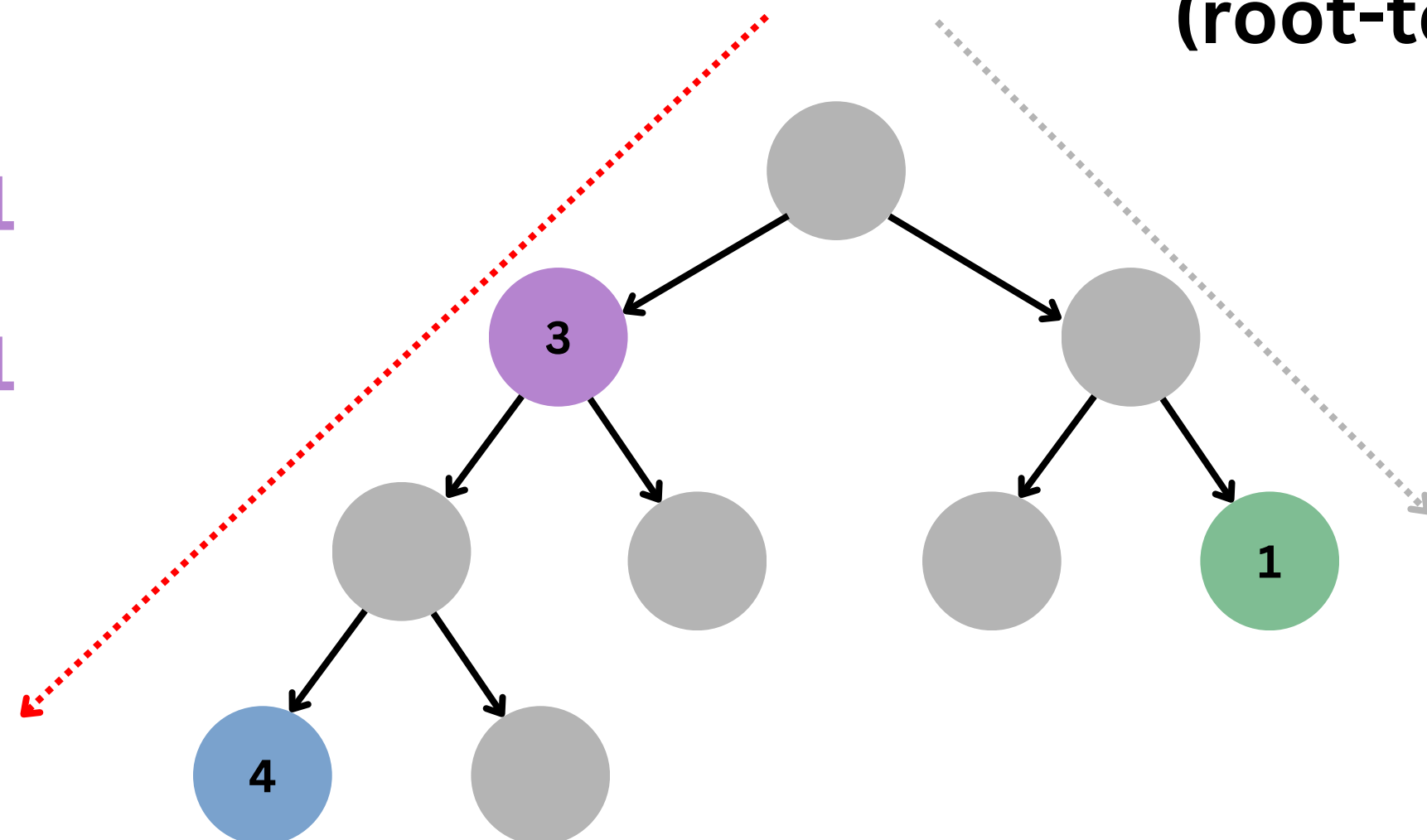
S ATGCTAGCTAGCTAGC

0010101 0010101 1001101

1001101 0010101 1001101

1110001 0010101

RTL
(root-to-leaf)





Kraken2

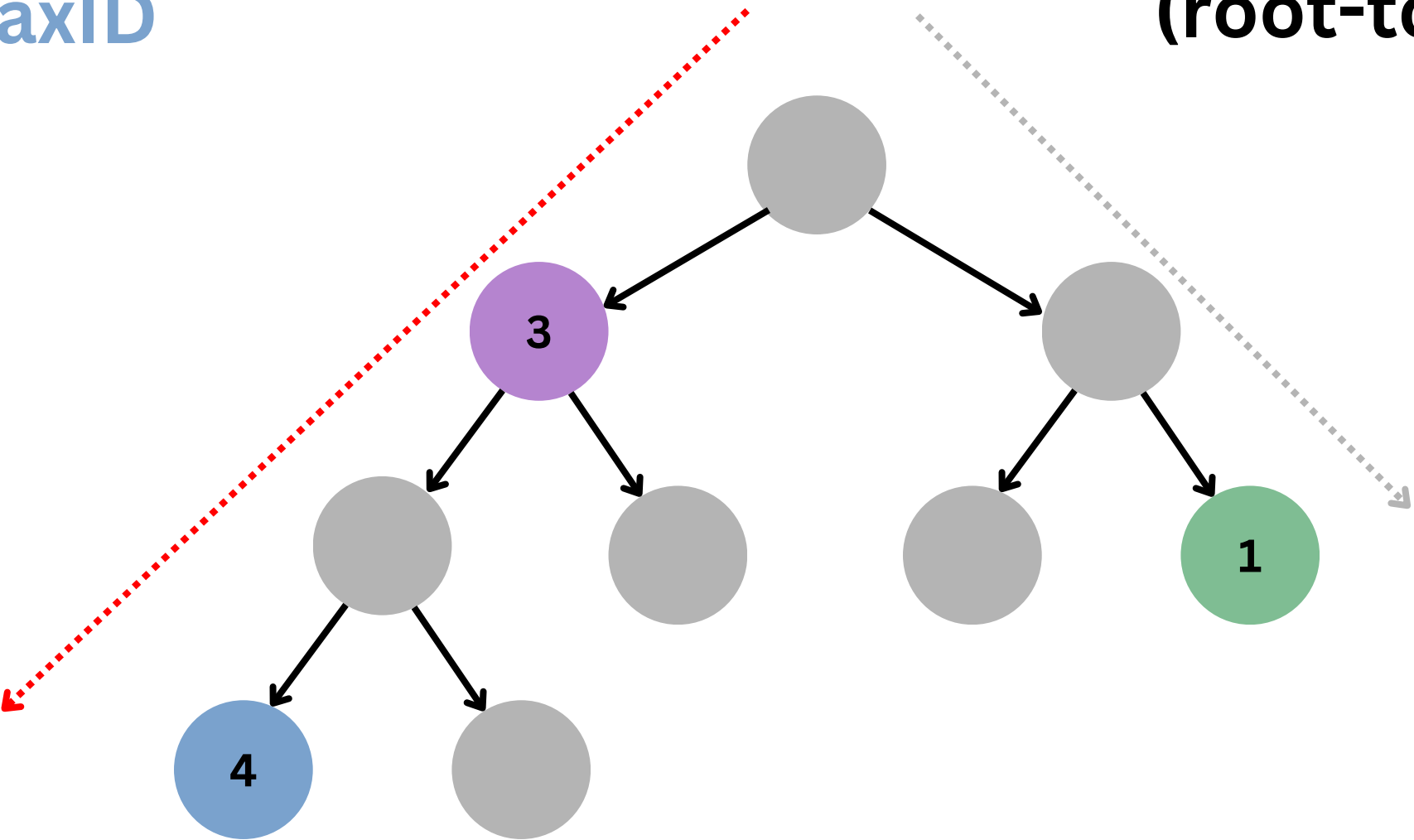
Kraken2-Classification of a sequence fragment

5. Asignación taxonómica mediante árbol de clasificación

S ATGCTAGCTAGCTAGC **TaxID**

0010101	0010101	1001101
1001101	0010101	1001101
1110001	0010101	

RTL
(root-to-leaf)





Kraken2-Reporte

83.56	514312	514312	U	0	unclassified
16.44	101180	0	R	1	root
16.44	101180	0	R1	131567	cellular organisms
16.44	101180	2775	D	2	Bacteria
13.99	86114	1	D1	1783270	FCB group
13.99	86112	0	D2	68336	Bacteroidetes/Chlorobi group
13.99	86103	8	P	976	Bacteroidetes
13.94	85798	2	C	200643	Bacteroidia
13.94	85789	19	O	171549	Bacteroidales
13.87	85392	0	F	815	Bacteroidaceae



Kraken2-Reporte

83.56	514312	514312	U	0	unclassified	Porcentaje de todos los reads que están asignados a ese clado
16.44	101180	0	R	1	root	
16.44	101180	0	R1	131567	cellular organisms	
16.44	101180	2775	D	2	Bacteria	
13.99	86114	1	D1	1783270	FCB group	
13.99	86112	0	D2	68336	Bacteroidetes/Chlorobi group	
13.99	86103	8	P	976	Bacteroidetes	
13.94	85798	2	C	200643	Bacteroidia	
13.94	85789	19	O	171549	Bacteroidales	
13.87	85392	0	F	815	Bacteroidaceae	



Kraken2-Reporte

83.56	514312	514312	U	0	unclassified	Número total de reads asignados directa o indirectamente a ese clado
16.44	101180	0	R	1	root	
16.44	101180	0	R1	131567	cellular organisms	
16.44	101180	2775	D	2	Bacteria	
13.99	86114	1	D1	1783270	FCB group	
13.99	86112	0	D2	68336	Bacteroidetes/Chlorobi group	
13.99	86103	8	P	976	Bacteroidetes	
13.94	85798	2	C	200643	Bacteroidia	
13.94	85789	19	O	171549	Bacteroidales	
13.87	85392	0	F	815	Bacteroidaceae	



Kraken2

Kraken2-Reporte

						Número de reads asignados directamente a ese taxón
83.56	514312	514312	U	0	unclassified	
16.44	101180	0	R	1	root	
16.44	101180	0	R1	131567	cellular organisms	
16.44	101180	2775	D	2	Bacteria	
13.99	86114	1	D1	1783270	FCB group	
13.99	86112	0	D2	68336	Bacteroidetes/Chlorobi group	
13.99	86103	8	P	976	Bacteroidetes	
13.94	85798	2	C	200643	Bacteroidia	
13.94	85789	19	O	171549	Bacteroidales	
13.87	85392	0	F	815	Bacteroidaceae	



Kraken2

Kraken2-Reporte

						Nivel taxonómico
83.56	514312	514312	U	0	unclassified	
16.44	101180	0	R	1	root	
16.44	101180	0	R1	131567	cellular organisms	
16.44	101180	2775	D	2	Bacteria	
13.99	86114	1	D1	1783270	FCB group	
13.99	86112	0	D2	68336	Bacteroidetes/Chlorobi group	
13.99	86103	8	P	976	Bacteroidetes	
13.94	85798	2	C	200643	Bacteroidia	
13.94	85789	19	O	171549	Bacteroidales	
13.87	85392	0	F	815	Bacteroidaceae	



Kraken2

Kraken2-Reporte

83.56	514312	514312	U	0	unclassified	Nivel taxonómico
16.44	101180	0	R	1	root	K=kingdom
16.44	101180	0	R1	131567	cellular organisms	P=phylum
16.44	101180	2775	D	2	Bacteria	C=class
13.99	86114	1	D1	1783270	FCB group	O=order
13.99	86112	0	D2	68336	Bacteroidetes/Chlorobi group	F=family
13.99	86103	8	P	976	Bacteroidetes	G=genus
13.94	85798	2	C	200643	Bacteroidia	S=species
13.94	85789	19	O	171549	Bacteroidales	U=unclassified
13.87	85392	0	F	815	Bacteroidaceae	R=root



Kraken2-Reporte

						Identificador numérico del taxón en NCBI Taxonomy
83.56	514312	514312	U	0	unclassified	
16.44	101180	0	R	1	root	
16.44	101180	0	R1	131567	cellular organisms	
16.44	101180	2775	D	2	Bacteria	
13.99	86114	1	D1	1783270	FCB group	
13.99	86112	0	D2	68336	Bacteroidetes/Chlorobi group	
13.99	86103	8	P	976	Bacteroidetes	
13.94	85798	2	C	200643	Bacteroidia	
13.94	85789	19	O	171549	Bacteroidales	
13.87	85392	0	F	815	Bacteroidaceae	



Kraken2

Kraken2-Reporte

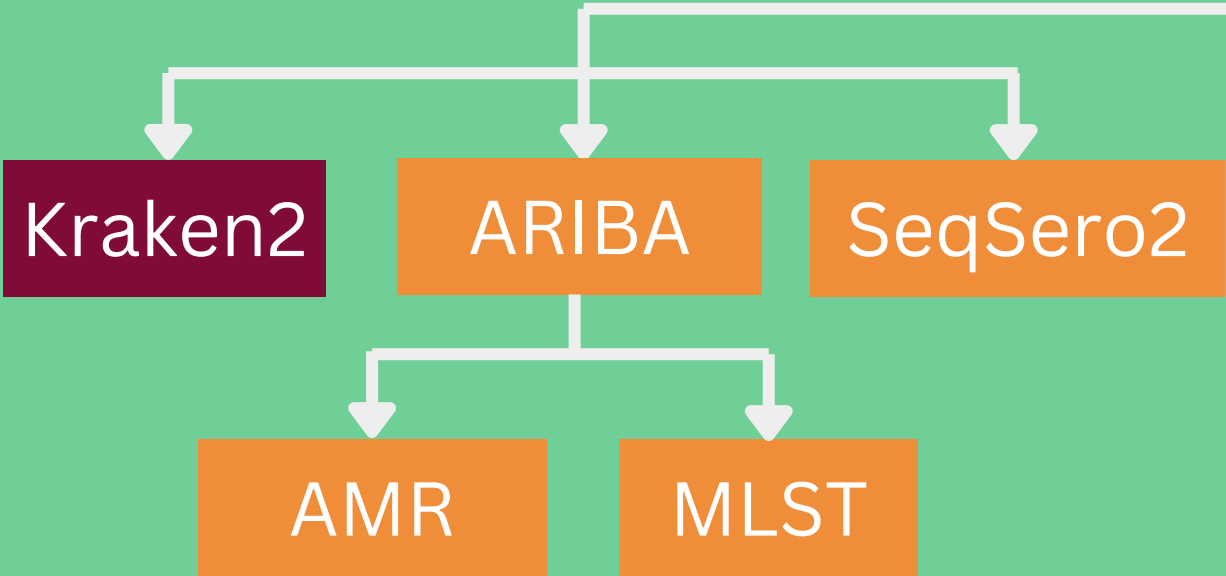
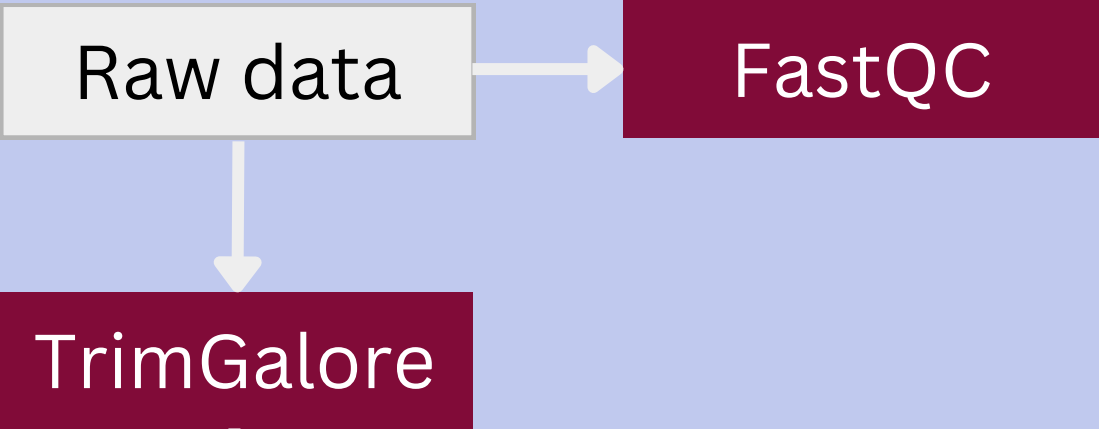
						Nombre científico del taxón
83.56	514312	514312	U	0	unclassified	
16.44	101180	0	R	1	root	
16.44	101180	0	R1	131567	cellular organisms	
16.44	101180	2775	D	2	Bacteria	
13.99	86114	1	D1	1783270	FCB group	
13.99	86112	0	D2	68336	Bacteroidetes/Chlorobi group	
13.99	86103	8	P	976	Bacteroidetes	
13.94	85798	2	C	200643	Bacteroidia	
13.94	85789	19	O	171549	Bacteroidales	
13.87	85392	0	F	815	Bacteroidaceae	

Preprocesamiento

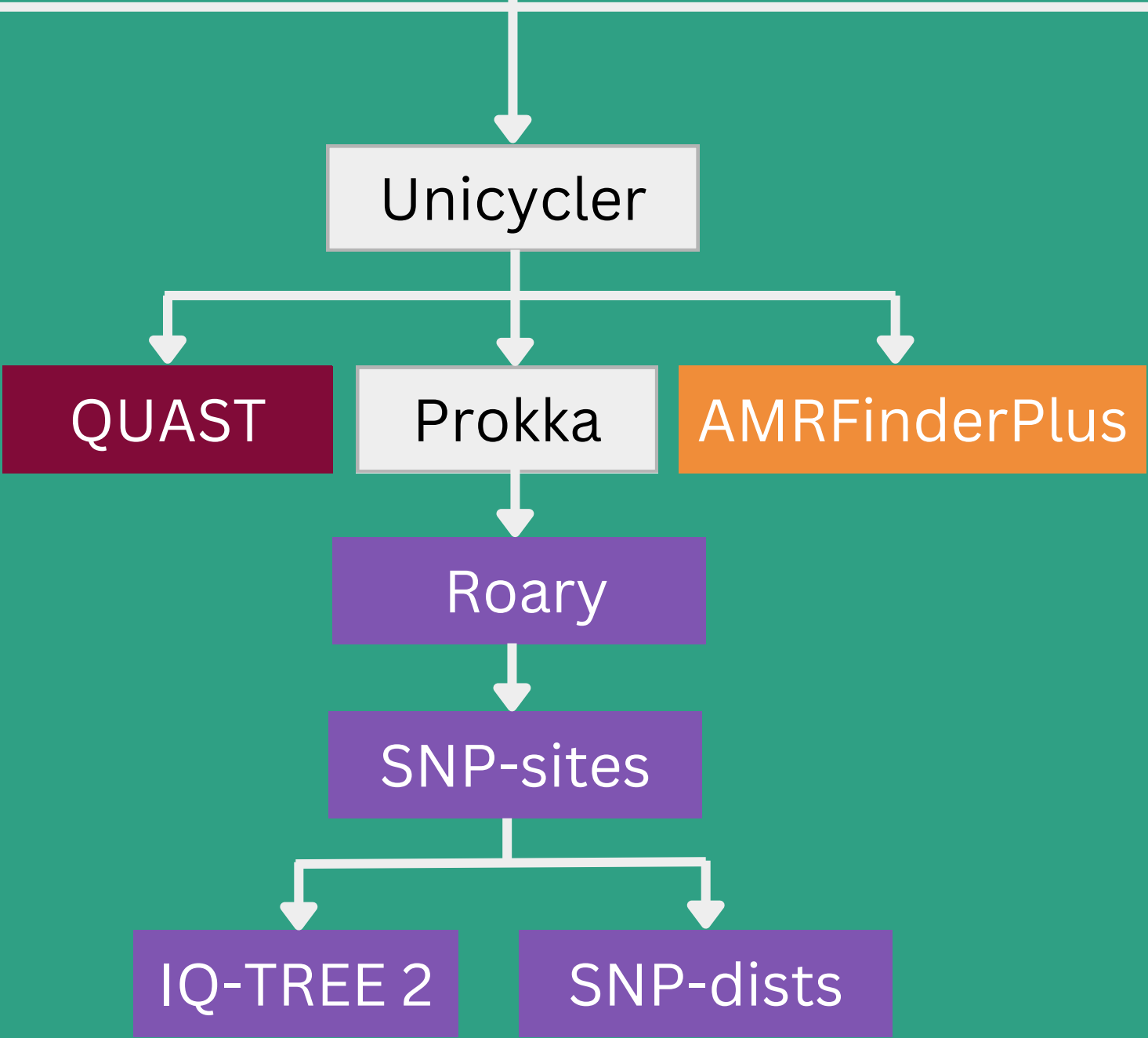
Control de calidad

Caracterización

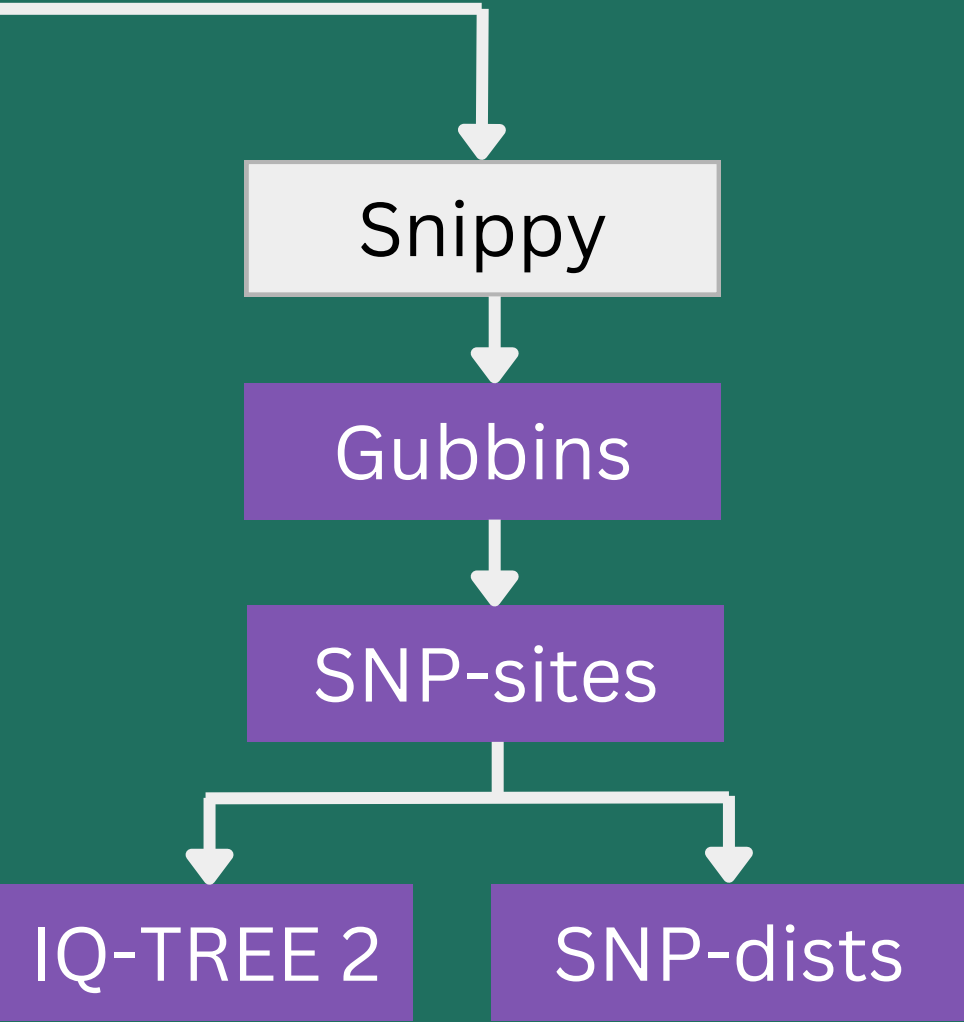
Comparación



Análisis a partir de lecturas



Análisis a partir de ensamblaje de novo



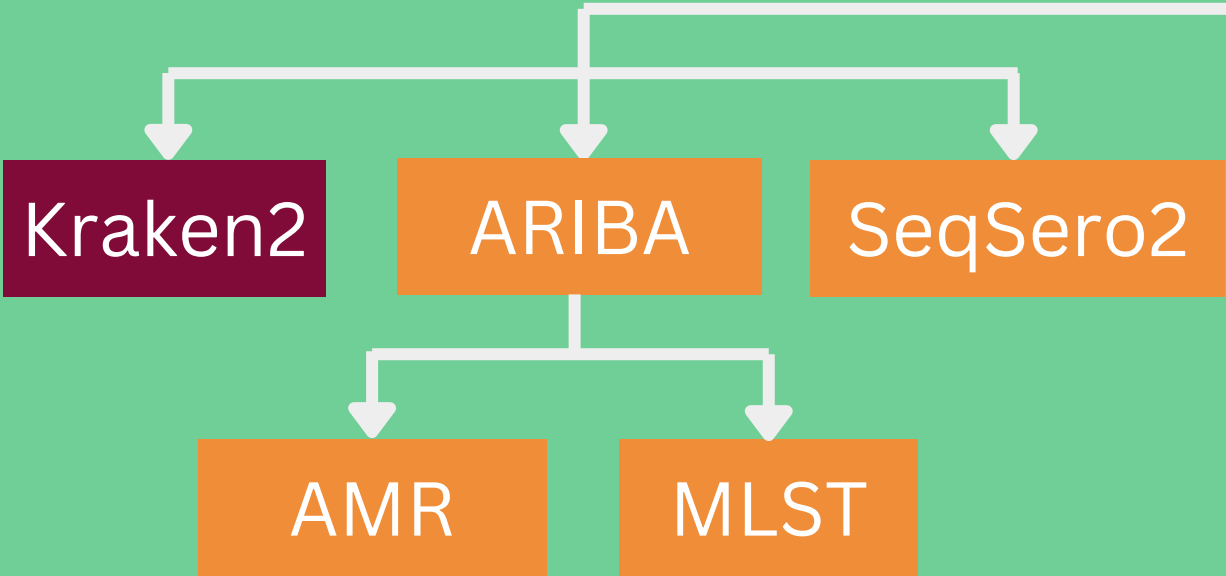
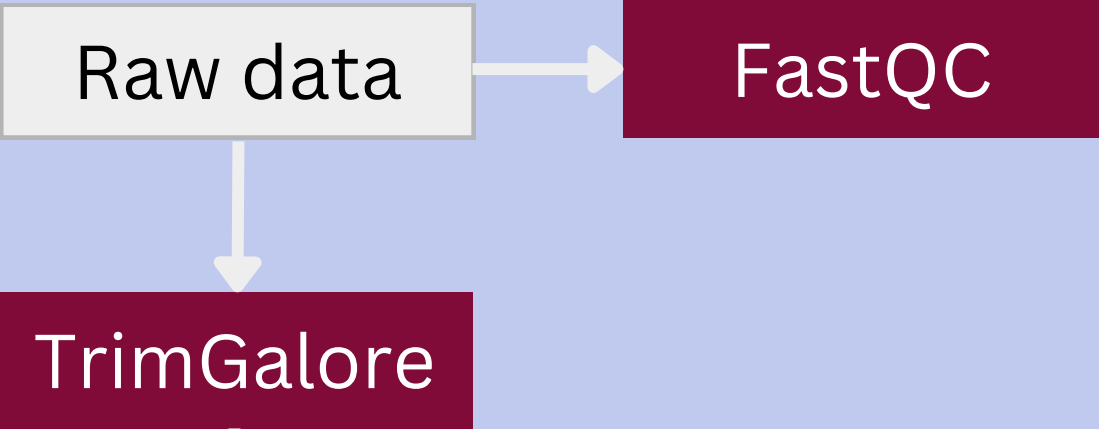
Análisis a partir de ensamblaje de referencia

Preprocesamiento

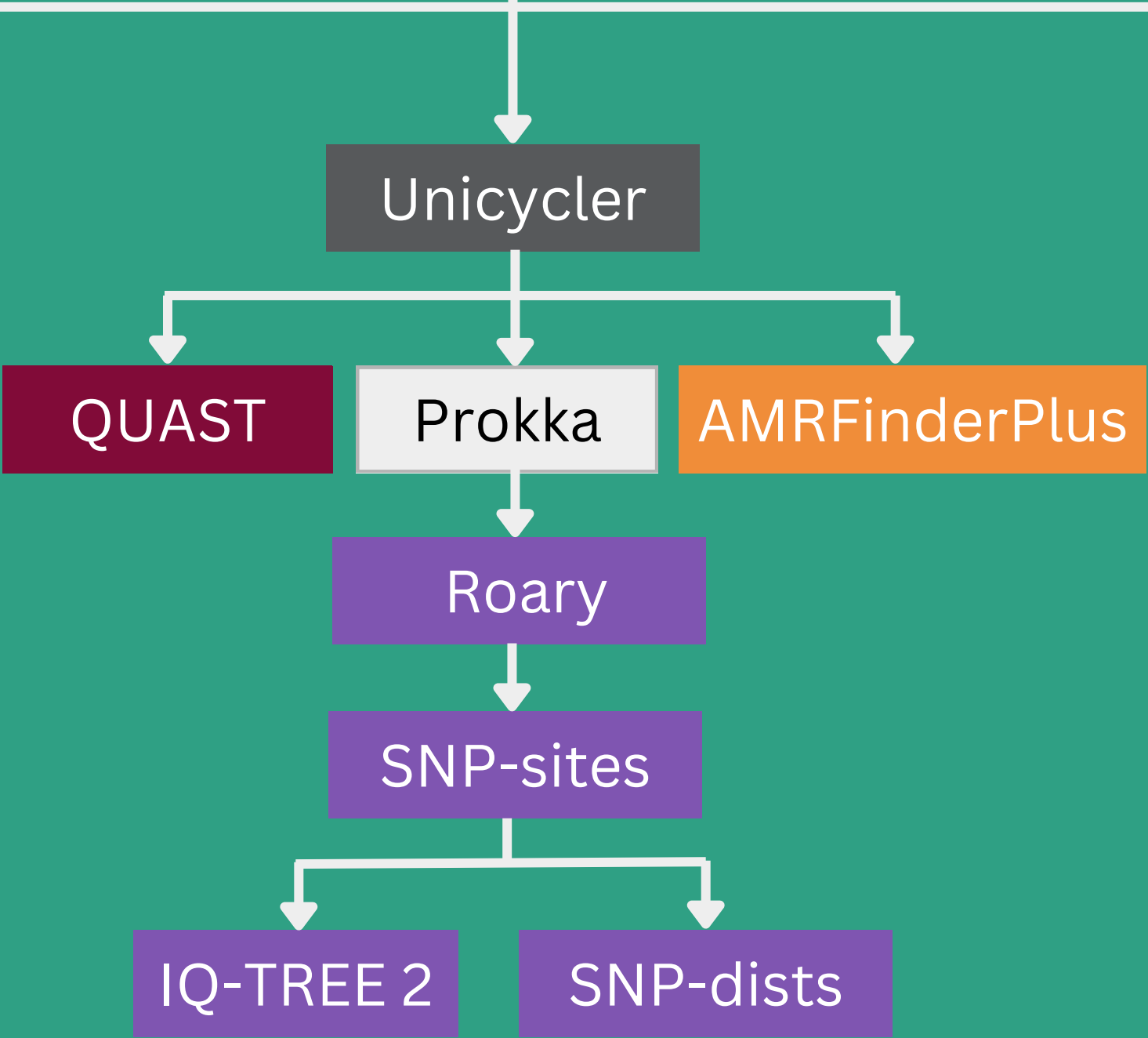
Control de calidad

Caracterización

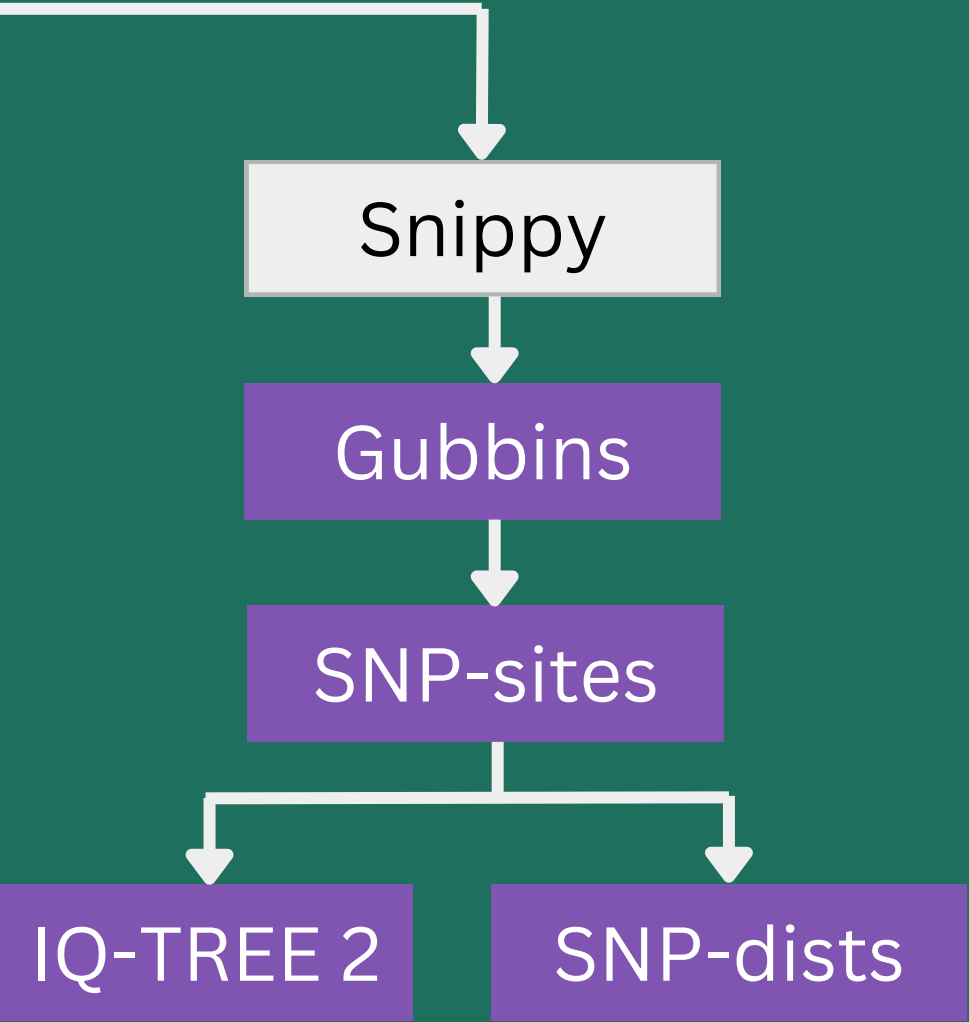
Comparación



Análisis a partir de lecturas

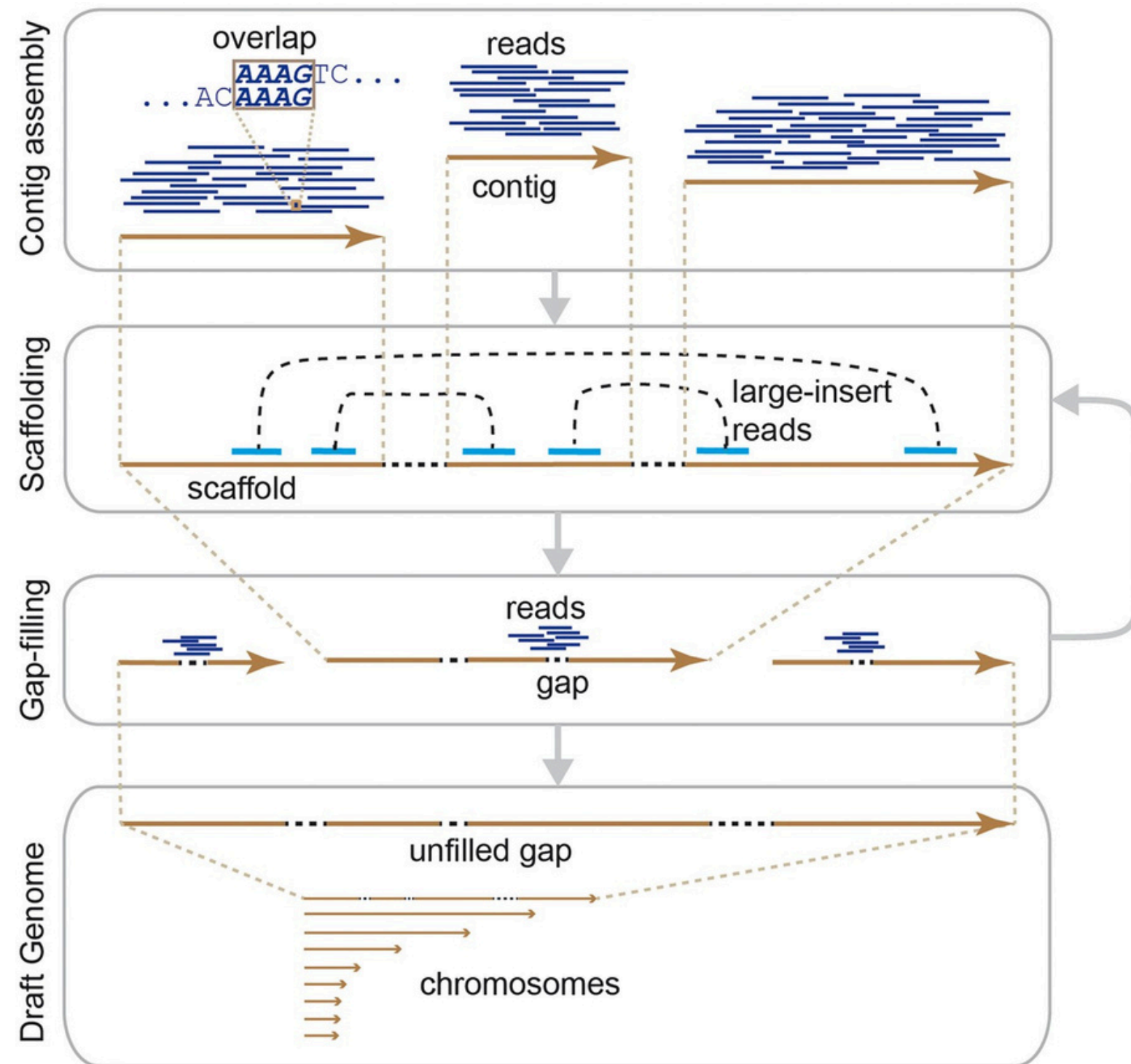


Análisis a partir de ensamblaje de novo

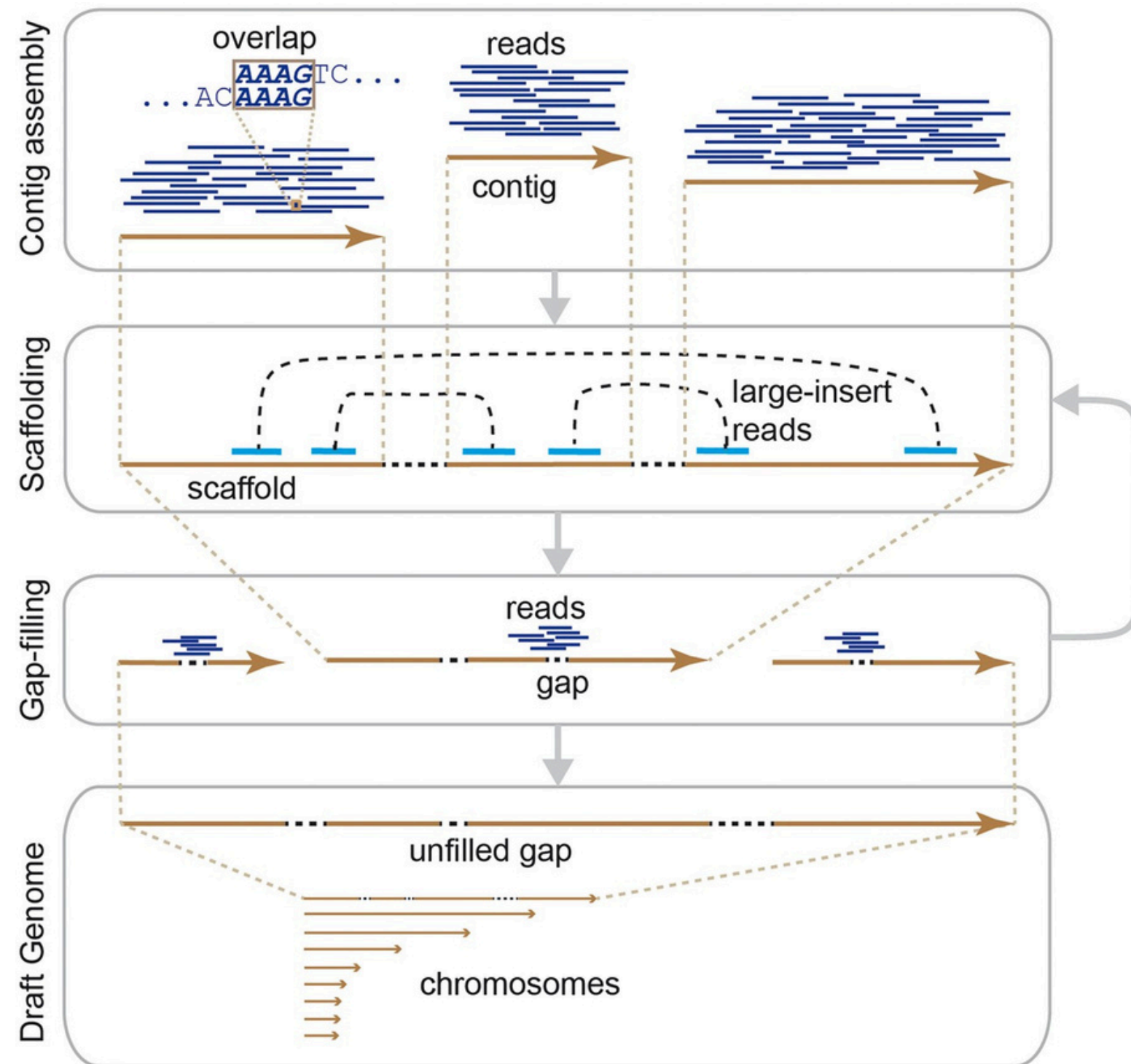


Análisis a partir de ensamblaje de referencia

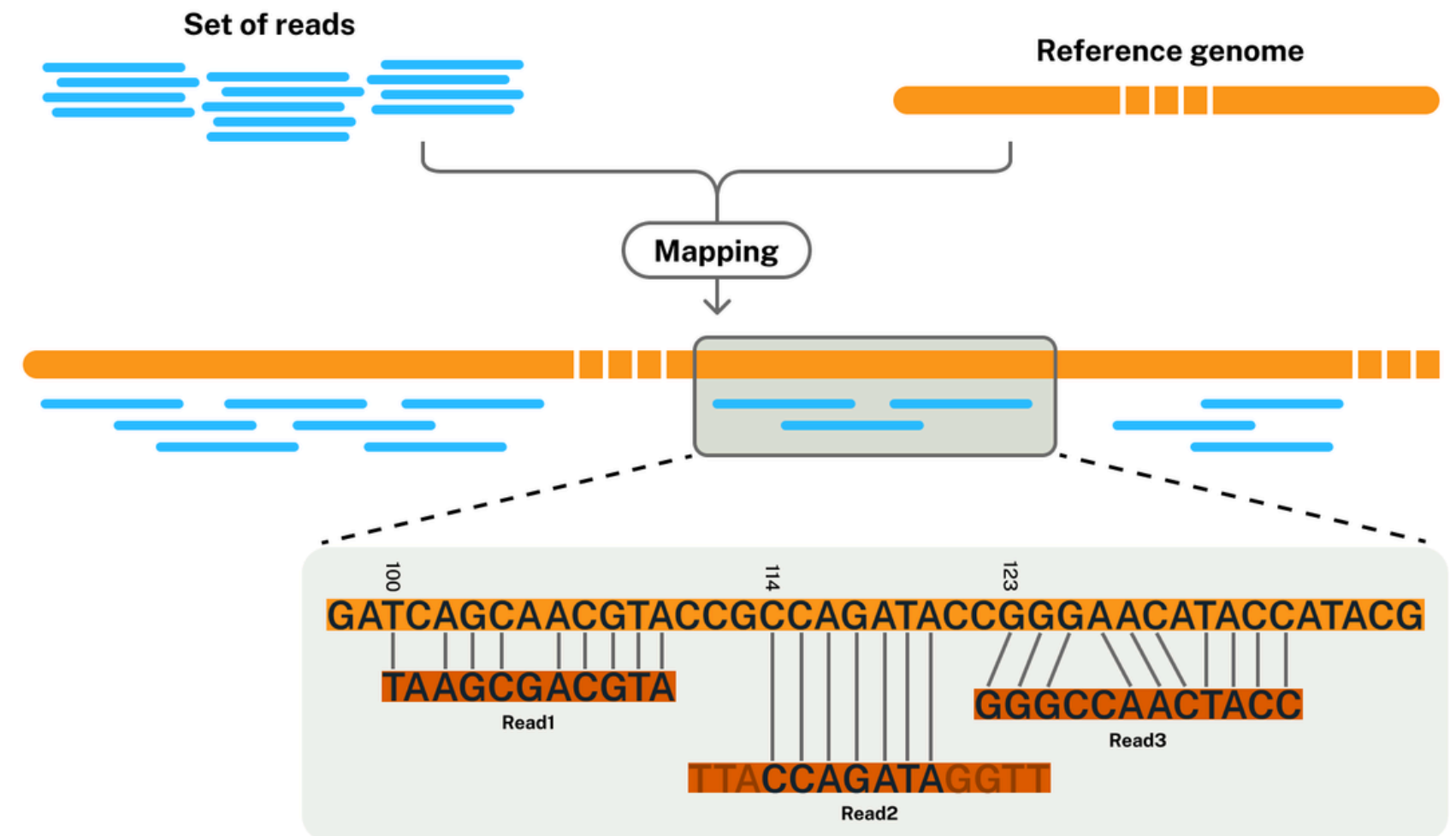
Ensamblaje *de novo*



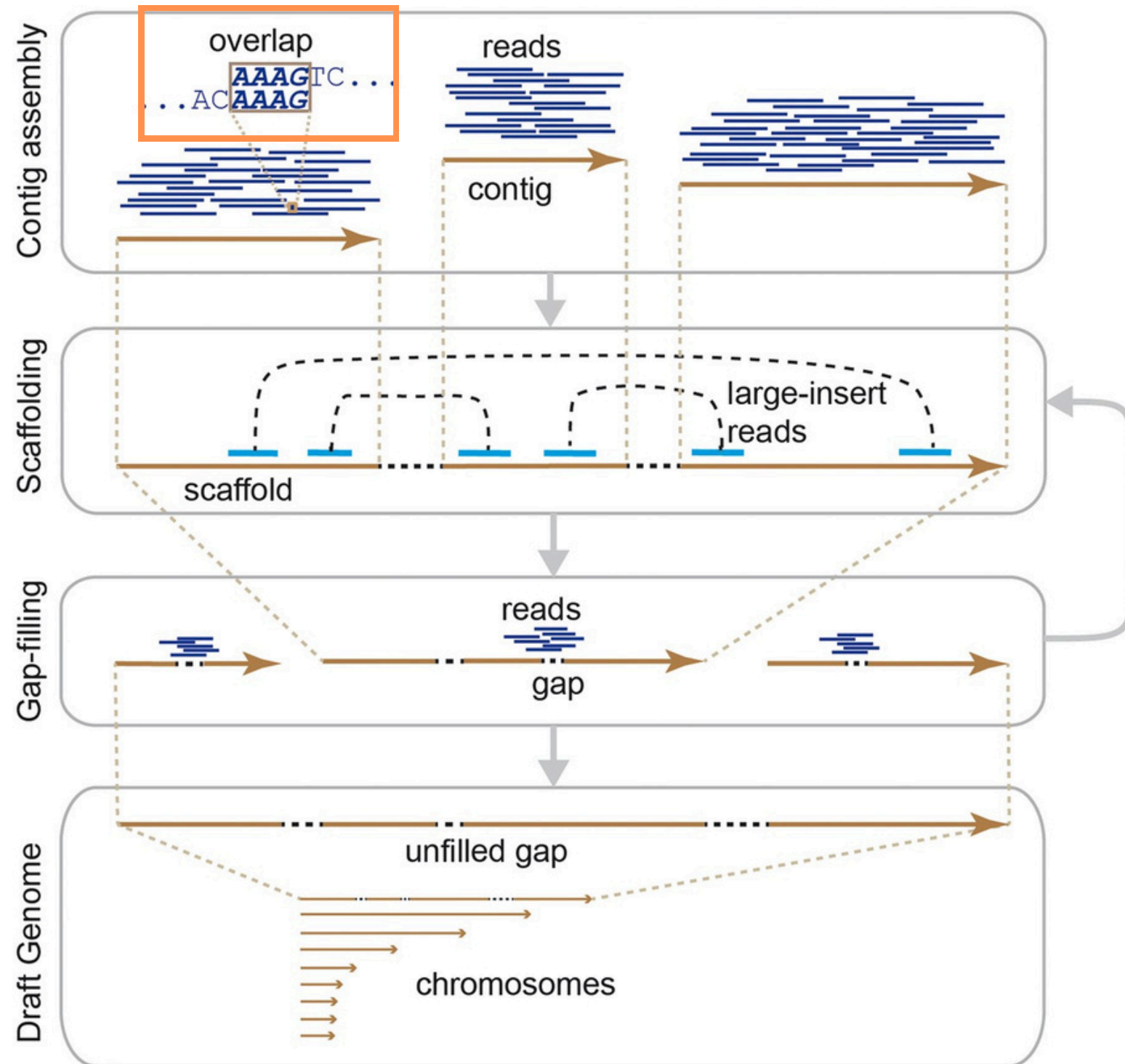
Ensamblaje *de novo*



Ensamblaje contra referencia



Ensamblaje *de novo*

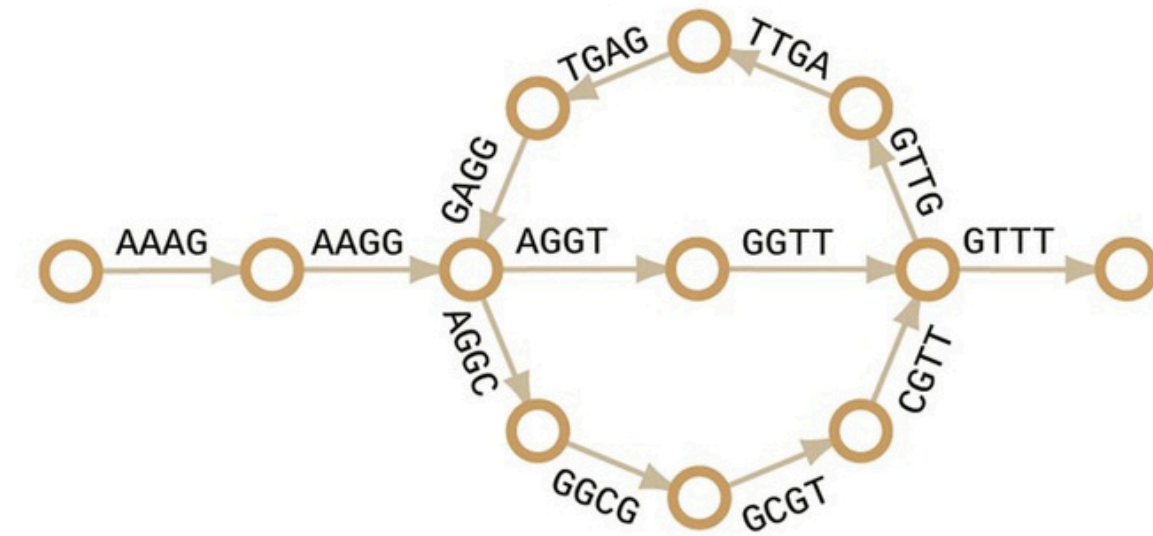


A Short read to k -mers ($k=4$)

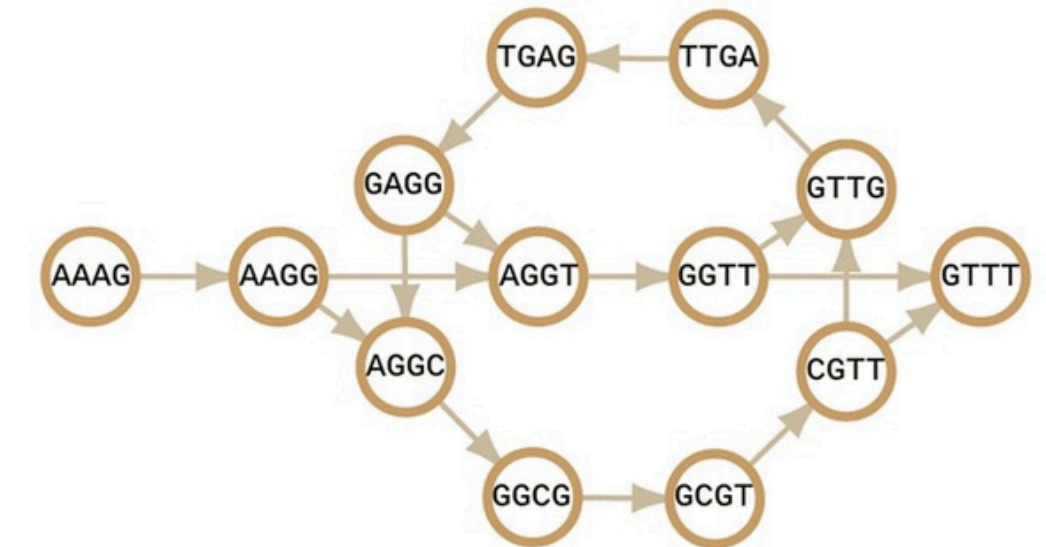
AAAGGCGTTGAGGTT

AAAG
AAGG
AGGC
GGCG
GCGT
CGTT
GTTG
TTGA
TGAG
GAGG
AGGT
GGTT

B Eulerian de Bruijn graph



C Hamiltonian de Bruijn graph

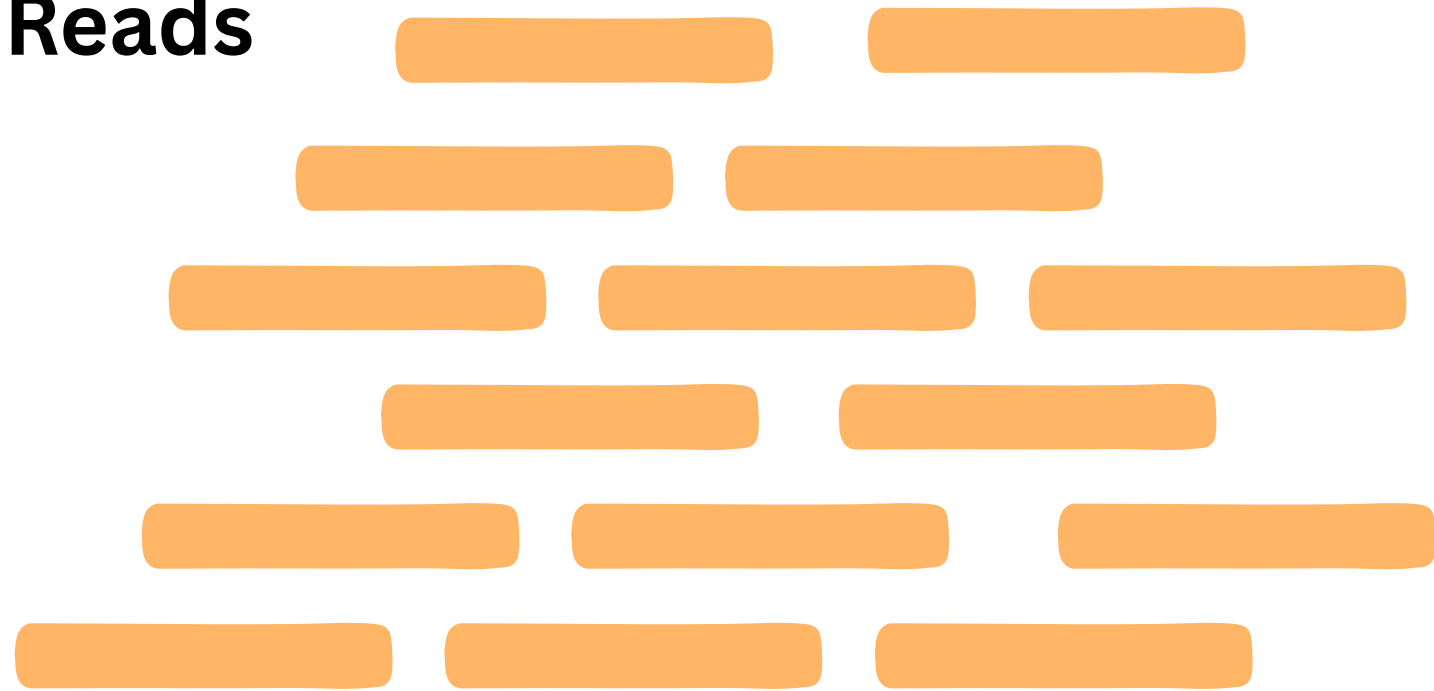




Unicycler

Ensamblaje

Reads



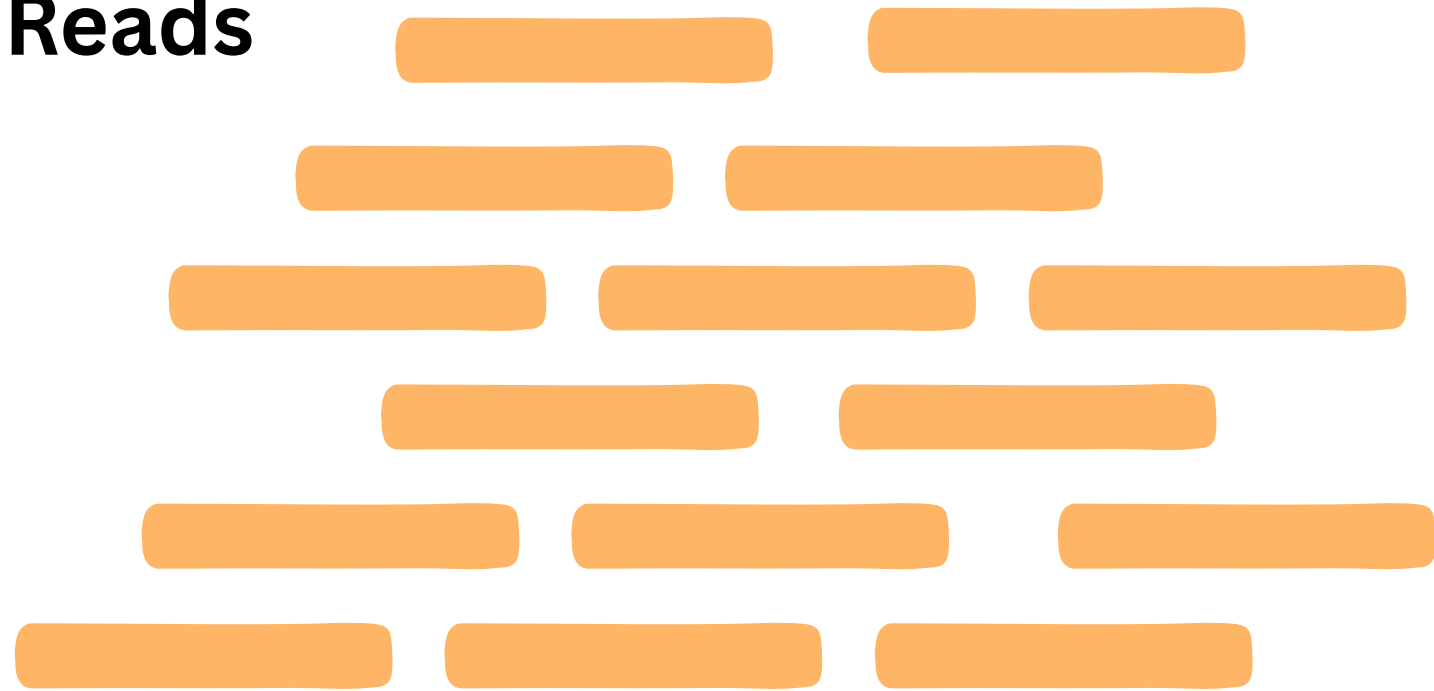
**Genoma de
referencia**



Unicycler

Ensamblaje

Reads

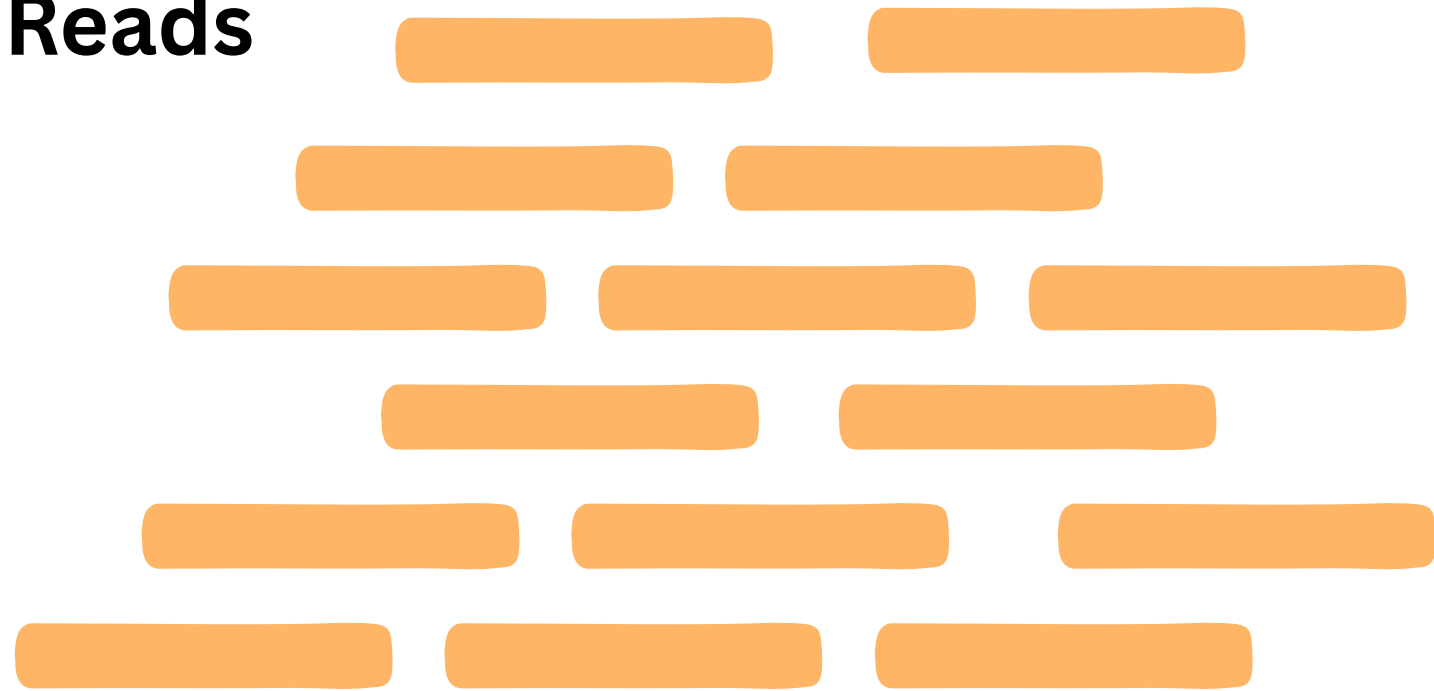




Unicycler

Ensamblaje

Reads



Construir el genoma



Ensamblaje: Secuenciación 'shotgun'

Input	GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTTT
-------	---

Ensamblaje: Secuenciación ‘shotgun’

Input	GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTTT
--------------	---

Copias	GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTTT
	GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTTT
	GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTTT
	GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTTT

Ensamblaje: Secuenciación ‘shotgun’

Input	GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTTT
--------------	---

Copias	GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTTT
	GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTTT
	GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTTT
	GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTTT

Fragmentos	GGCGTCTA TATCTCGG CTCTAGGCCCTC ATTTTTTT
	GGC GTCTATAT CTCGGCTCTAGGCCCTCA TTTTTT
	GGCGTC TATATCT CGGCTCTAGGCCCT CATTTTTTT
	GGCGTCTAT ATCTCGGCTCTAG GCCCTCA TTTTTT



Unicycler

Ensamblaje

**Reconstruir
esto**

CTAGGCCCTCAATTTT
CTCTAGGCCCTCAATTTT
GGCTCTAGGCCCTCAATTTT
CTCGGCTCTAGCCCCTCAATTT
TATCTCGACTCTAGGCCCTCA
TATCTCGACTCTAGGCC
TCTATATCTCGGCTCTAGG
GGCGTCTATATCTCG
GGCGTCGATATCT
GGCGTCTATATCT
GGCGTCTATATCTCGGCTCTAGGCCCTCAATTTT



TCTATATCTCGGCTCTAGG

TATCTCGACTCTAGGCC



```
TCTATATCTCGGCTCTAGG
      ||||| |||||
      TATCTCGACTCTAGGCC
```



Unicycler

Primer ley del ensamblaje

Ensamblaje

El sufijo del read A similar al prefijo del read B

```
TCTATATCTCGGCTCTAGG
      ||||| |||||
      TATCTCGACTCTAGGCC
```

A y B podrían superponerse en el genoma

```
TCTATATCTCGGCTCTAGG
GGCGTCTATATCTCGGCTCTAGGCCCTCATTTTTT
      TATCTCGACTCTAGGCC
```



Unicycler

Segunda ley del ensamblaje

Ensamblaje

Mayor cobertura conduce a superposiciones más numerosas y más prolongadas

Menos cobertura

```
CTAGGCCCTCAATTTT
CTCGGCTCTAGCCCCTCATTTT
TCTATATCTCGGCTCTAGG
GGCGTCGATATCT
GGCGTCTATATCTCGGCTCTAGGCCCTCATTTT
```

Más cobertura

```
CTAGGCCCTCAATTTT
GGCTCTAGGCCCTCATTTT
CTCGGCTCTAGCCCCTCATTTT
TATCTCGACTCTAGGCCCTCA
TCTATATCTCGGCTCTAGG
GGCGTCTATATCTCG
GGCGTCTATATCT
```



Unicycler

De Bruijn graph

Genoma

AAABBBBA



Unicycler

De Bruijn graph

Genoma

AAABBBBA

k

3-mers

AAA

De Bruijn graph

Genoma

AAABBBBA

k

3-mers

AAA

AAB

ABB

BBB

BBB

BBA



Unicycler

De Bruijn graph

Genoma

AAABBBBA

k

3-mers

AAA

AAB

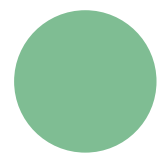
ABB

BBB

BBB

BBA

Nodos = k-1 mers





Unicycler

De Bruijn graph

Genoma

AAABBBBA

k

3-mers

AAA

AAB

ABB

BBB

BBB

BBA

Nodos = k-1 mers

Aristas = k-mers



De Bruijn graph

Genoma

AAABBBBA

k
3-mers

AAA
AAB
ABB
BBB
BBB
BBA

k-1
2-mers

AA	AA
AA	AB
AB	BB
BB	BB
BB	BB
BB	BA

Nodos = k-1 mers

Aristas = k-mers



Unicycler



De Bruijn graph

Genoma

AAABBBBA

k
3-mers

AAA
AAB
ABB
BBB
BBB
BBA

k-1
2-mers

AA	AA
AA	AB
AB	BB
BB	BB
BB	BB
BB	BA

Nodos = k-1 mers

Aristas = k-mers





Unicycler

De Bruijn graph

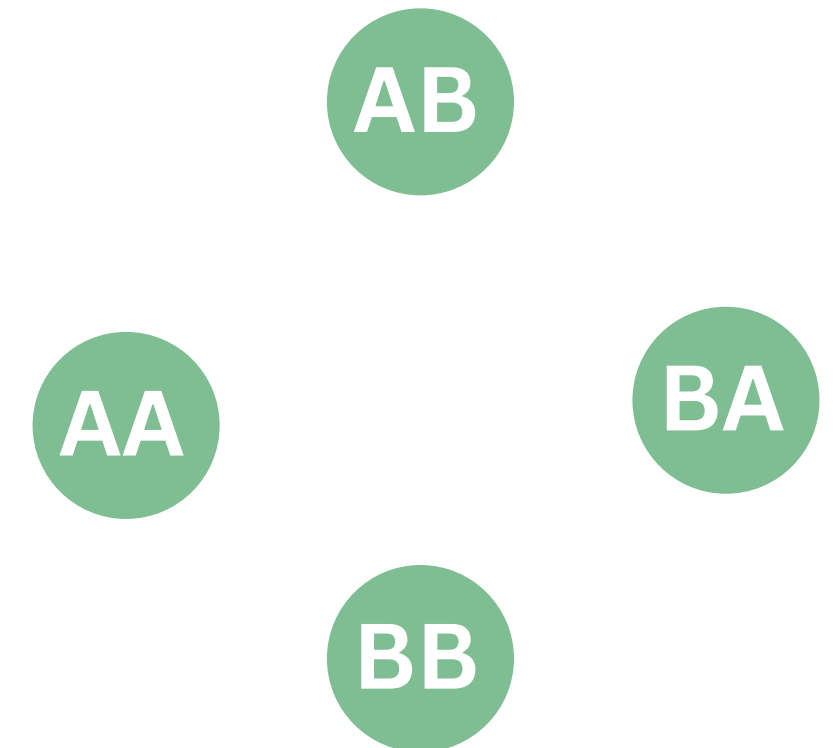
Genoma

AAABBBBA

k
3-mers

AAA
AAB
ABB
BBB
BBB
BBA

k-1
2-mers



Nodos = k-1 mers
Aristas = k-mers





Unicycler

De Bruijn graph

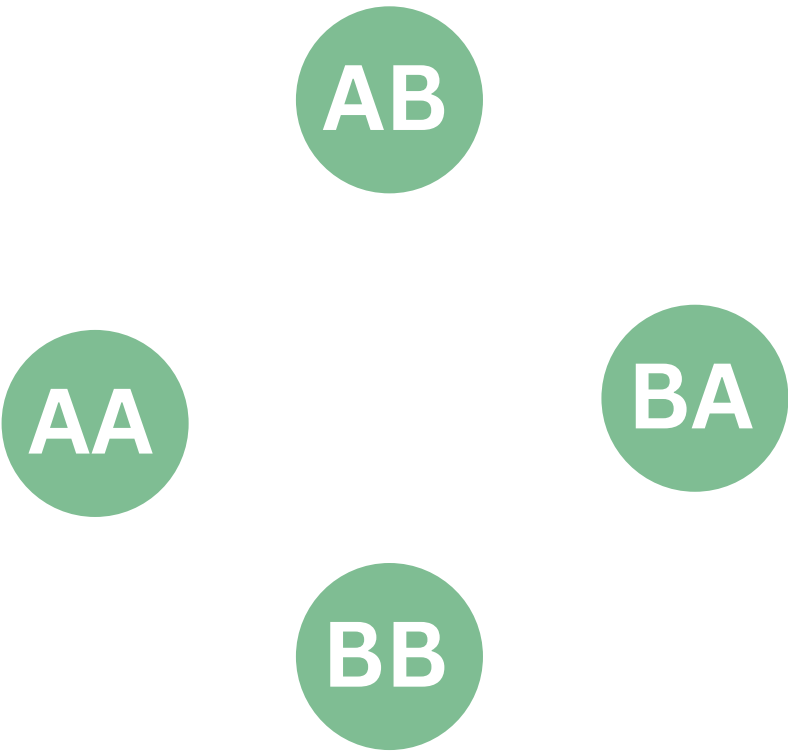
Genoma

AAABBBBA

k
3-mers

k-1
2-mers

AAA
AAB
ABB
BBB
BBB
BBA



Nodos = k-1 mers
Aristas = k-mers



AAA : AA → AA

Unicycler



De Bruijn graph

Genoma

AAABBBBA

k
3-mers

AAA

AAB

ABB

BBB

BBB

BBA

k-1
2-mers

AB

AA

BA

BB

Nodos = k-1 mers

Aristas = k-mers



AAA : AA → AA



Unicycler

De Bruijn graph

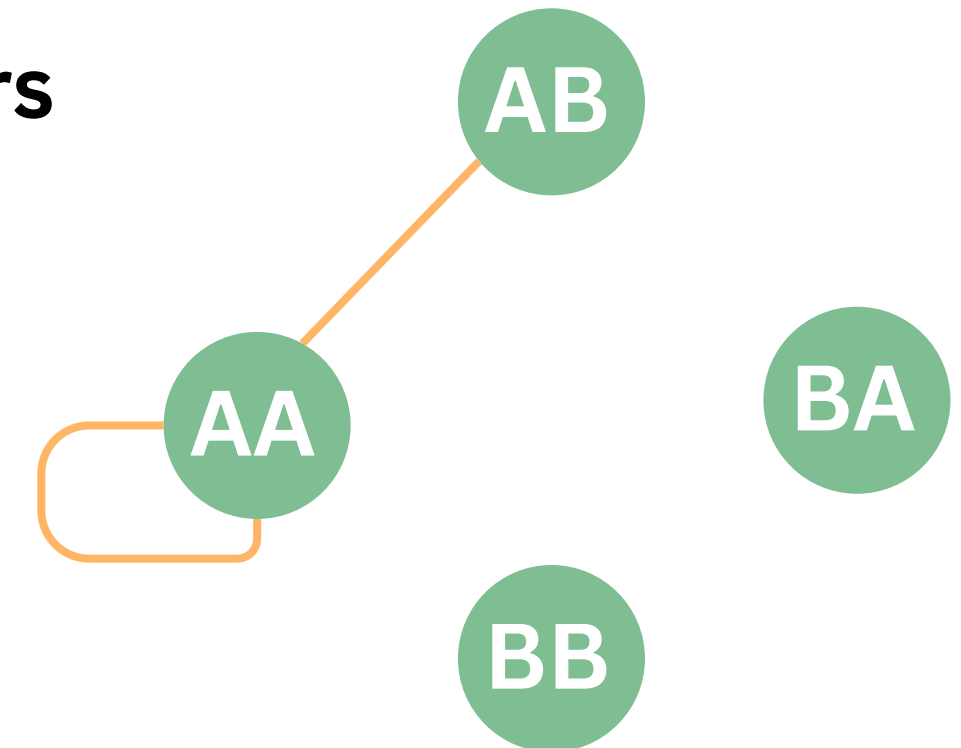
Genoma

AAABBBBA

k
3-mers

k-1
2-mers

AAA
AAB
ABB
BBB
BBB
BBA



Nodos = k-1 mers
Aristas = k-mers



AAB : AA → AB



Unicycler

De Bruijn graph

Genoma

AAABBBBA

k
3-mers

k-1
2-mers

AAA
AAB
ABB
BBB
BBB
BBA

Nodos = k-1 mers
Aristas = k-mers

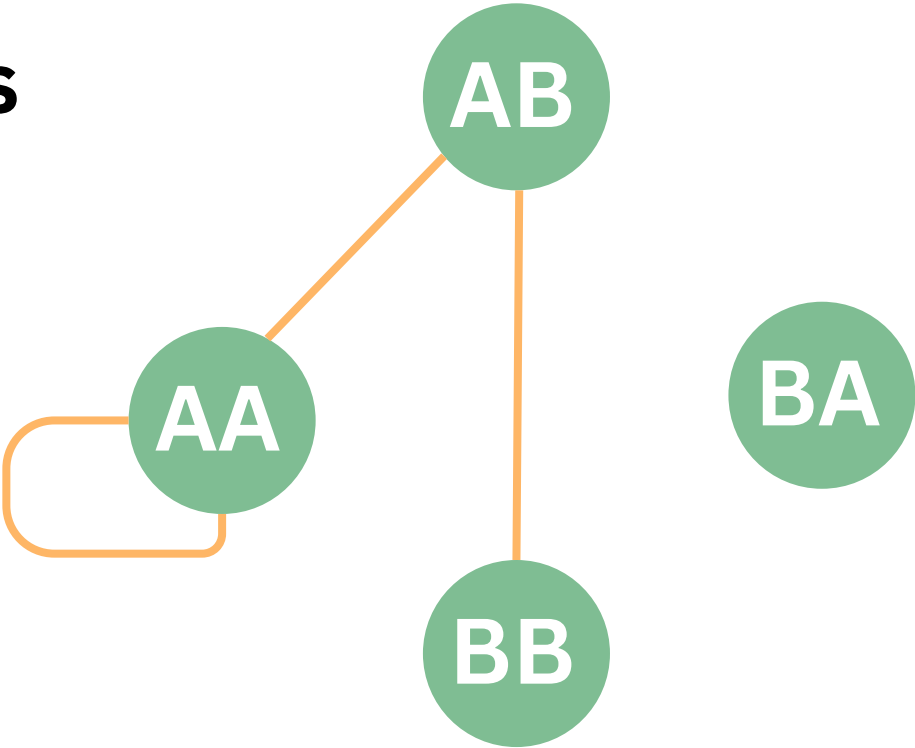


ABB: AB → BB





Unicycler

De Bruijn graph

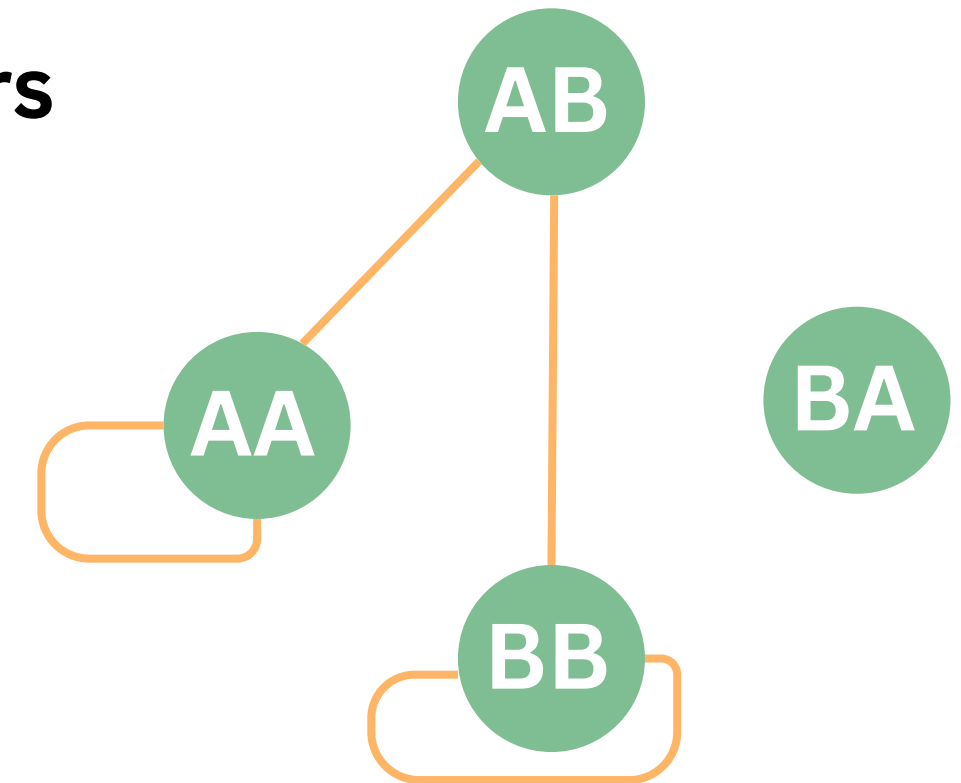
Genoma

AAABBBBA

k
3-mers

k-1
2-mers

AAA
AAB
ABB
BBB
BBB
BBA



BBB: BB → BB

Nodos = k-1 mers
Aristas = k-mers





Unicycler

De Bruijn graph

Genoma

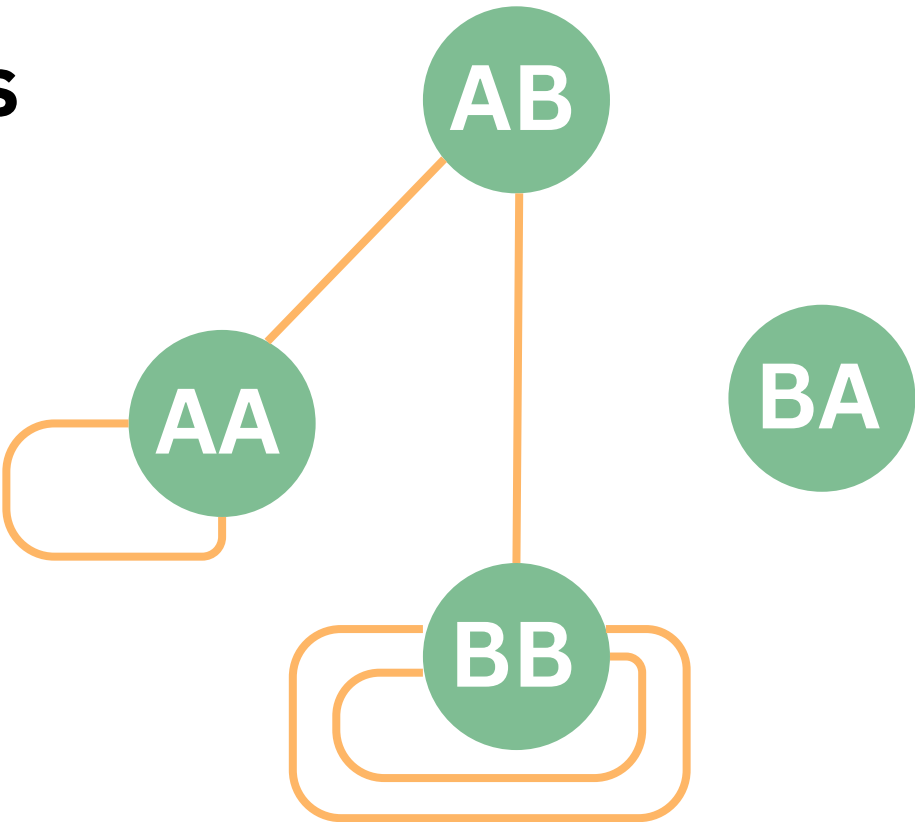
AAABBBBA

k
3-mers

k-1
2-mers

AAA
AAB
ABB
BBB
BBB
BBA

Nodos = k-1 mers
Aristas = k-mers



BBB: BB → BB



Unicycler

De Bruijn graph

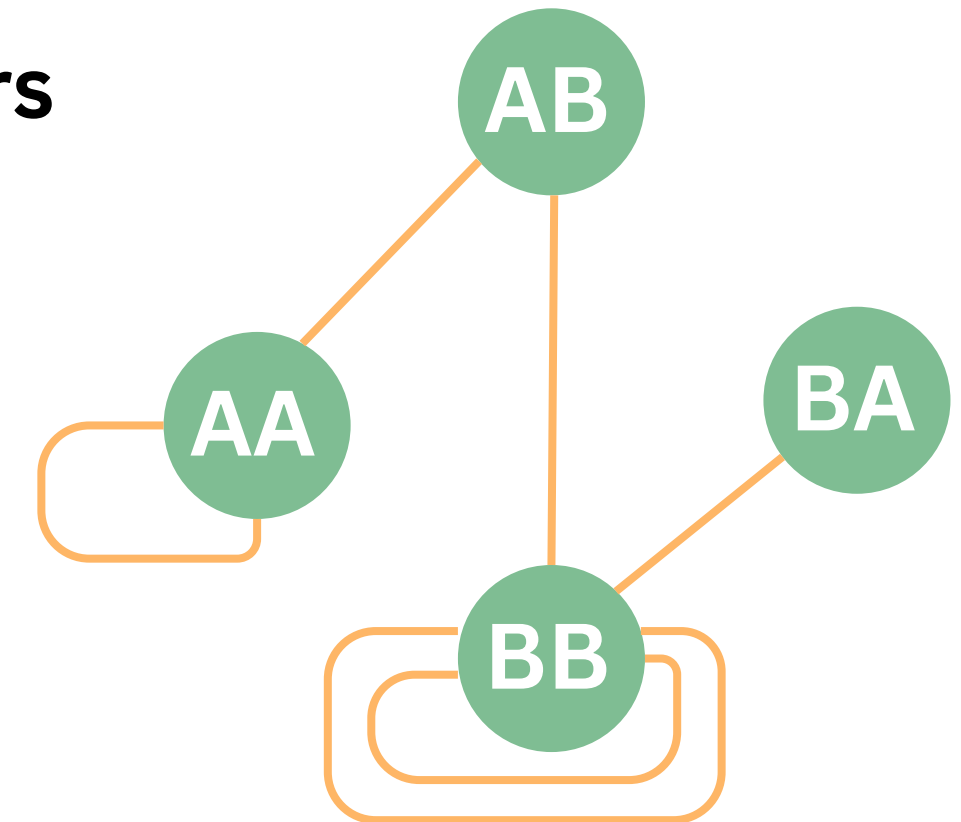
Genoma

AAABBBBA

k
3-mers

AAA
AAB
ABB
BBB
BBB
BBA

k-1
2-mers



BBA: BB → BA

Nodos = k-1 mers
Aristas = k-mers





Unicycler

De Bruijn graph

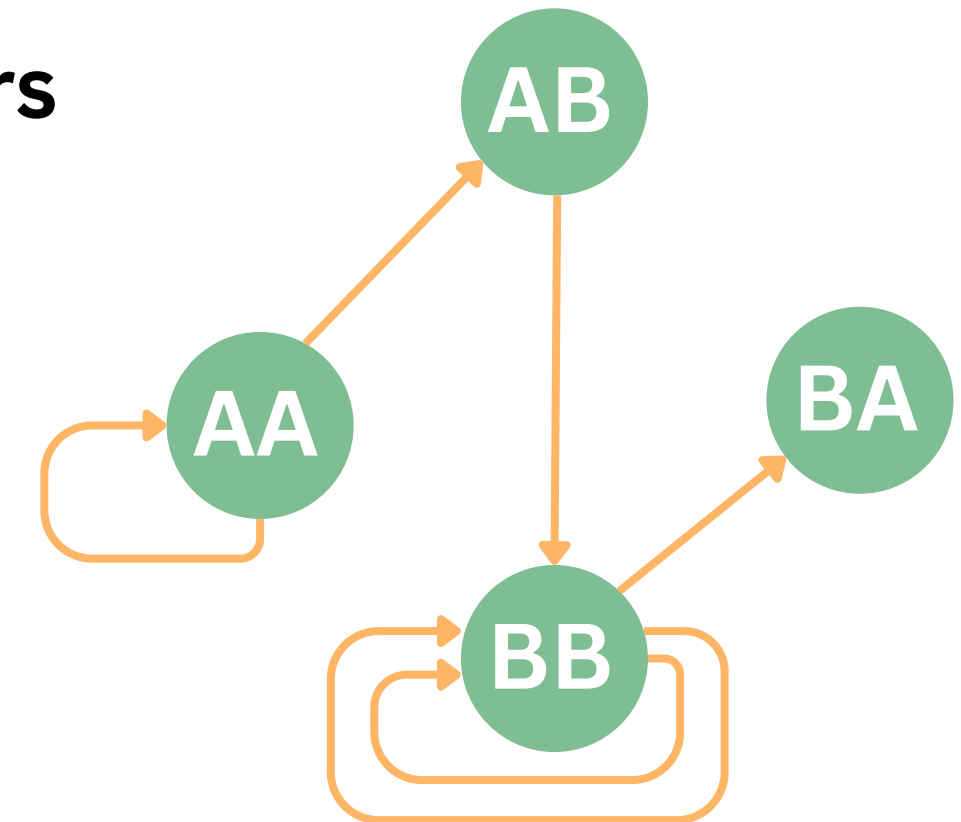
Genoma

k
3-mers

AAABBBBA

k-1
2-mers

AAA
AAB
ABB
BBB
BBB
BBA



Nodos = k-1 mers
Aristas = k-mers



Un recorrido que atraviesa cada arista exactamente una vez permite reconstruir el genoma: recorrido euleriano



Unicycler

De Bruijn graph

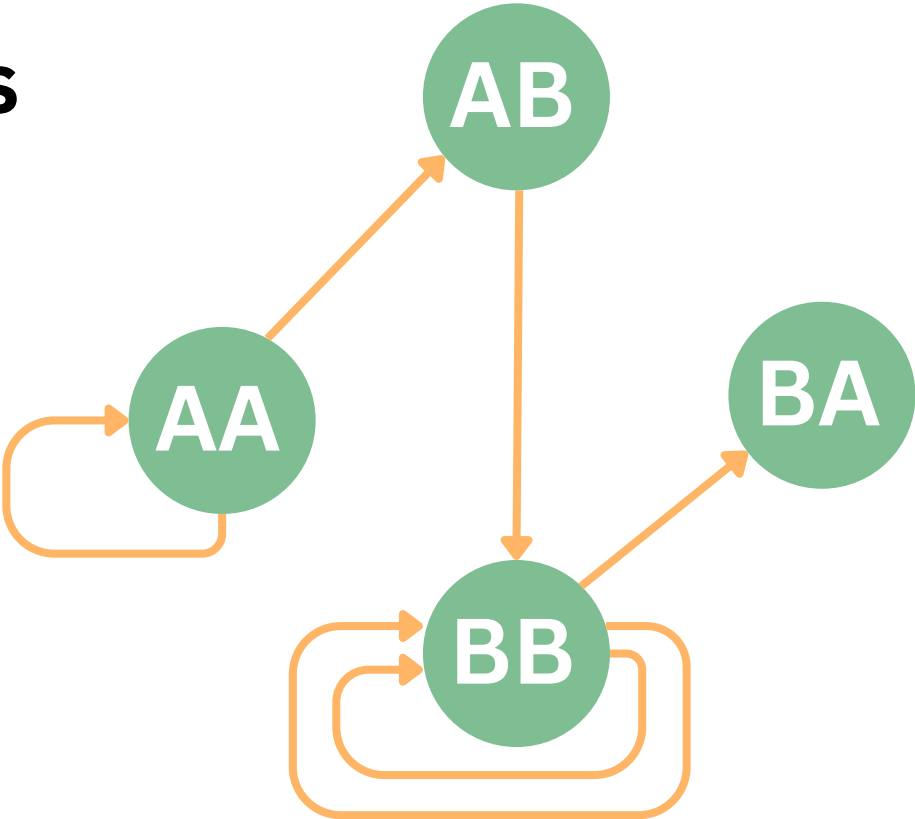
Genoma

AAABBBBA

k
3-mers

AAA
AAB
ABB
BBB
BBB
BBA

k-1
2-mers



Nodos = k-1 mers
Aristas = k-mers



AAABBBBA



Unicycler

Tercera ley del ensamblaje

De Bruijn graph

Las secuencias repetidas dificultan el ensamblaje del genoma, la posibilidad de ensamblarlo sin errores depende de la longitud de las lecturas y de los patrones repetitivos presentes en el genoma.





Unicycler

Tercera ley del ensamblaje

De Bruijn graph

Las secuencias repetidas dificultan el ensamblaje del genoma, la posibilidad de ensamblarlo sin errores depende de la **longitud de las lecturas** y de los patrones repetitivos presentes en el genoma.





Unicycler

De Bruijn graph

Tercera ley del ensamblaje

Genoma

ATGATGCTA

k
3-mers

k-1
2-mers



Unicycler

De Bruijn graph

Tercera ley del ensamblaje

Genoma	ATGATGCTA			
k	k-1			
3-mers	2-mers			
ATG	AT	TG		
TGA	TG	GA		
GAT	GA	AT		
ATG	AT	TG		
TGC	TG	GC		
GCT	GC	CT		
CTA	CT	TA		



Genoma

ATGATGCTA

k
3-mers

ATG
TGA
GAT
ATG
TGC
GCT
CTA

k-1
2-mers



Genoma

ATGATGCTA

k
3-mers

ATG

TGA

GAT

ATG

TGC

GCT

CTA

k-1
2-mers

AT

TG

GC

TA

GA

CT



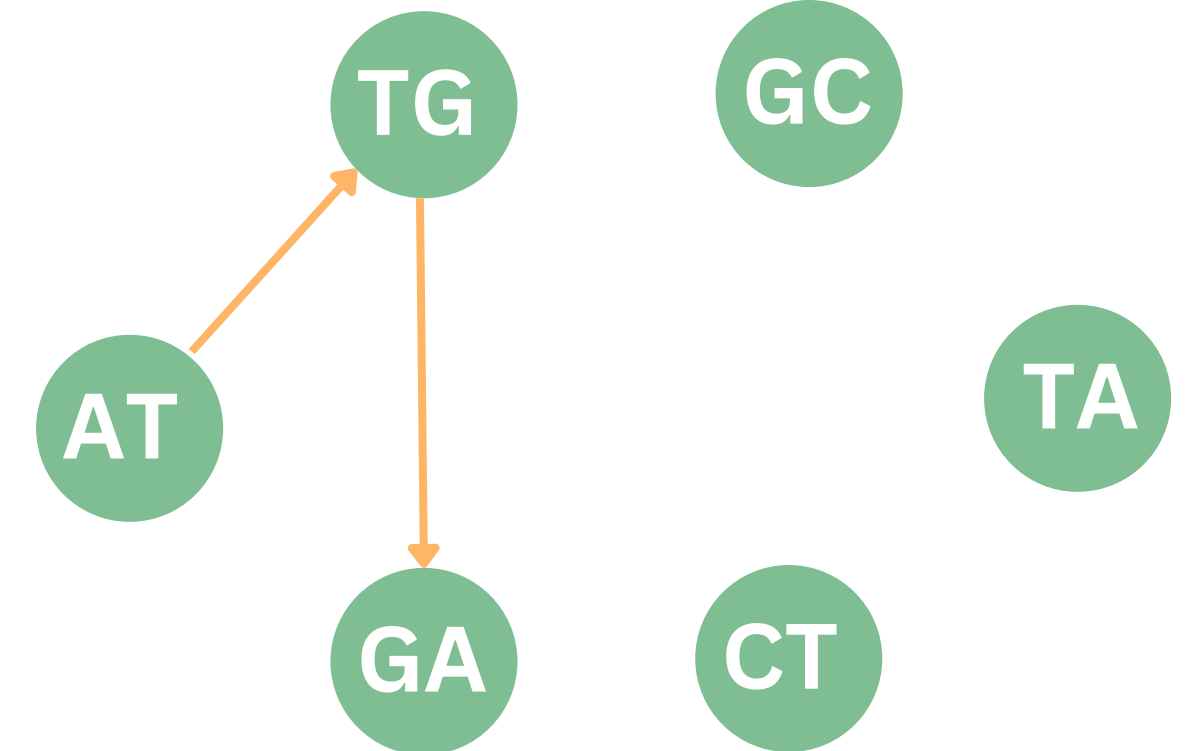
Genoma

ATGATGCTA

k
3-mers

ATG
TGA
GAT
ATG
TGC
GCT
CTA

k-1
2-mers



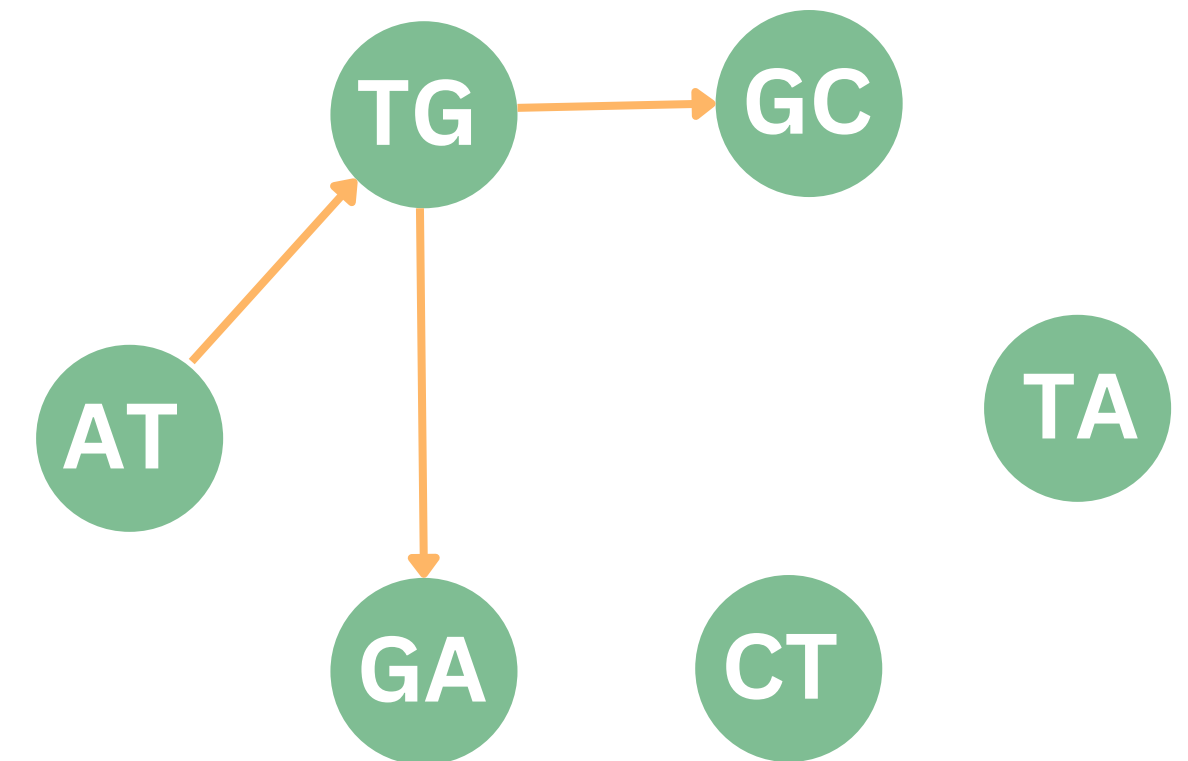
Genoma

ATGATGCTA

k
3-mers

ATG
TGA
GAT
ATG
TGC
GCT
CTA

k-1
2-mers



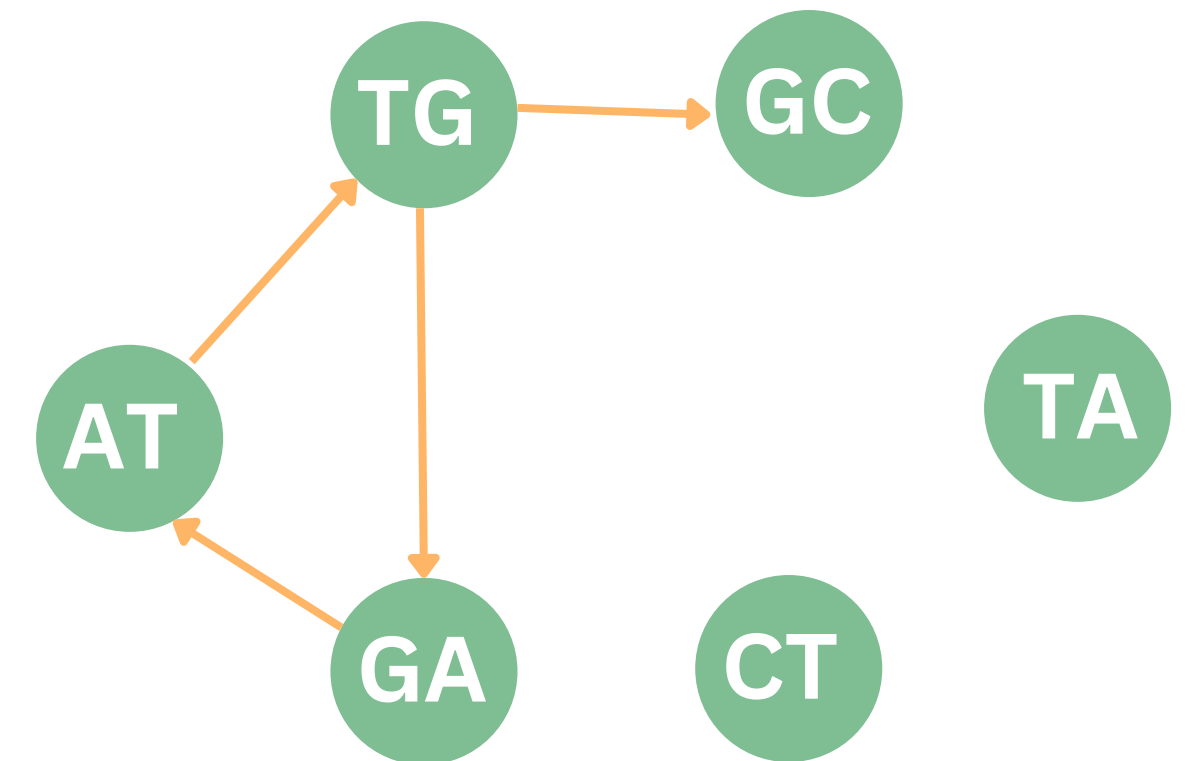
Genoma

ATGATGCTA

k
3-mers

ATG
TGA
GAT
ATG
TGC
GCT
CTA

k-1
2-mers



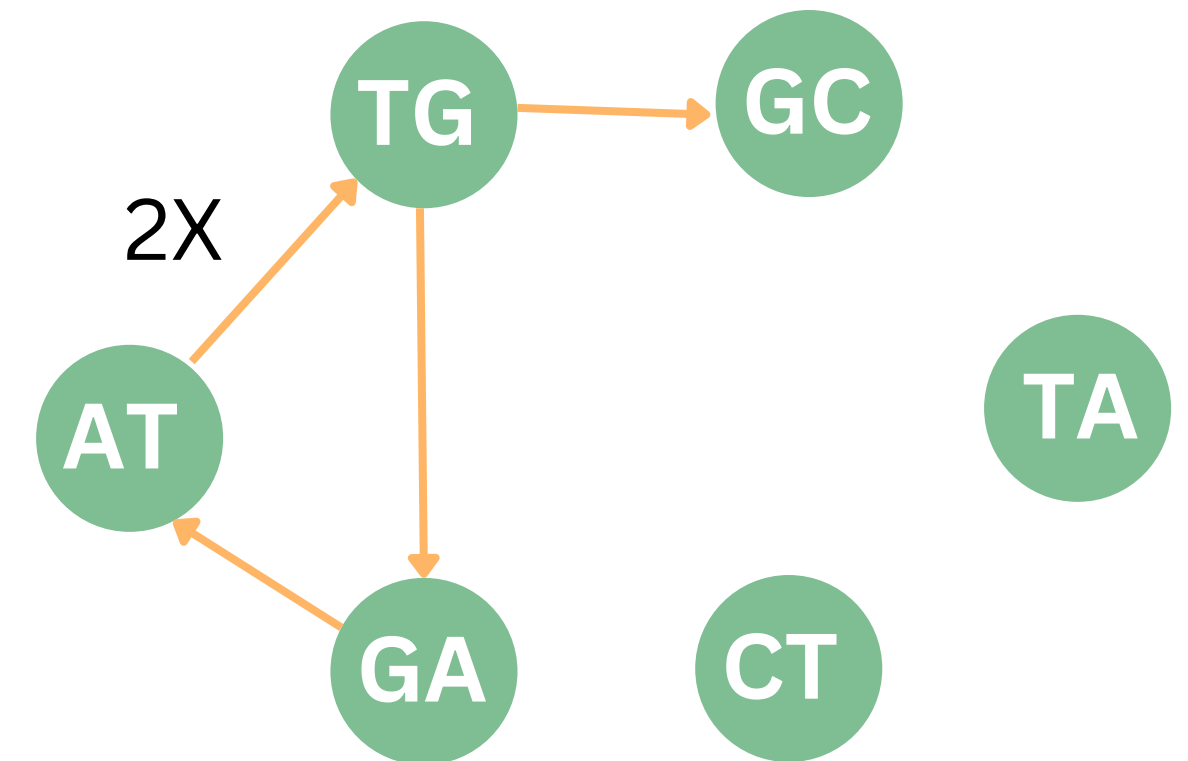
Genoma

ATGATGCTA

k
3-mers

ATG
TGA
GAT
ATG
TGC
GCT
CTA

k-1
2-mers



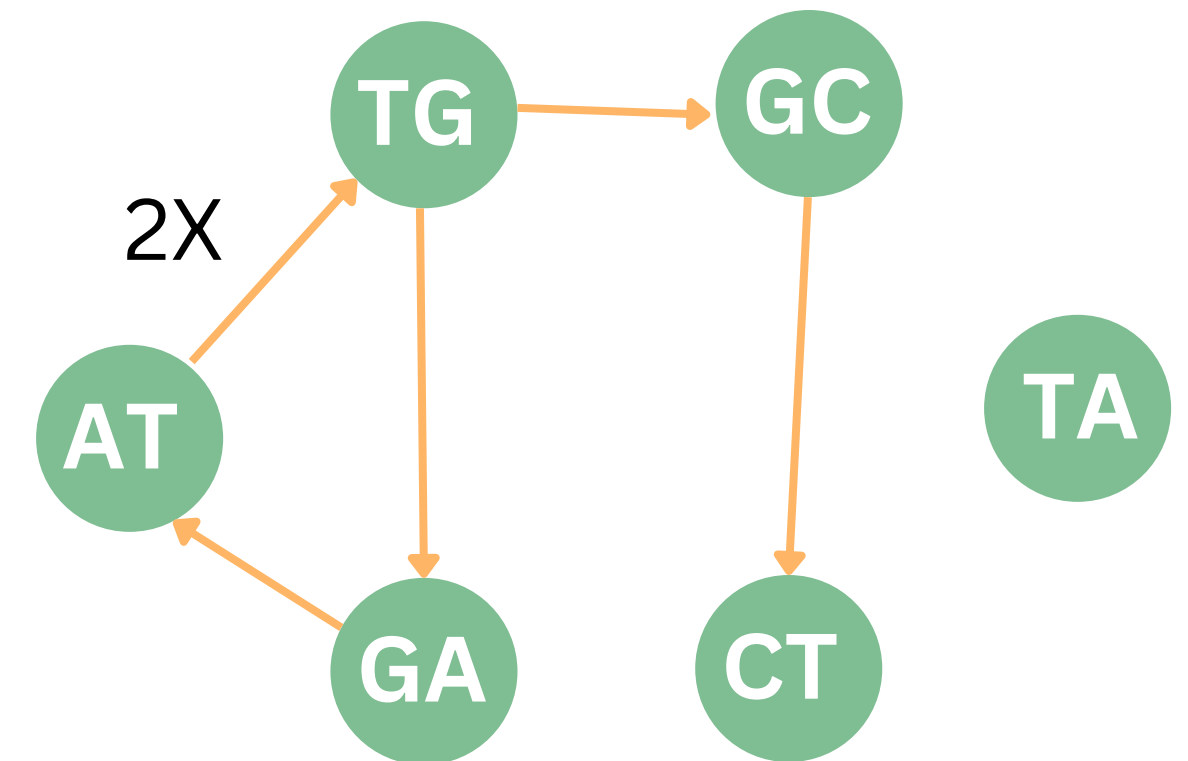
Genoma

ATGATGCTA

k
3-mers

ATG
TGA
GAT
ATG
TGC
GCT
CTA

k-1
2-mers



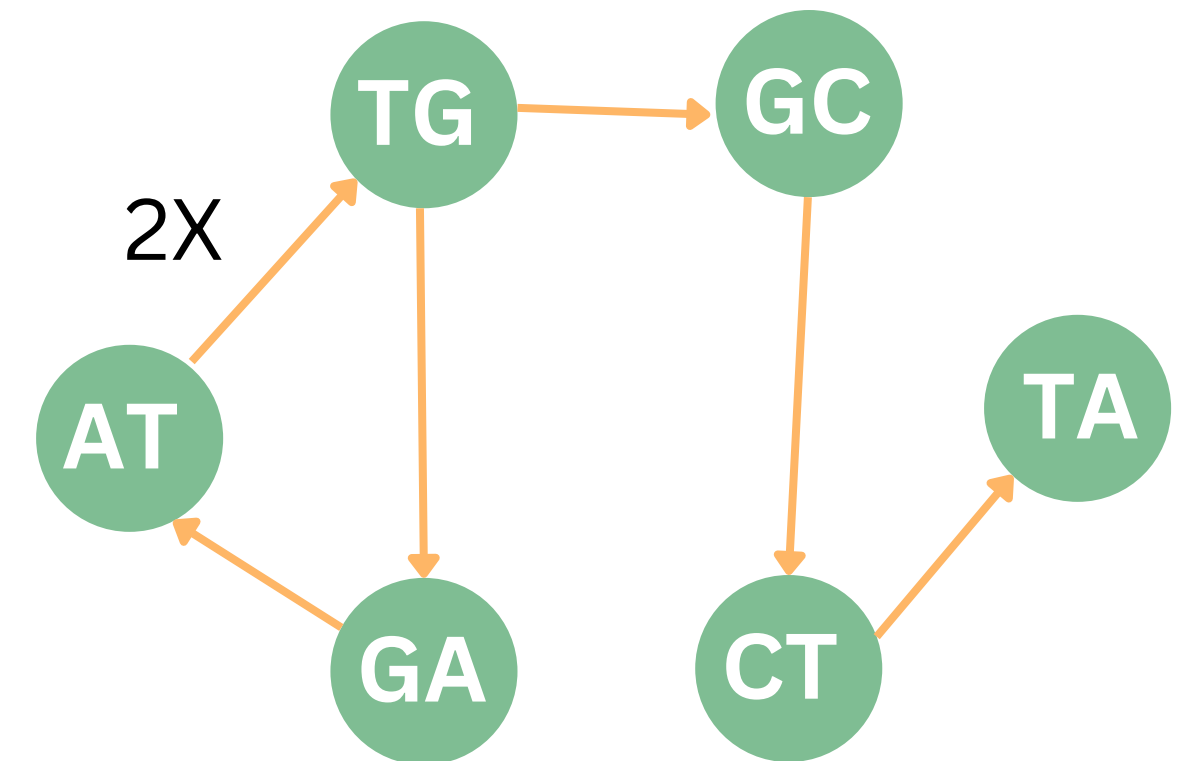
Genoma

ATGATGCTA

k
3-mers

ATG
TGA
GAT
ATG
TGC
GCT
CTA

k-1
2-mers



ATGATGCTA

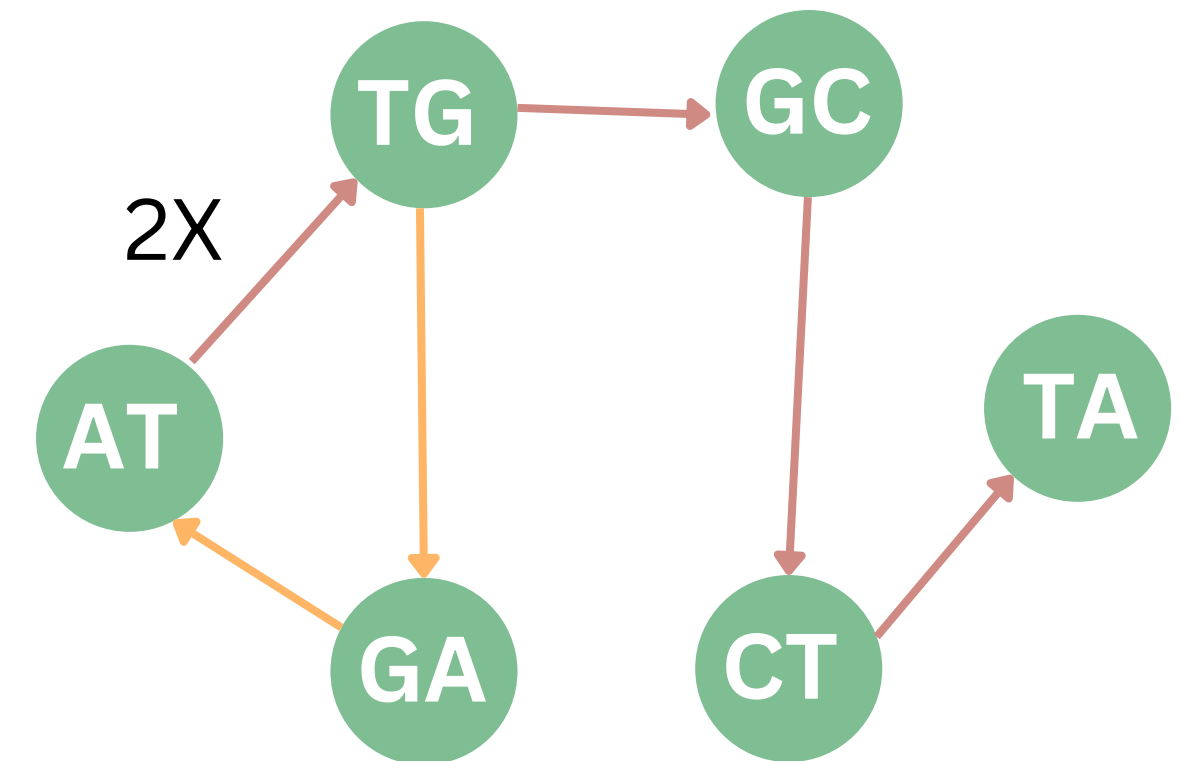
Genoma

ATGATGCTA

k
3-mers

ATG
TGA
GAT
ATG
TGC
GCT
CTA

k-1
2-mers



ATGCTATG



Secuencia



Short reads



Secuencias de alta
calidad y profundidad

Contigs



Unicycler



Secuencia



Long reads

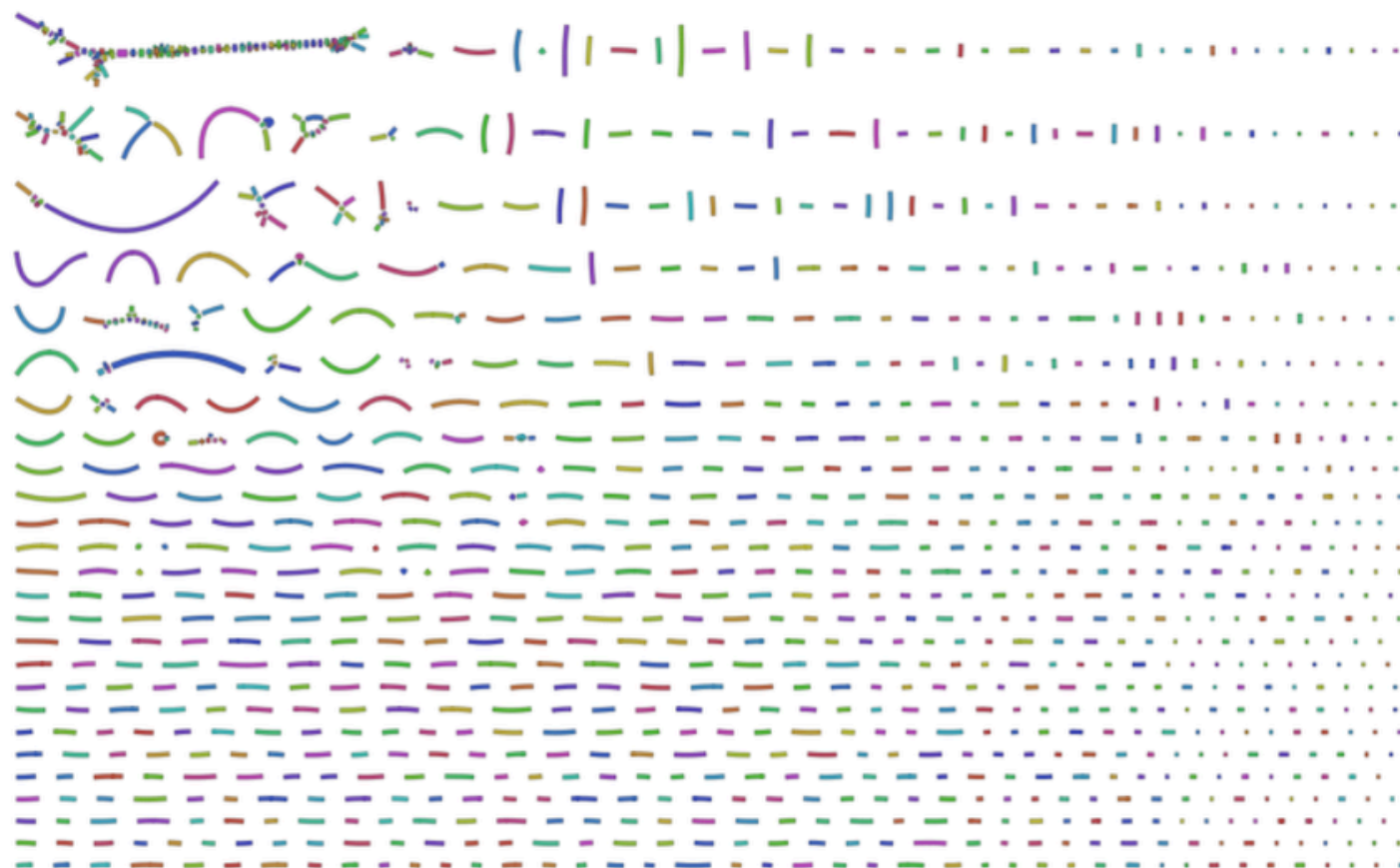


Relativamente
propensas a errores,
Menor profundidad

1 Contig

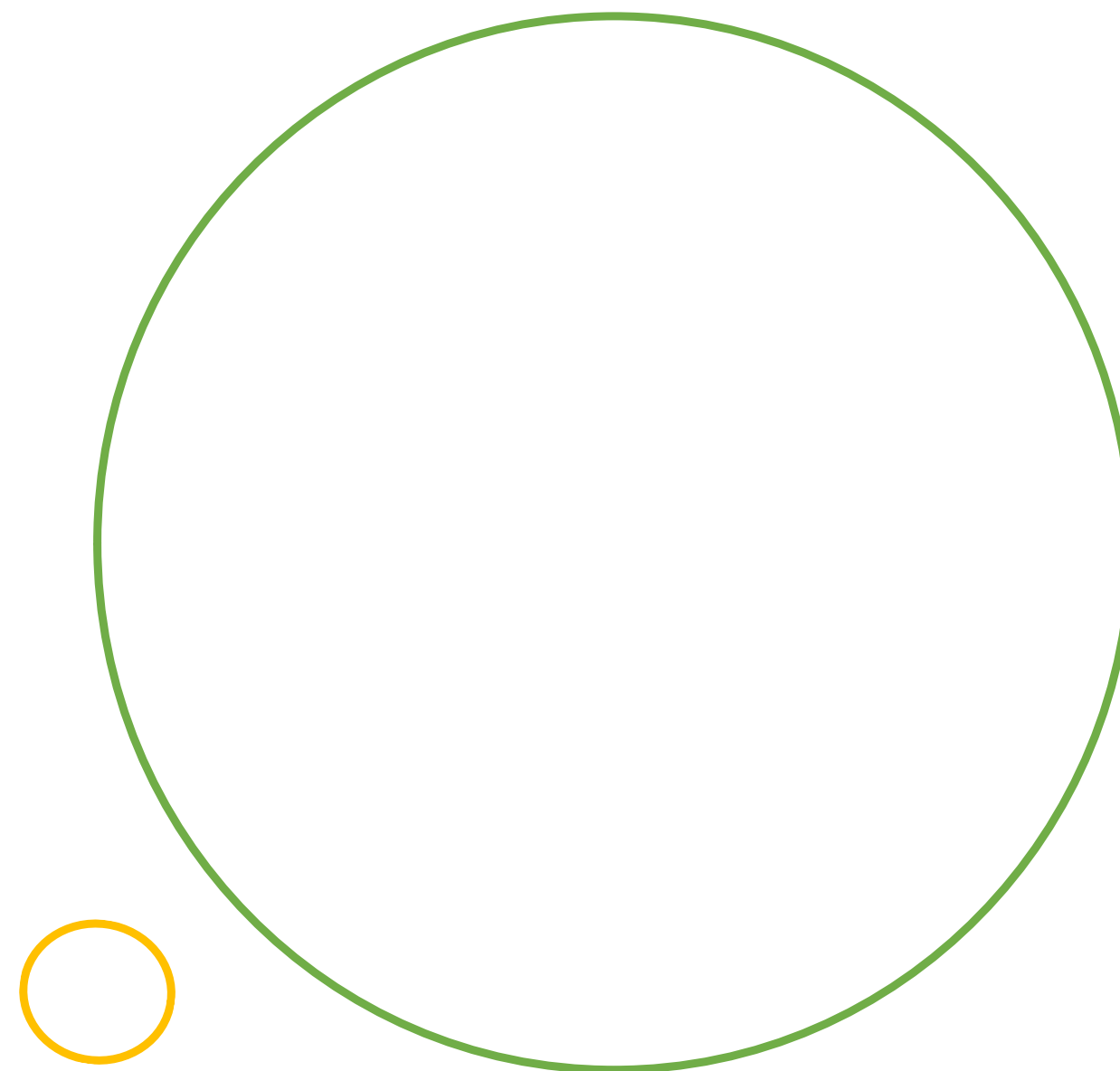


Draft



Muchos contigs

Completo



1 Contig por replicón

Unicycler



Secuencia



Short reads



Long reads



1 Contig





Ensamblajes híbridos

Secuencia



Short reads



Long reads



1 Contig





Unicycler

A. Short read assembly

Ensamblaje de las lecturas cortas usando SPAdes, que construye un Grafo de De Bruijn

1. Divide las lecturas en k-mers
2. Construye el grafo
3. Prueba muchos valores de k

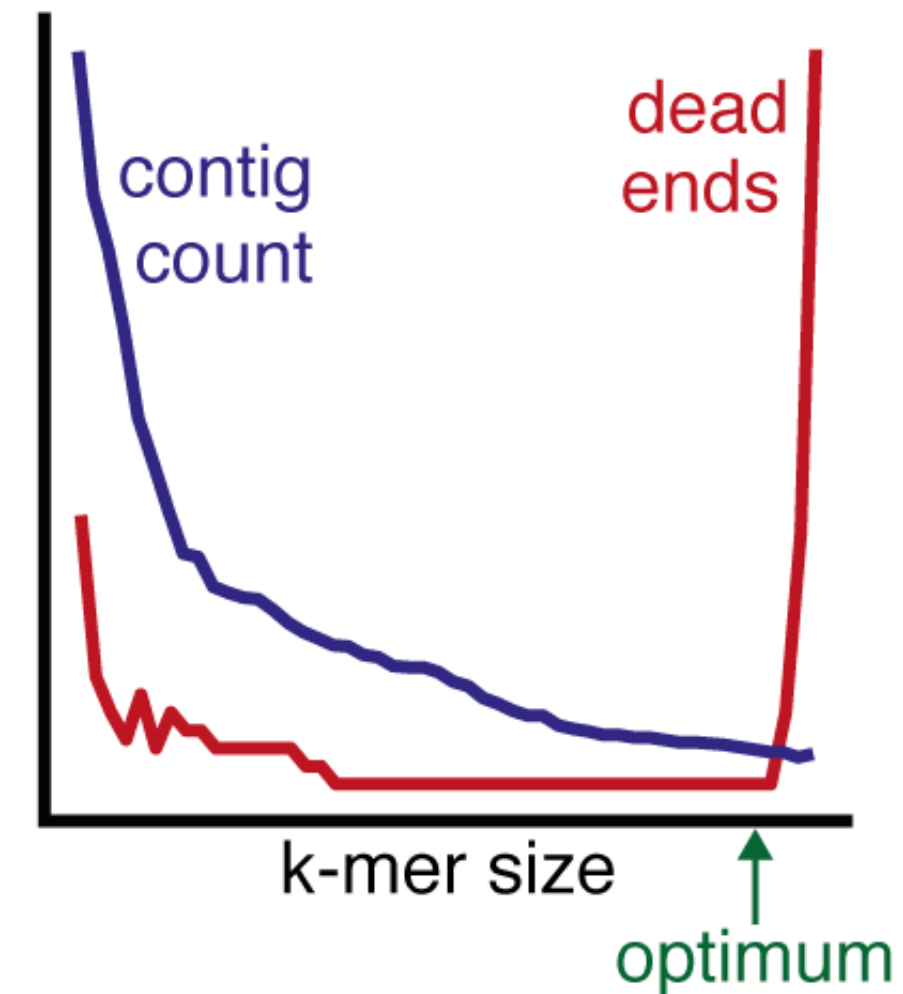
k-mer pequeños: grafos más conectados

k-mer grandes: mayor capacidad de resolver regiones repetitivas

A. Short read assembly

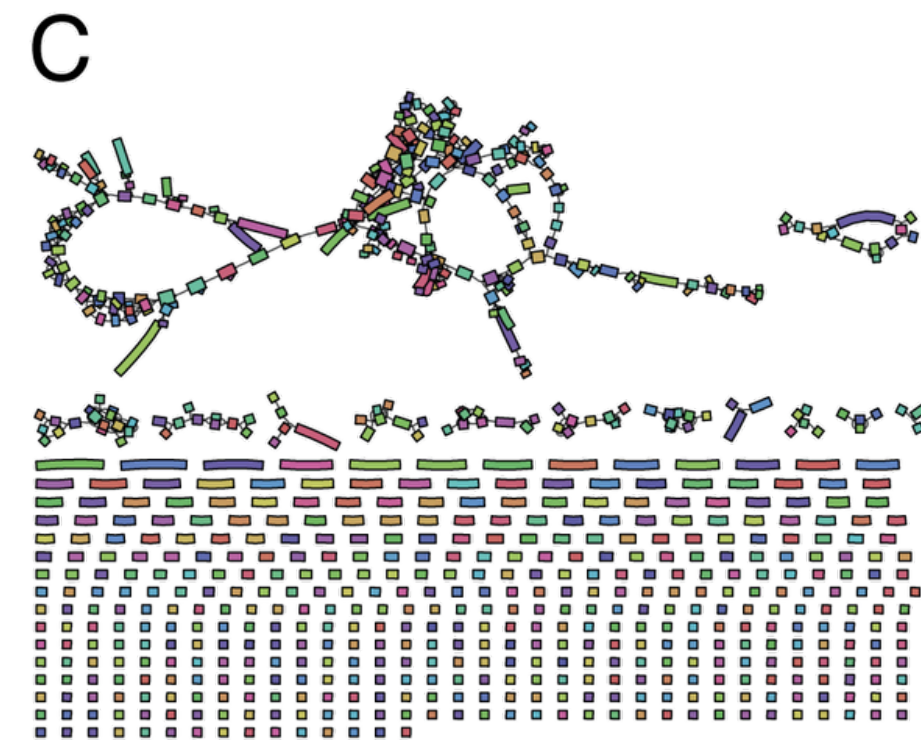
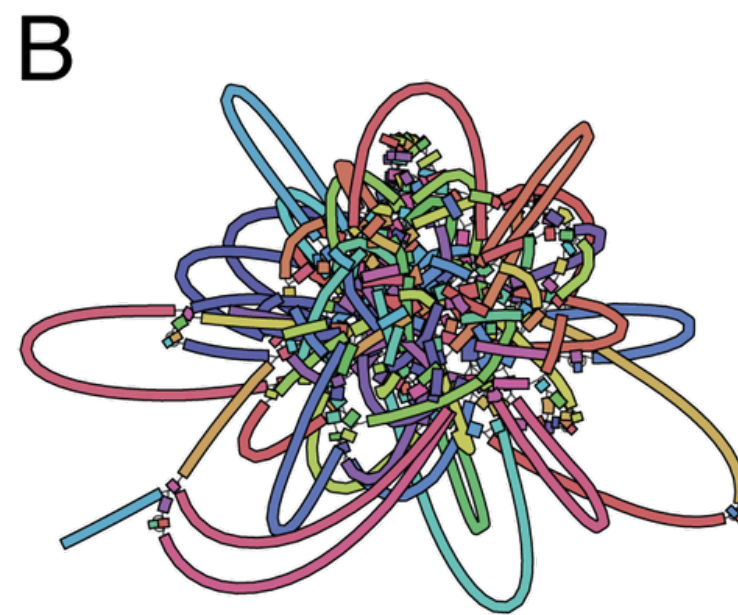
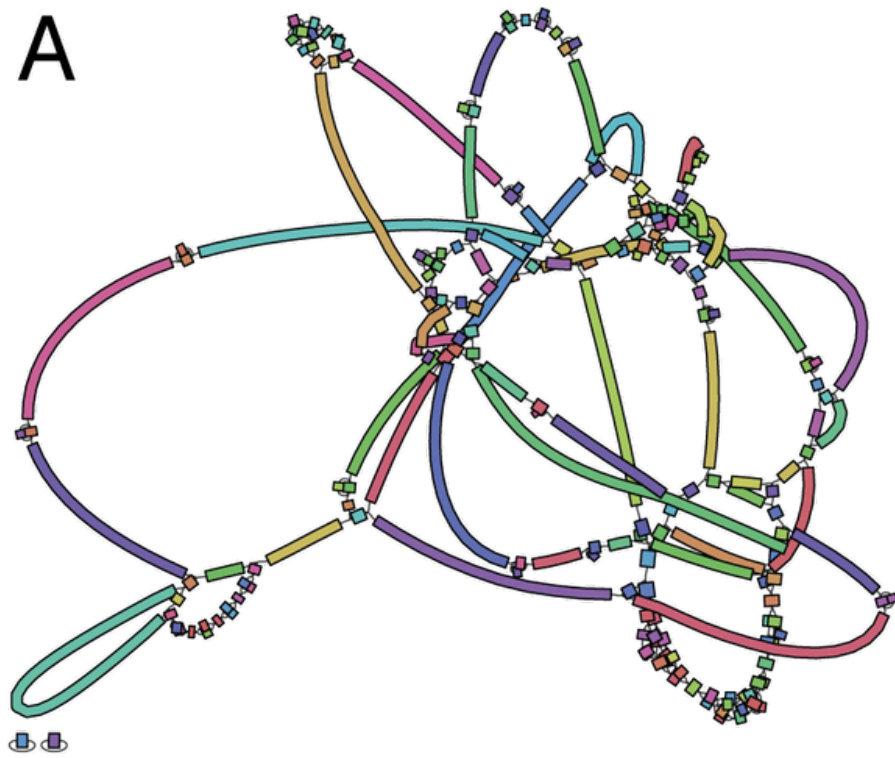
Ensamblaje de las lecturas cortas usando SPAdes, que construye un Grafo de De Bruijn

1. Divide las lecturas en k-mers
2. Construye el grafo
3. Prueba muchos valores de k



A. Short read assembly

Ensamblaje de las lecturas cortas usando SPAdes, que construye un Grafo de De Bruijn

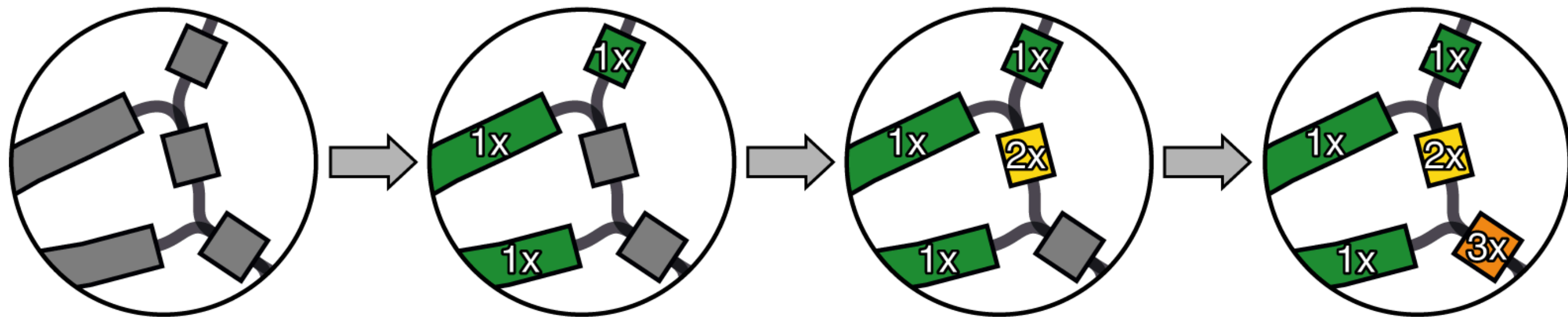


Unicycler

B. Multiplicity

Cuántas veces aparece cada contig en el genoma

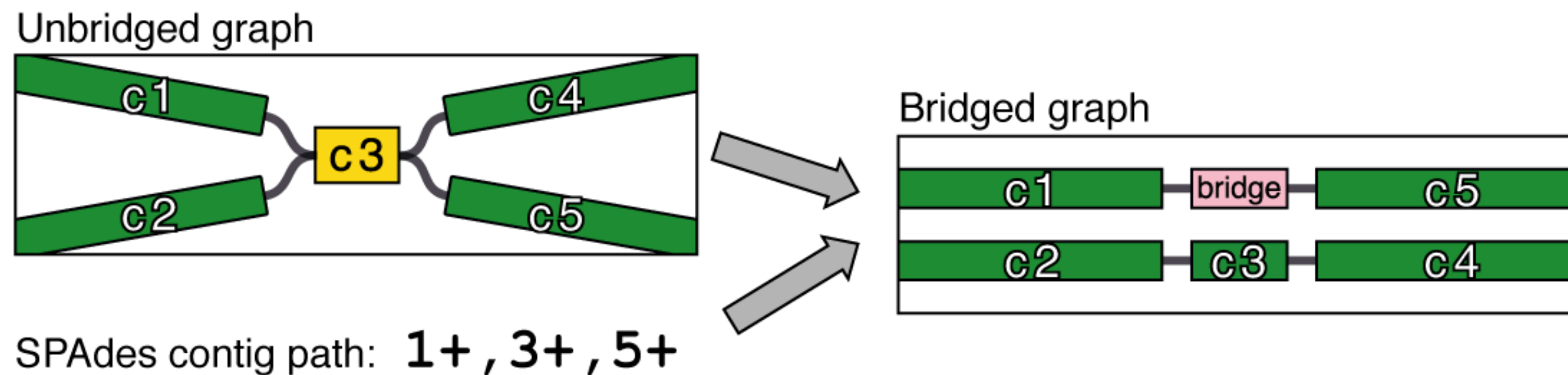
- profundidad de cobertura (depth)
- conexiones del grafo



Distinguir entre región única vs repetición

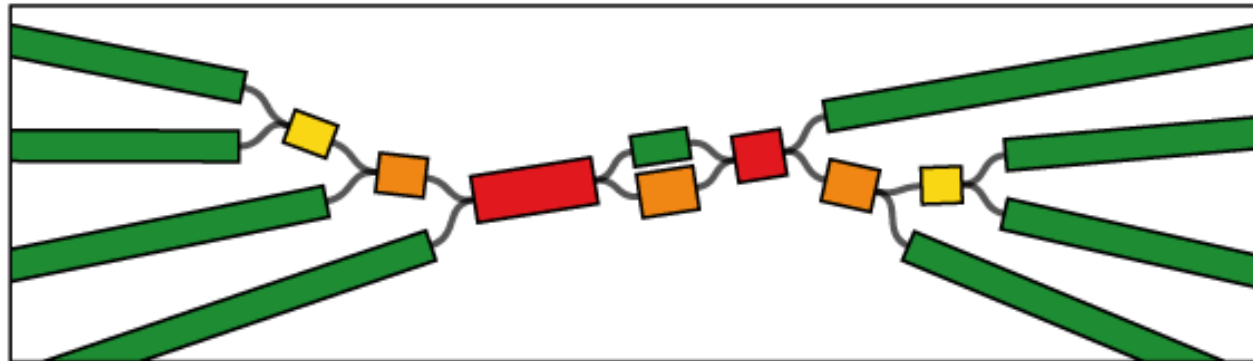
C. Short read bridging

Construye conexiones (bridges) entre contigs cuando existe evidencia de que están adyacentes en el genoma

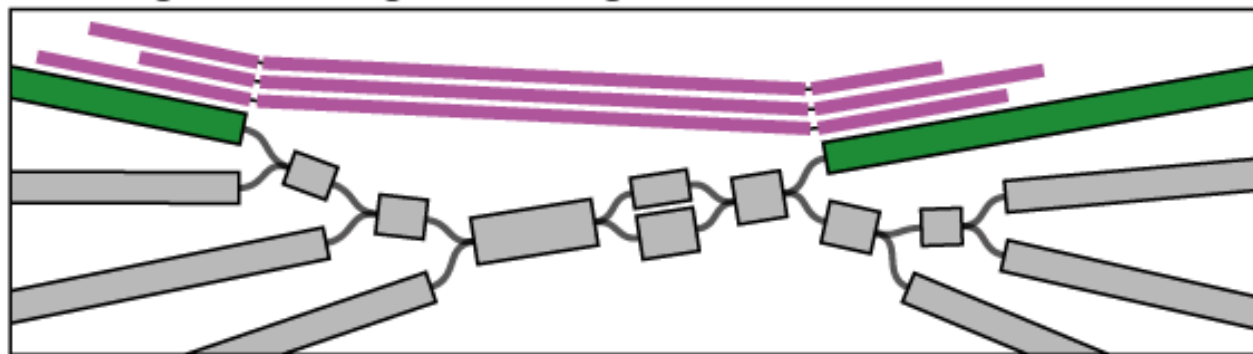


D. Direct long-read bridging

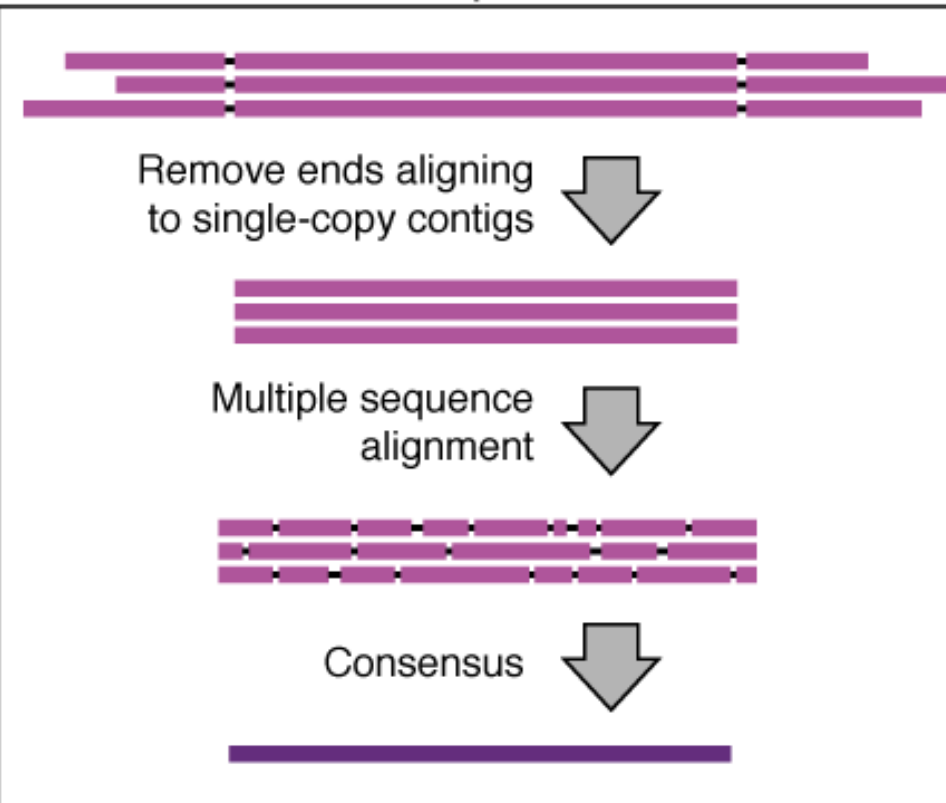
Repeat region in unbridged graph



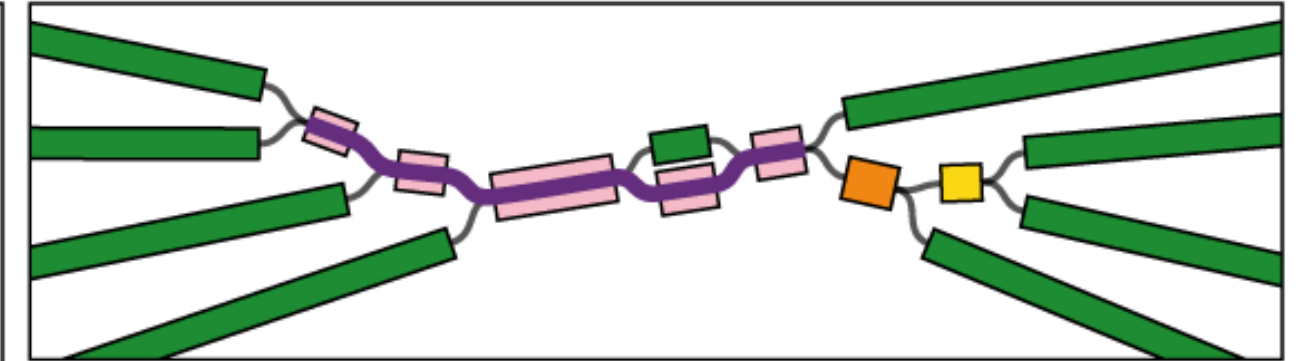
Semi-global long read alignment



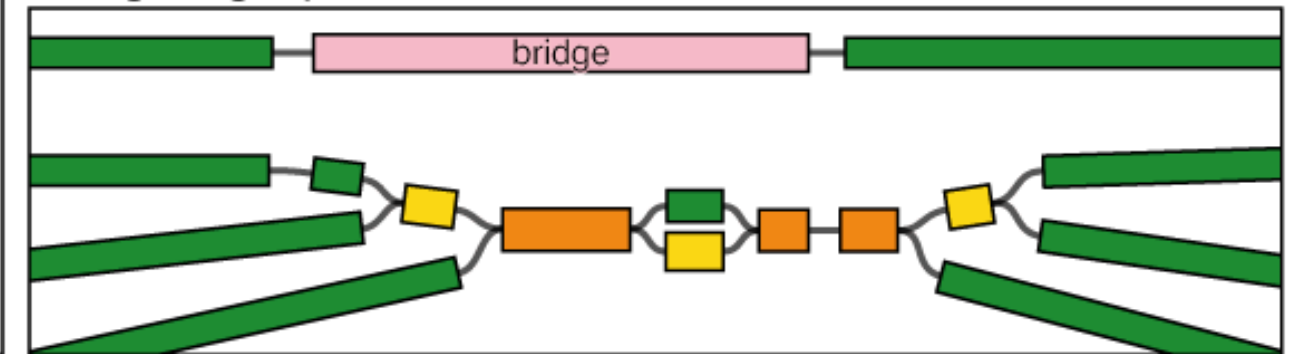
Consensus read sequence



Path finding



Bridged graph



```
>LC817487.1  Salmonella  sp
```

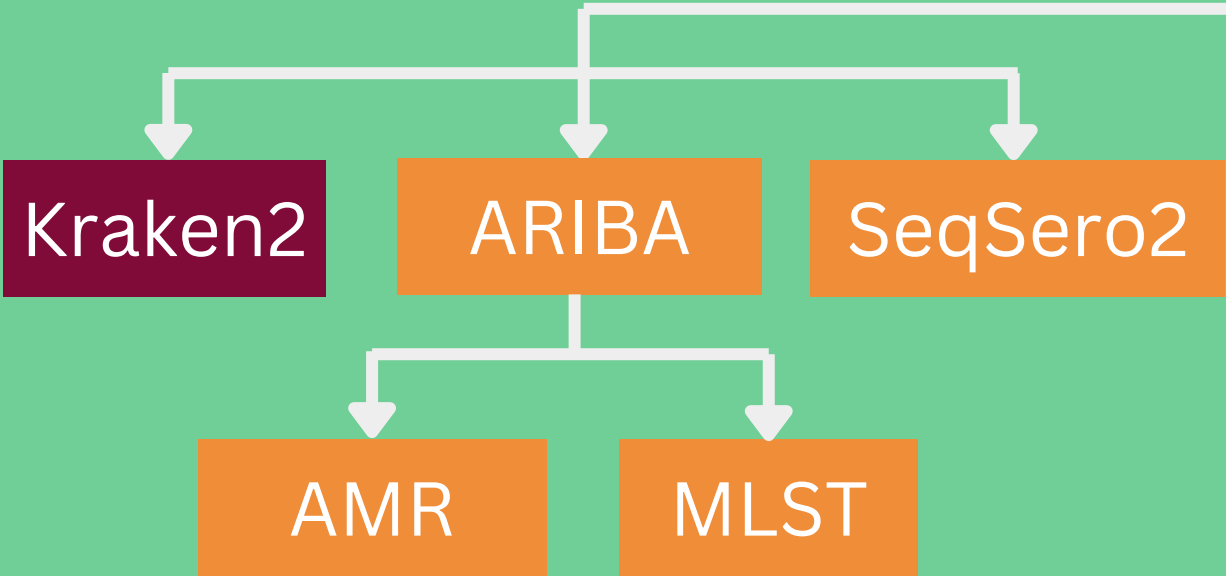
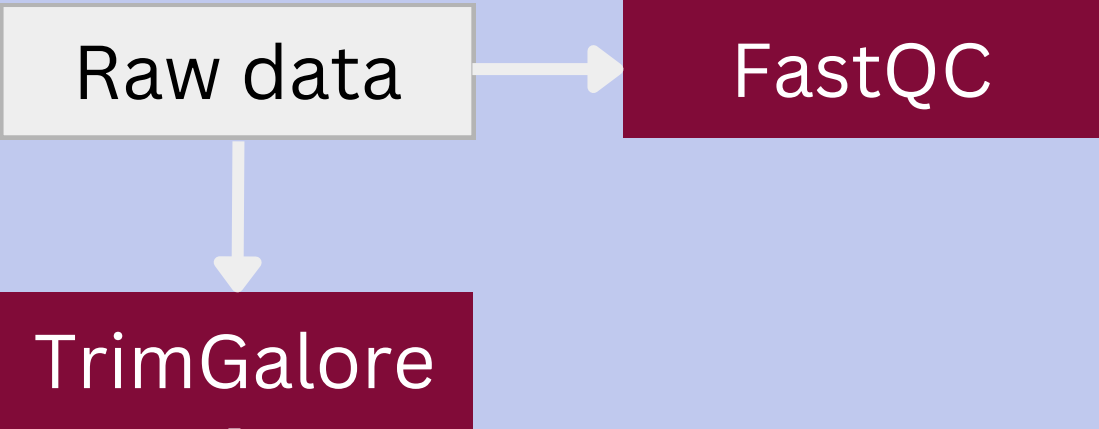
```
CTGGTGGTGCTGGGGCCGTAATGGTTCCGACTATTCCGCCCGCCGTGCTGGCCGCCTGTTTACG  
CGCTGACTGCTGTGAAATCTGGACTGACGTCGATGGCGTGTATACCTGTGACCCGCGGCCAGG  
TGCCGGACGCCAGGCTGCTGAAATCGATGTCCTACCAGGAAGCGATGGAACTCTCTTACTTC  
GGCGCCAAAGTCCTTCACCCTCGCACCATACGCCCATCGCCCAGTTCCAGATCCCCTGTCT  
GATTAAAAATACCGGCAATCCGCAGGCGCCAGGAACGCTGATCGGCGCGGTCCAGCGACGATG  
ATAATCTGCCGGTTAAAGGGATCTCTAACCTTAACAACATGGCGATGTTTAGCGTCTCCGGC  
CCGGGAATGAA
```

Preprocesamiento

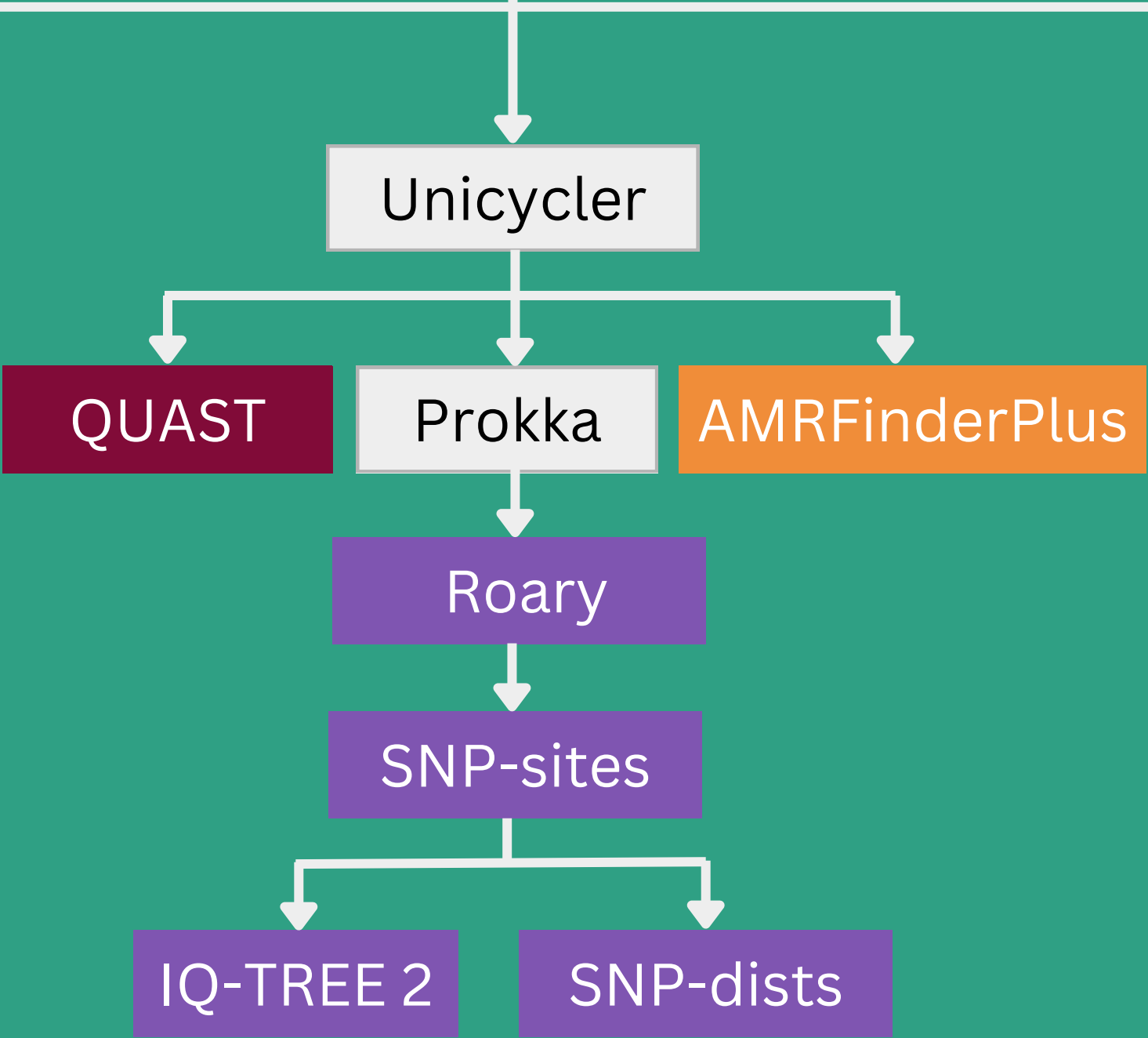
Control de calidad

Caracterización

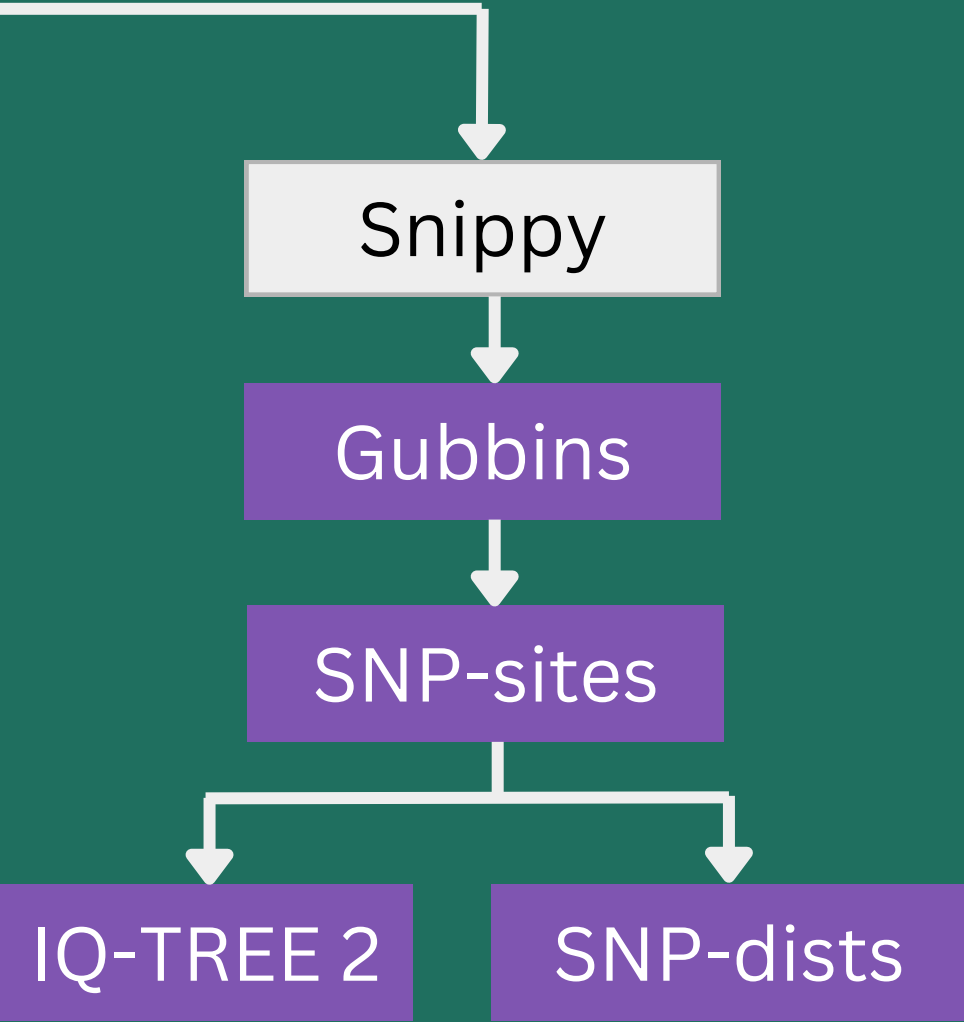
Comparación



Análisis a partir de lecturas



Análisis a partir de ensamblaje de novo



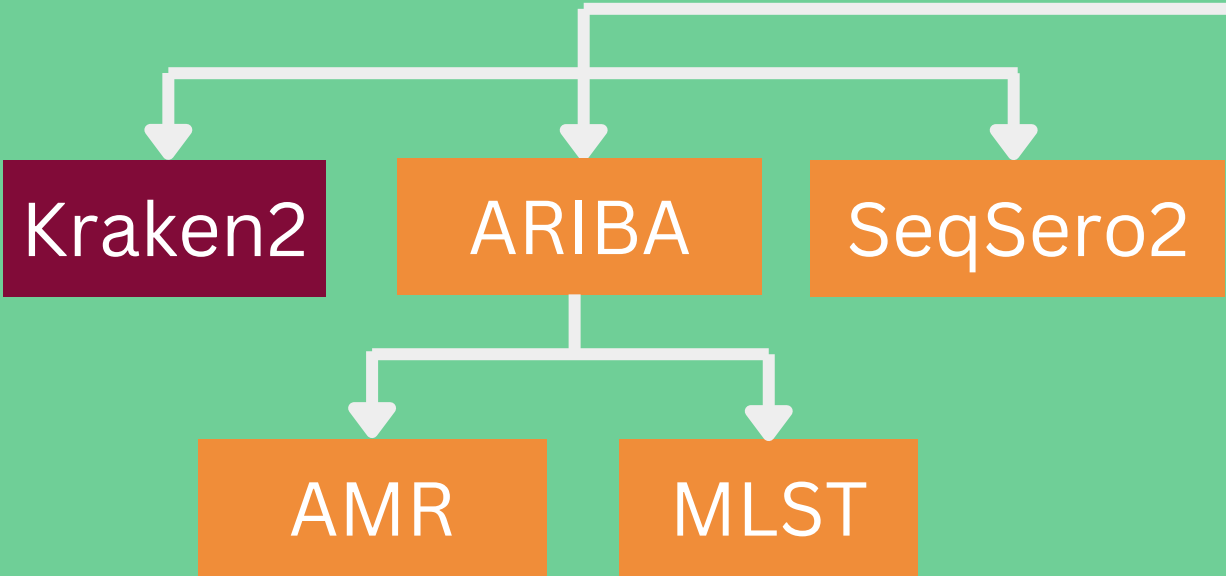
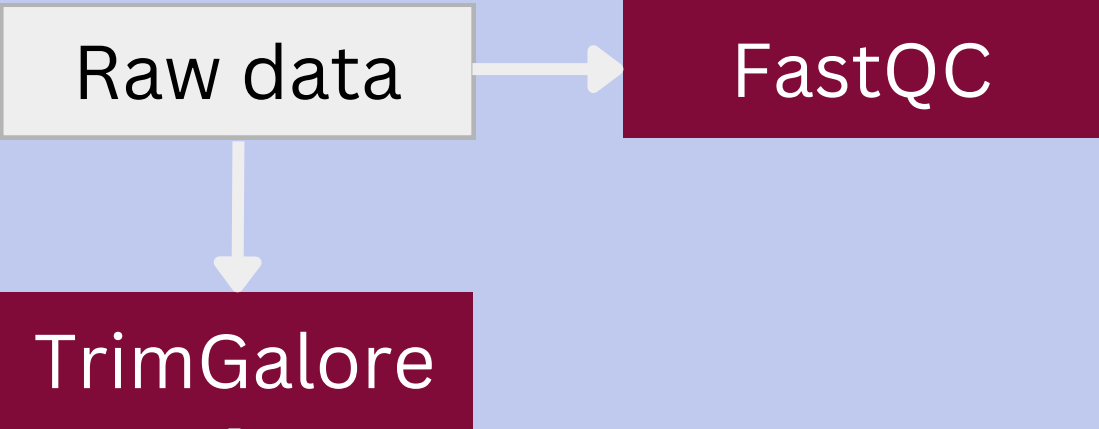
Análisis a partir de ensamblaje de referencia

Preprocesamiento

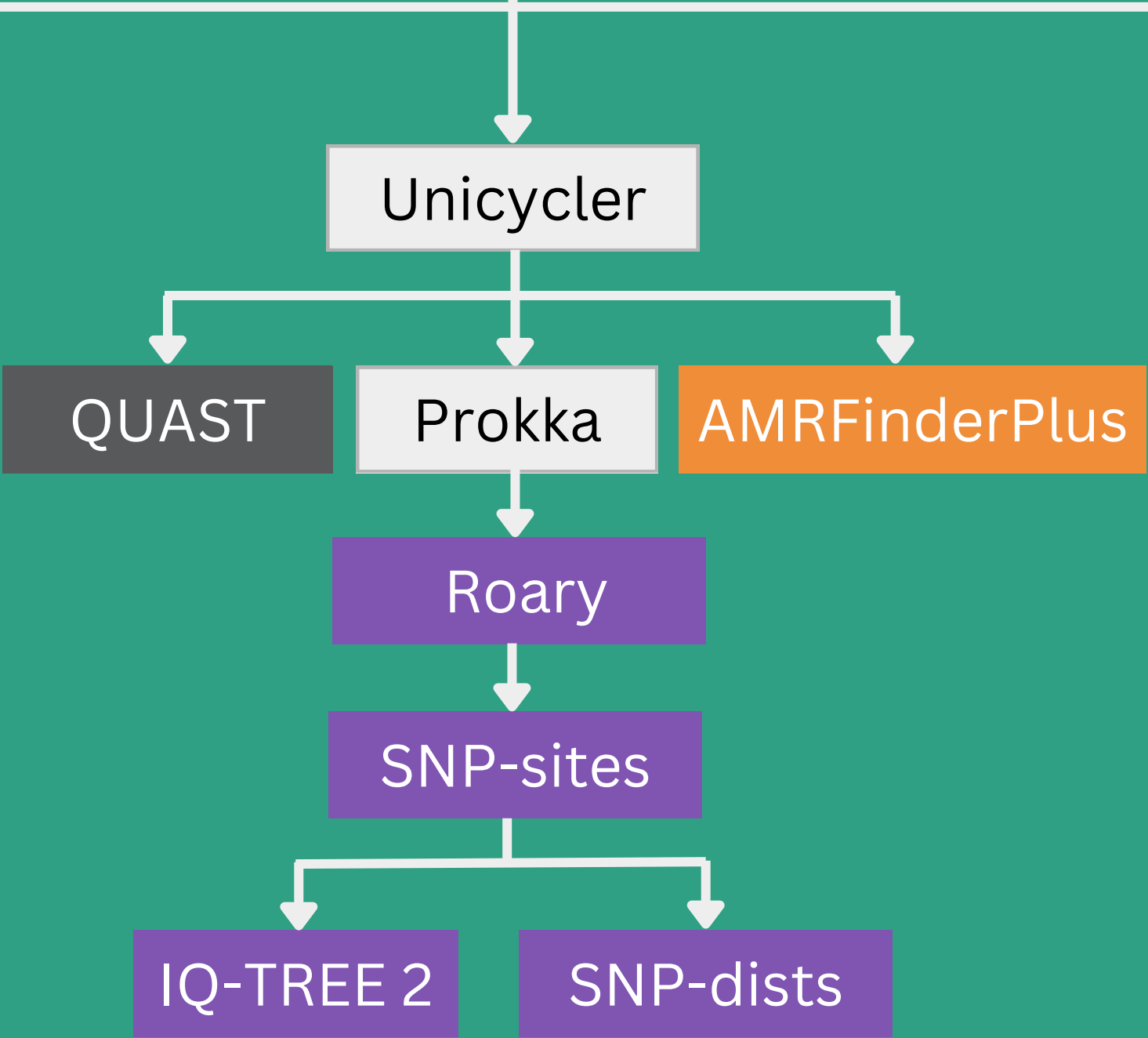
Control de calidad

Caracterización

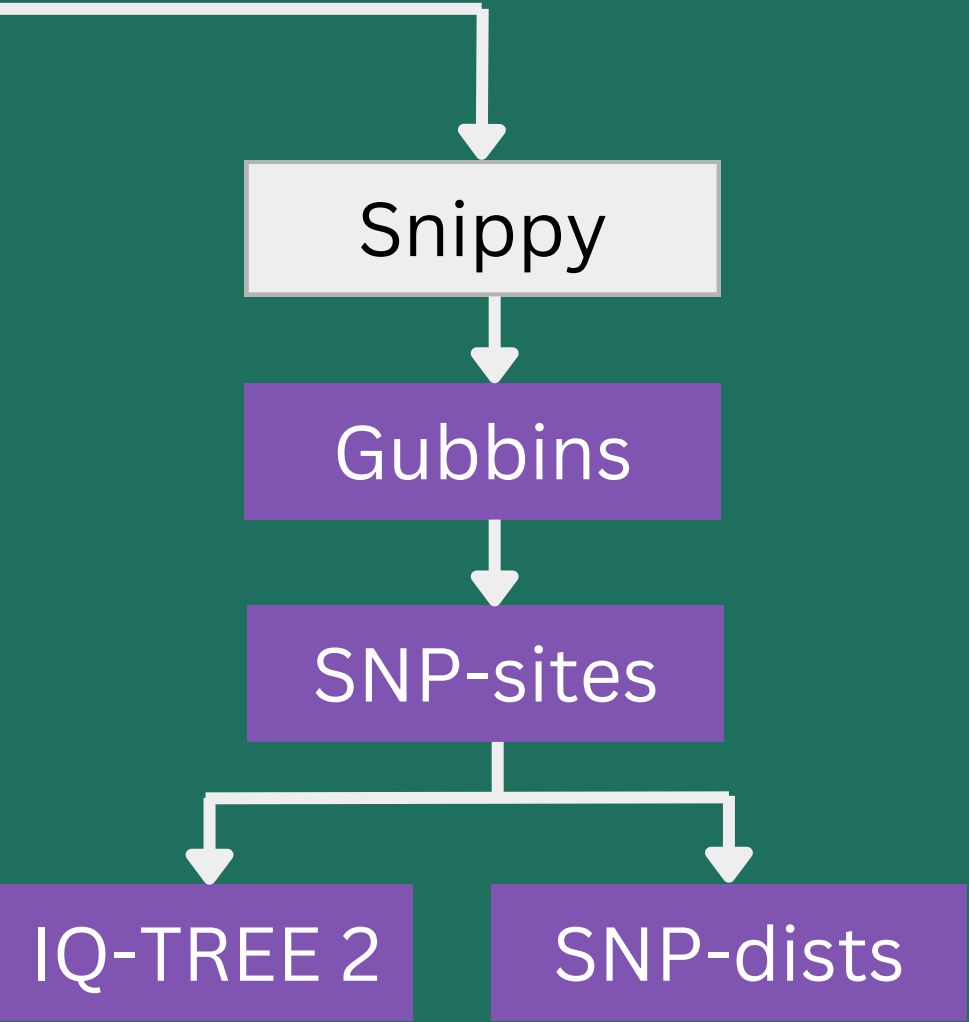
Comparación



Análisis a partir de lecturas



Análisis a partir de ensamblaje de novo



Análisis a partir de ensamblaje de referencia



QUAST

No. of contigs: Número total de contigs en el ensamble

Largest contig: Longitud en pb del contig más largo en el ensamblaje

Total length: Número total de bases en el ensamble

%GC: Proporción de nucleótidos G + C en el ensamble en relación al largo total del ensamble

N50: Longitud del contig más corto con el que se alcanza a cubrir el 50% del genoma

L50: Número de contigs con los que se llega al N50



QUAST

↓ **No. of contigs:** Número total de contigs en el ensamble

↑ **Largest contig:** Longitud en pb del contig más largo en el ensamblaje

Total length: Número total de bases en el ensamble

%GC: Proporción de nucleótidos G + C en el ensamble en relación al largo total del ensamble

↑ **N50:** Longitud del contig más corto con el que se alcanza a cubrir el 50% del genoma

↓ **L50:** Número de contigs con los que se llega al N50



N50 y L50

OMb

1Mb

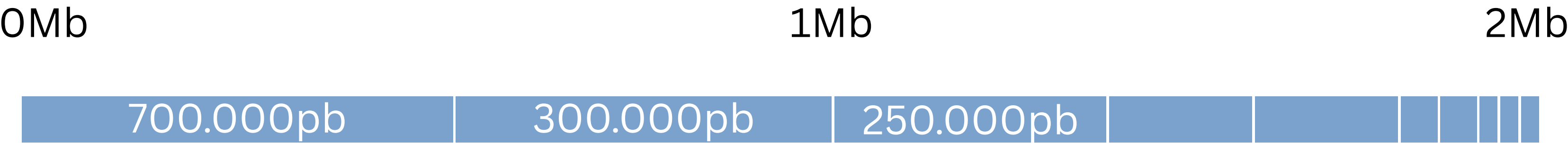
2Mb





QUAST

N50 y L50



N50: 300.000

L50: 2



Cobertura

Porcentaje del **genoma** sobre el cual se alinea (mapea) al menos UNA **lectura** de secuenciación.

Reads



Ref

Alta cobertura, secuenciación superficial



Cobertura

Porcentaje del **genoma** sobre el cual se alinea (mapea) al menos UNA **lectura** de secuenciación.

Cuánto del genoma de referencia se observa

Reads



Ref



Alta cobertura, secuenciación superficial

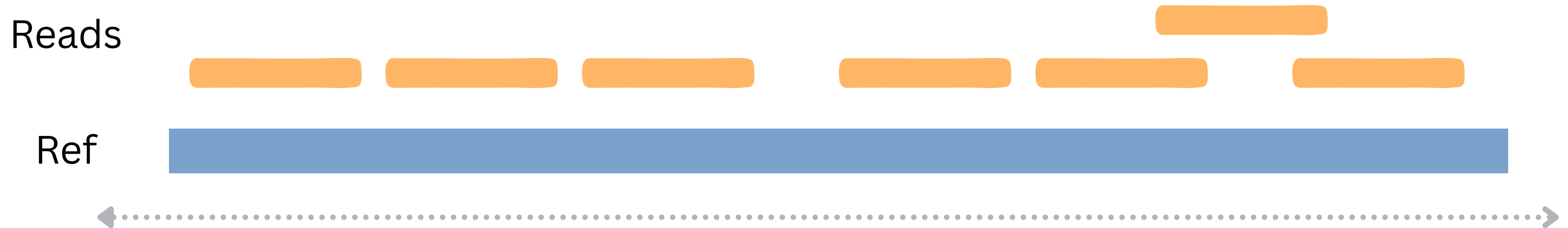


Cobertura

Porcentaje del **genoma** sobre el cual se alinea (mapea) al menos UNA **lectura** de secuenciación.

$$C = \frac{N * 2 * L}{G}$$

N= total de reads
L= longitud de read
G= tamaño del genoma

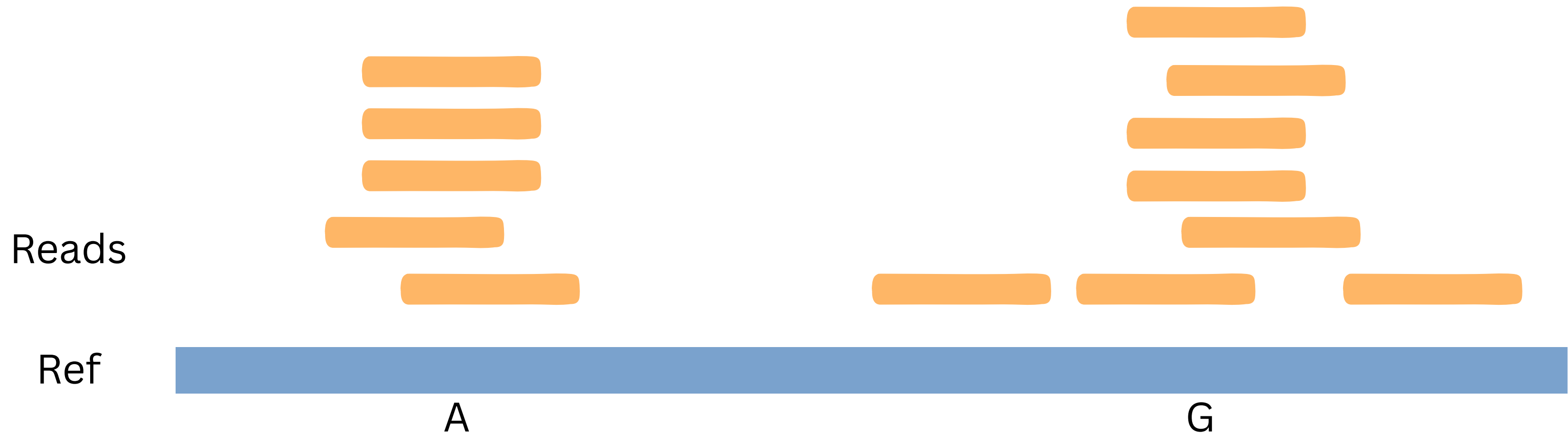


Alta cobertura, secuenciación superficial



Profundidad

Número de **lecturas** de secuenciación que se alinean en una región del **genoma**.

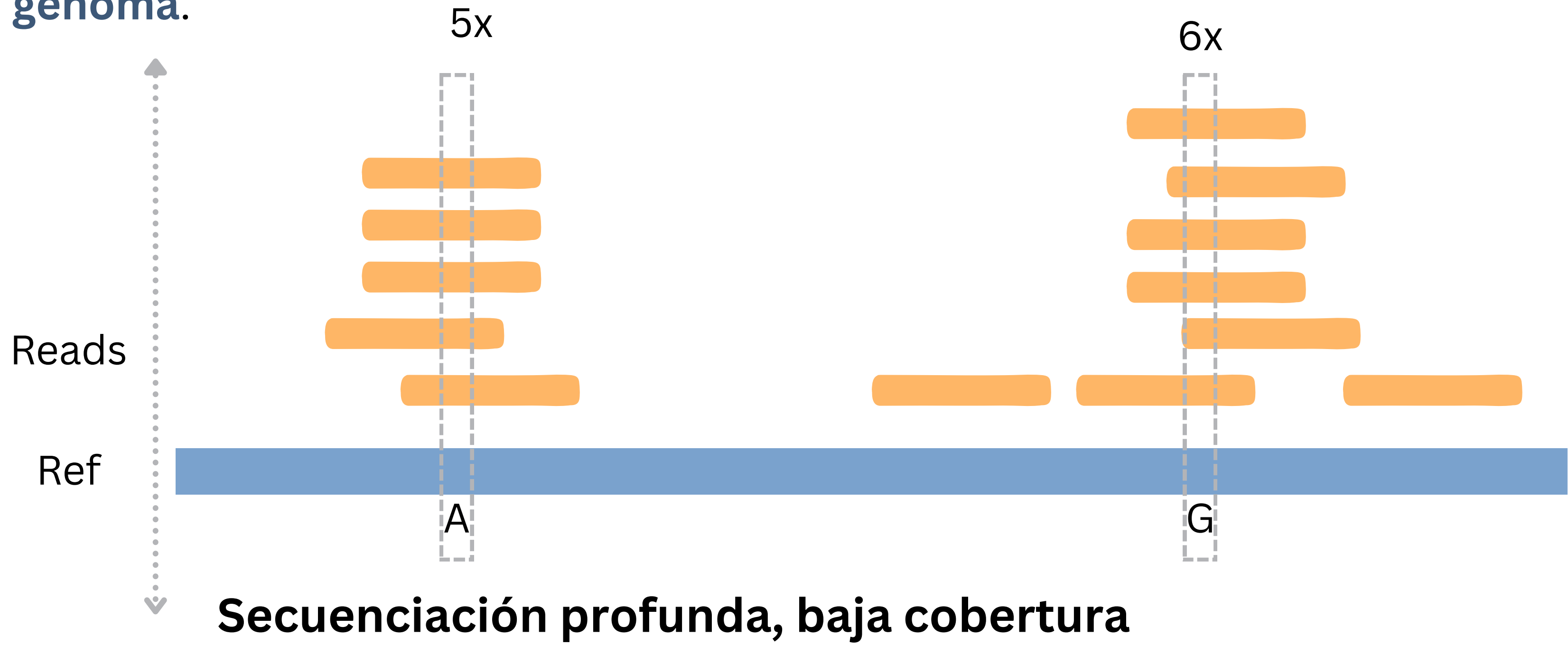


Secuenciación profunda, baja cobertura



Profundidad

Número de **lecturas** de secuenciación que se alinean en una región del **genoma**.

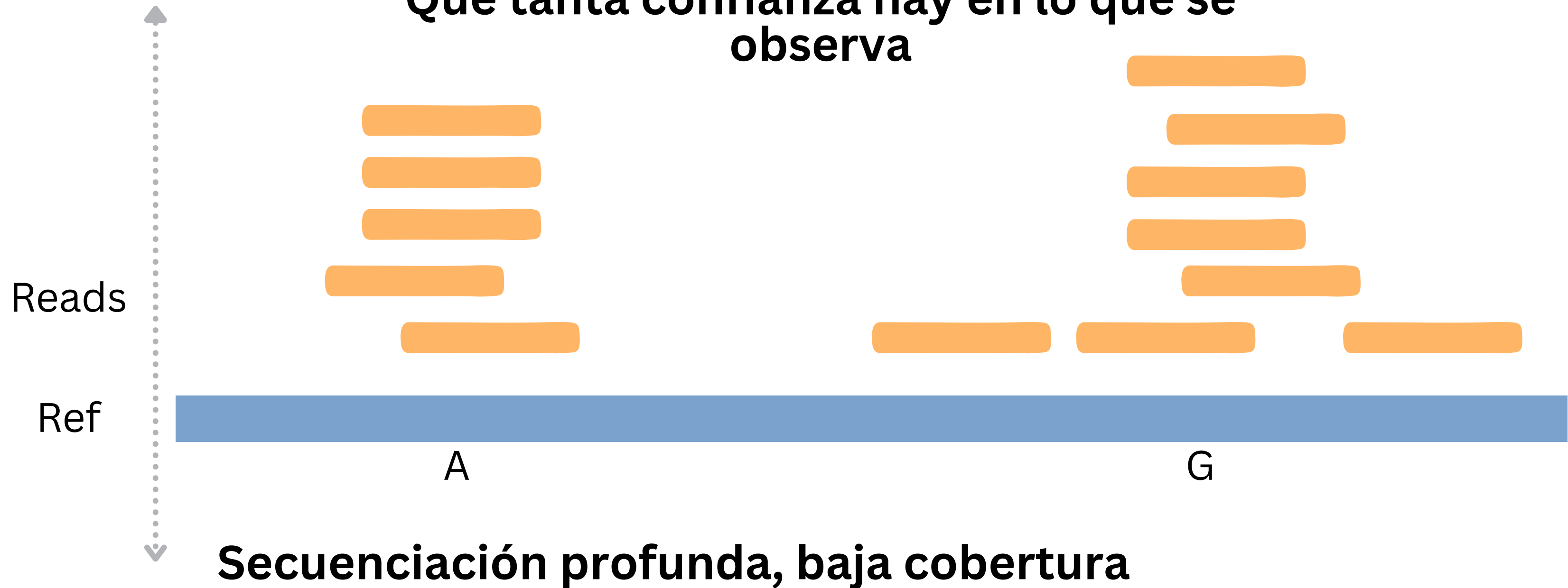




Profundidad

Número de **lecturas** de secuenciación que se alinean en una región del **genoma**.

Qué tanta confianza hay en lo que se observa



Capacitación de secuenciación genómica y análisis bioinformático

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